Big Ideas Math Middle School: Two-Method Block & SC-6 Workbook

45 Textbook Sections × 2 Full Practice Sheets (Practice A: Key Ideas + Practice B: Modeling Real Life) • 90 Two-Method Worked Examples

1. AUTHENTIC BIG IDEAS METHODS

Every worksheet uses the exact vocabulary and two core methods taught in Big Ideas Math: Factor Trees & Division Ladders, LCD Fraction Towers & Reciprocals, Tape Diagrams & Ratio Tables, Area Models & Balance Scales.

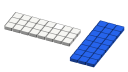
2. TWO WAYS = DOUBLE POINTS (+20)

Every Worked Example models TWO ways to solve: Method 1 (+10 pts — 1st Token) and Method 2 (+20 pts DOUBLE — 2nd Token) alongside step-by-step physical Block Kit instructions and SC-6 verification.

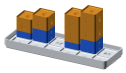
3. 2 SHEETS PER SECTION (900 PROBLEMS)

Each section includes Practice A (Key Ideas & Error Analysis) and Practice B (Modeling Real Life Applications) with generous open workspace to work out every problem and write your final answer.

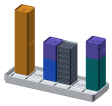
HANDS-ON MANIPULATIVE BLOCK KITS & SIMPLIFIED 2-PIECE QUEST-01 SCOREBOARD (+10 / +20 TOKENS)



COUNT-01: Array Tiles



PRIME-01: Prime Towers



FRAC-01: 84mm Fractions



ALGB-01: Algebra Stage



TIME-01: Geared Clock

UNIT / CHAPTER & BIG IDEAS ALIGNMENT	SECTIONS & WORKSHEETS (PRACTICE A & B)	TWO-METHOD DOUBLE POINTS FOCUS
Unit 1: Numerical Expressions & Factors Big Ideas Math Grade 6 — Chapter 1 • Counting & Array Studio (COUNT-01) + Prime Factor Studio (PRIME-01)	[ ] Sec 1.1 (pp.2-3): Whole Number Operations & Division with Remainders [ ] Sec 1.2 (pp.4-5): Powers and Exponents [ ] Sec 1.3 (pp.6-7): Order of Operations [ ] Sec 1.4 (pp.8-9): Prime Factorization [ ] Sec 1.5 (pp.10-11): Greatest Common Factor (GCF) & Least Common Multiple (L..	**Arrays vs Div Check; Trees vs Ladders** Kit: Counting & Array Studio (COUNT-01) **[ ] 10 Sheets (100 Problems)**
Unit 2: Fractions & Decimals Big Ideas Math Grade 6 — Chapter 2 • Fraction Studio (FRAC-01) + Place Value Studio (PLACE-01)	[ ] Sec 2.1 (pp.12-13): Equivalent Fractions, Mixed Numbers & LCD Addition [ ] Sec 2.2 (pp.14-15): Multiplying Fractions and Mixed Numbers [ ] Sec 2.3 (pp.16-17): Dividing Fractions and Mixed Numbers [ ] Sec 2.4 (pp.18-19): Adding, Subtracting, and Multiplying Decimals [ ] Sec 2.5 (pp.20-21): Dividing Whole Numbers and Decimals	**LCD Towers vs Cross-Mult; Reciprocals** Kit: Fraction Studio (FRAC-01) **[ ] 10 Sheets (100 Problems)**
Unit 3: Ratios, Rates & Percents Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Counting & Array Tiles (COUNT-01) + Fraction Studio (FRAC-01)	[ ] Sec 3.1 (pp.22-23): Ratios and Tape Diagrams [ ] Sec 3.2 (pp.24-25): Using Ratio Tables and Double Number Lines [ ] Sec 3.3 (pp.26-27): Rates, Unit Rates, and Comparing Rates [ ] Sec 3.4 (pp.28-29): Percents, Fractions, and Decimals [ ] Sec 3.5 (pp.30-31): Solving Percent Problems, Discounts & Conversions	**Tape Diagrams vs Ratio Tables & Rates** Kit: Counting & Array Tiles (COUNT-01) **[ ] 10 Sheets (100 Problems)**
Unit 4: Algebraic Expressions & Properties Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Algebra Boxes & Factoring Stage (ALGB-01) + Counting Tiles (COUNT-01)	[ ] Sec 4.1 (pp.32-33): Writing and Evaluating Algebraic Expressions [ ] Sec 4.2 (pp.34-35): Simplifying Expressions by Combining Like Terms [ ] Sec 4.3 (pp.36-37): The Distributive Property [ ] Sec 4.4 (pp.38-39): Factoring Expressions Using the GCF	**Substitution, Like Terms & Area vs GCF** Kit: Algebra Boxes & Factoring Stage (AL.. **[ ] 8 Sheets (80 Problems)**
Unit 5: Areas of Polygons & Elapsed Time Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Counting & Array Tiles (COUNT-01) + Planetary Geared Clock (TIME-01)	[ ] Sec 5.1 (pp.40-41): Areas of Parallelograms and Rectangles [ ] Sec 5.2 (pp.42-43): Areas of Triangles [ ] Sec 5.3 (pp.44-45): Areas of Trapezoids and Composite Figures [ ] Sec 5.4 (pp.46-47): Polygons in the Coordinate Plane & Elapsed Time	**Cut-and-Slide Grids vs Polygon Formulas** Kit: Counting & Array Tiles (COUNT-01) **[ ] 8 Sheets (80 Problems)**
Unit 6: Integers, Rational Numbers & The Coordinate Plane Big Ideas Math Grade 6 — Chapter 6 • Ten-Frame Bicolor Counters (BASE-10) + Number Rack (RACK-01)	[ ] Sec 6.1 (pp.48-49): Positive and Negative Integers & Opposites [ ] Sec 6.2 (pp.50-51): Comparing and Ordering Integers & Rational Numbers [ ] Sec 6.3 (pp.52-53): Absolute Value and Magnitude [ ] Sec 6.4 (pp.54-55): The Four-Quadrant Coordinate Plane & Reflections [ ] Sec 6.5 (pp.56-57): Distances in the Coordinate Plane	**Number Lines vs Opposites & |x| Sums** Kit: Ten-Frame Bicolor Counters (BASE-10) **[ ] 10 Sheets (100 Problems)**
Unit 7: Equations & Inequalities Big Ideas Math Grade 6 — Chapter 7 • Algebra Boxes & Stage (ALGB-01) + Number Rack (RACK-01)	[ ] Sec 7.1 (pp.58-59): Writing Equations in One Variable & Testing Solutions [ ] Sec 7.2 (pp.60-61): Solving Equations Using Addition or Subtraction [ ] Sec 7.3 (pp.62-63): Solving Equations Using Multiplication or Division [ ] Sec 7.4 (pp.64-65): Writing and Graphing Inequalities [ ] Sec 7.5 (pp.66-67): Writing Equations in Two Variables (y = kx and y = x + b)	**Mental Inspection vs Inverse Operations** Kit: Algebra Boxes & Stage (ALGB-01) **[ ] 10 Sheets (100 Problems)**
Unit 8: Surface Area & Volume of 3D Solids Big Ideas Math Grade 6 — Chapter 8 • Counting & Array Tiles (COUNT-01) + Gridfinity 3x1 Storage Boxes	[ ] Sec 8.1 (pp.68-69): Three-Dimensional Figures, Nets & Rectangular Prisms [ ] Sec 8.2 (pp.70-71): Surface Areas of Triangular Prisms and Pyramids [ ] Sec 8.3 (pp.72-73): Volumes of Rectangular Prisms [ ] Sec 8.4 (pp.74-75): Circles, Cones, and the Pythagorean Theorem	**Unfolded 2D Nets vs Prism/Cone Formulas** Kit: Counting & Array Tiles (COUNT-01) **[ ] 8 Sheets (80 Problems)**
Unit 9: Statistical Measures & Data Displays Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Counting & Array Tiles (COUNT-01) + Probability Dice Studio	[ ] Sec 9.1 (pp.76-77): Statistical Questions, Dot Plots, and the Mean [ ] Sec 9.2 (pp.78-79): Measures of Center: Median, Mode, and Range [ ] Sec 9.3 (pp.80-81): Quartiles, Interquartile Range (IQR) & Box Plots [ ] Sec 9.4 (pp.82-83): Mean Absolute Deviation (MAD)	**Fair-Share Leveling vs IQR & MAD Sums** Kit: Counting & Array Tiles (COUNT-01) **[ ] 8 Sheets (80 Problems)**
Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equati.. Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Ten-Frame Zero-Pair Counters (BASE-10) + Algebra Stage (ALGB-01) + Probability	[ ] Sec 10.1 (pp.84-85): Adding and Subtracting Rational Numbers [ ] Sec 10.2 (pp.86-87): Multiplying and Dividing Rational Numbers [ ] Sec 10.3 (pp.88-89): Solving Two-Step Equations (ax + b = c) [ ] Sec 10.4 (pp.90-91): Theoretical Probability, Combinations & Permutations	**Zero-Pair Chips vs Rules & Sample Space** Kit: Ten-Frame Zero-Pair Counters (BASE-.. **[ ] 8 Sheets (80 Problems)**

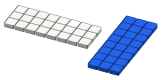
BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 1.1 — Practice A: Whole Number Operations & Division with Remainders

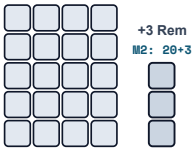
Key Vocabulary: dividend • divisor • quotient • remainder

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)



M1: 5×4 Array (20)



DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Deal 23 COUNT-01 tiles into rows of 4 -> 5 full rows (20 tiles) + 3 leftover tiles (5 R 3).

Block Way 2 (+20 pts): Push the 5×4 = 20 tile array together and snap the 3 remainder tiles back on (20 + 3 = 23).

Both Ways Proof: Dealing out rows (Division) and re-joining 5×4 + 3 (Multiplication Check) use the exact same 23 tiles!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Divide 23 ÷ 4 and find the quotient and remainder.

METHOD 1 (+10 pts — 1st Token): Equal-Groups Array Model

Build equal rows of 4 tiles from 23 tiles: 5 full rows use 5 × 4 = 20 tiles, leaving 3 tiles left over -> 5 R 3.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Division Algorithm & Check

Divide 23 ÷ 4 = 5 with remainder 23 - 20 = 3. Check: (Quotient × Divisor) + Remainder = (5 × 4) + 3 = 23.

SC-6 Keystrokes: 23 ENTER 4 g ÷R

Answer: 5 R 3

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 1.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Equal-Rows Array Model | Method 2: Division Algor..

1 Arrange 31 tiles into rows of 6. Find Quotient & Remainder.

Ans: _____

2 Divide 47 ÷ 5. Write the quotient and remainder, and check your answer by multiplying.

Ans: _____

3 Divide 58 ÷ 7. Find the quotient and remainder, then check using (Quotient × Divisor) + Remainder.

Ans: _____

4 Divide 85 ÷ 9. How many full groups of 9 and how many left?

Ans: _____

5 Divide 146 ÷ 12. Find the Quotient and Remainder.

Ans: _____

6 Divide 250 ÷ 36. Find the quotient and remainder, and verify with the Division Algorithm.

Ans: _____

7 Find the missing dividend: Quotient = 14, Divisor = 8, Remainder = 5.

Ans: _____

8 Error Analysis: Sam says 75 ÷ 8 = 8 R 11. Explain why a remainder of 11 is impossible and find the correct quotient.

Ans: _____

9 A 3D spool has 300g of PLA. Each box takes 42g. How many full boxes can you print, and how much PLA is left?

Ans: _____

10 115 students ride vans that hold 14 kids each. How many vans must be booked so nobody is left behind?

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 1.1 — Practice B: Modeling Real Life (Whole Number Operations & Division with Remainders)

Textbook Methods: Method 1: Equal-Rows Array Model | Method 2: Division Algorithm & Multiplication Check

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: INTERPRETING THE REMAINDER (135 ÷ 8) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A robotics tournament has 135 students riding in vans that hold 8 students each. Find 135 ÷ 8 and how many vans are needed.

METHOD 1 (+10 pts — 1st Token): Long Division & Remainder Meaning

Divide 135 ÷ 8: 8 × 16 = 128, with remainder 135 - 128 = 7 (16 R 7). Since 7 students cannot be left behind, round up to 17 vans.

COUNT-01 Way 1: Deal 135 into rows of 8 -> 16 full rows (128) + 7 leftover tiles in a 17th partial row.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Multiplication Check & Capacity Bounds

Check 16 vans: 16 × 8 = 128 seats (7 seats short). Check 17 vans: 17 × 8 = 136 seats (fits all 135 students with 1 empty seat!).

COUNT-01 Way 2: Compare 16 full rows of 8 (128 seats, 7 short) vs. 17 full rows of 8 (136 seats, 1 spare seat!).

SC-6 Keystrokes: 135 ENTER 8 g ÷R f LASTX × +

Answer: 16 R 7 -> 17 vans needed

SECTION 1.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Drone Battery Packs: A robotics team has 53 lithium cells. Each drone battery pack uses 6 cells. Find the quotient and remainder. How many full packs can they build?

Ans: _____

2 Arcade Prize Tickets: You win 94 tickets and trade them for mini-figures that cost 8 tickets each. How many mini-figures do you get, and how many tickets remain?

Ans: _____

3 Camp Canoe Fleet: 78 campers are crossing a lake in canoes that hold 4 campers each. Divide 78 ÷ 4. How many canoes are needed so every camper has a seat?

Ans: _____

4 Cookie Bakery Tray: A baker places 110 warm cookies onto cooling racks that hold 12 cookies per rack. How many racks are completely filled, and how many cookies go on the last rack?

Ans: _____

5 3D Printer Spool: A filament spool has 450 meters of PLA. Each chess knight uses 16 meters. How many complete knights can be printed, and how much PLA is left over?

Ans: _____

6 Trading Card Binder: You have 165 collectible cards and put 9 cards into each plastic binder page. How many pages are completely full, and how many pages do you need in all?

Ans: _____

7 Division Algorithm Check: When a mystery number of marbles is divided into bags of 15, the quotient is 12 and the remainder is 7. Use (Quotient × Divisor) + Remainder to find the total marbles.

Ans: _____

8 Theme Park Coaster: A roller coaster train holds 24 riders. There are 200 people in line. How many full trains run, and how many riders are on the final train?

Ans: _____

9 Pizza Party Slices: Each large pizza is cut into 8 slices. If 67 students each eat 1 slice, how many whole pizzas must the school order, and how many slices will be left over?

Ans: _____

10 Maker Lab Drawer Bins: You pack 325 screws into Gridfinity bins that hold 28 screws each. Find the quotient and remainder, and check your answer by multiplying.

Ans: _____

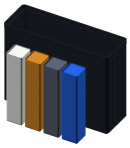
BLOCK KIT: COUNT-01 • Counting & Array Tiles

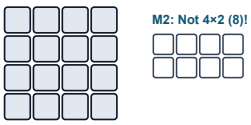
Section 1.2 — Practice A: Powers and Exponents

Key Vocabulary: power • base • exponent • perfect square • perfect cube

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)





DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Lay 4 COUNT-01 tiles per row and repeat 4 rows to build a 4×4 = 16 square array (4^2).

Block Way 2 (+20 pts): Stack 4 of those 4×4 square layers high ($4^3 = 64$), and place a 4×2 = 8 array beside it to prove $4^2 \neq 4 \times 2$!

Both Ways Proof: Repeated factor groups (Method 1) physically form a 2D Square (x^2) and 3D Cube (x^3) on the baseplate!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Write 4^2 and 4^3 as repeated multiplication and find their values. Compare 4^2 to 4×2.

METHOD 1 (+10 pts — 1st Token): Write as Repeated Factors

Identify Base = 4, Exponent = 2 or 3. Expand: $4^2 = 4 \times 4 = 16$, and $4^3 = 4 \times 4 \times 4 = 16 \times 4 = 64$.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Perfect Square / Cube Model

Count the area of a 4 × 4 square (16 units²) and volume of a 4 × 4 × 4 cube (64 units³), or use [x^2] and [g][x^3].

SC-6 Keystrokes: `4` `x²` `4` `g` `x³`

Answer: $4^2 = 16$, $4^3 = 64$

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 1.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Repeated Multiplication Factors | Method 2: 2D Sq..

1 Write 5^2 as repeated multiplication and find how many tiles are in a 5 × 5 square.

Ans: _____

2 Evaluate 7^2 and compare it to 7 × 2.

Ans: _____

3 Evaluate 3^3 (a 3 × 3 × 3 cube of blocks).

Ans: _____

4 Evaluate the power 5^3 by writing the repeated multiplication factors.

Ans: _____

5 Write 2^5 as repeated multiplication and find the value of the power.

Ans: _____

6 Write 6^4 as repeated multiplication and evaluate the power.

Ans: _____

7 Evaluate 10^4 and 1^4. How many zeros are in 10^4?

Ans: _____

8 Error Analysis: Mia wrote 2^4 = 8 because 2 × 4 = 8. Describe and correct her error.

Ans: _____

9 A cell doubles every hour (2^n). How many cells are there after 6 hours (2^6)? Solve 2 ways!

Ans: _____

10 Modeling Real Life: A giant Rubik's display cube is built from 216 unit cubes. What is the edge length of the cube?

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 1.2 — Practice B: Modeling Real Life (Powers and Exponents)

Textbook Methods: Method 1: Repeated Multiplication Factors | Method 2: 2D Square / 3D Cube Array & Power Key

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: DOUBLING & TRIPLING GROWTH (EVALUATE 3^4) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A video clip is shared by 3 friends in Round 1, and triples every round for 4 rounds (3^4). How many shares happen in Round 4?

METHOD 1 (+10 pts — 1st Token): Step-by-Step Repeated Multiplication
Write four factors of 3: 3^4 = 3 × 3 × 3 × 3. Multiply in pairs: (3 × 3) × (3 × 3) = 9 × 9 = 81 shares.

COUNT-01 Way 1: Build a 3×3 square (9) and another 3×3 square (9), then form a 9×9 = 81 array.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Constant Multiplier Ladder ([g] [REP])
Round 1 = 3, Round 2 = 9, Round 3 = 27, Round 4 = 81. Verify on SC-6 with 3 [ENTER] 4 [g] [y^x] -> 81.

COUNT-01 Way 2: Triple your tile stack each round: 3 tiles -> 9 tiles -> 27 tiles -> 81 tiles!

SC-6 Keystrokes: 3 ENTER 4 g y^x

Answer: 81 shares

SECTION 1.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Pixel Art Canvas: A retro game sprite is drawn on an 8 × 8 square grid (8^2). Write 8^2 as repeated multiplication and find the total number of pixels.

Ans: _____

2 Minecraft Storage Room: You dig a cubic room measuring 6 blocks long, 6 blocks wide, and 6 blocks tall (6^3). How many blocks do you mine in all?

Ans: _____

3 Cell Division Lab: A single bacterium splits into 2 bacteria every hour. After 6 hours, there are 2^6 bacteria. Evaluate 2^6.

Ans: _____

4 Tournament Bracket: A tennis tournament has 4^4 players in the opening round. Write 4^4 as repeated factors and find how many players entered.

Ans: _____

5 Compare Squares: A patio has a 12 ft × 12 ft main deck (12^2) and a 5 ft × 5 ft grill pad (5^2). What is the combined area of both squares?

Ans: _____

6 Kindness Chain Text: On Day 1, 5 students each invite 5 friends. Each day the number multiplies by 5. How many students are invited on Day 4 (5^4)?

Ans: _____

7 Perfect Square Garden: A square community garden has an area of 196 square meters. What is the side length of the square garden?

Ans: _____

8 Cube vs. Square Showdown: Which is greater: 2^5 or 5^2? Evaluate both powers and find their difference.

Ans: _____

9 Metric Data Byte: One kilobyte in binary memory is $2^{1□}$ bytes. Evaluate $2^{1□}$ by squaring 2^5 (32 × 32) or using the power key.

Ans: _____

10 Sculpture Pedestal: A sculptor stacks a 3^3-inch cube on top of a 7^3-inch cube. What is the combined volume of the two cubes?

Ans: _____

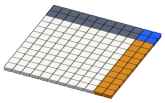
BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 1.3 — Practice A: Order of Operations

Key Vocabulary: numerical expression • evaluate • order of operations • grouping symbols

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)



M1 ZONE 1 TILES:
8 + 4 = 12

M2 ZONE 2 / SPLIT:
9 - 3 = 6

M1 = M2 ==> 72

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): In Left Zone of COUNT-01 build (8 + 4) = 12 tiles; in Right Zone build (9 - 3) = 6 tiles -> 12 × 6 = 72.

Block Way 2 (+20 pts): Build the 12×6 array as two connected chunks: (8 × 6 = 48) + (4 × 6 = 24) = 72 tiles!

Both Ways Proof: Solving the two parentheses zones first (12 × 6) matches splitting the array into 48 + 24 = 72!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Evaluate (8 + 4) × (9 - 3) using the Big Ideas Order of Operations.

METHOD 1 (+10 pts — 1st Token): PEMDAS Step-by-Step Rewrite

1. Grouping Symbols: (8 + 4) × (9 - 3) = 12 × 6. 2. Multiply: 12 × 6 = 72.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Area Split / Stack Chunks

Solve Left Chunk (8 + 4 = 12) and Right Chunk (9 - 3 = 6) separately, then multiply the two chunks: 12 × 6 = 72.

SC-6 Keystrokes: 8 ENTER 4 + 9 ENTER 3 - ×

Answer: 72

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 1.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: PEMDAS Line-by-Line Funnel | Method 2: Inside-Out..

1 Evaluate 5 + 2 × 3. Multiply before adding!

Ans: _____

2 Evaluate (14 + 6) ÷ (7 - 2). Solve both parentheses first!

Ans: _____

3 Evaluate 30 - 4 × 6.

Ans: _____

4 Evaluate (9 + 3) × (15 - 8).

Ans: _____

5 Evaluate 50 - (3 + 4)². Square the sum before subtracting from 50!

Ans: _____

6 Evaluate 120 - 4 × (3 + 2)².

Ans: _____

7 Fraction Bar Grouping: Evaluate (24 + 12) / (9 - 3).

Ans: _____

8 Evaluate 3^3 + 15 ÷ 3 - 2 × 4.

Ans: _____

9 Evaluate the numerical expression 10 + (15 - 3)² ÷ 9 using the order of operations.

Ans: _____

10 Movie tickets: 2 adults at $12 each and 4 kids at $8 each, minus a $10 coupon. Write an expression and find total cost.

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 1.3 — Practice B: Modeling Real Life (Order of Operations)

Textbook Methods: Method 1: PEMDAS Line-by-Line Funnel | Method 2: Inside-Out Grouping Chunks

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: MULTI-STEP TICKET ORDER (40 - 3 × (2 + 3)² ÷ 5) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Evaluate 40 - 3 × (2 + 3)² ÷ 5 using the Order of Operations.

METHOD 1 (+10 pts — 1st Token): Standard PEMDAS Funnel

1. Parentheses: (2+3) = 5. 2. Exponent: 5² = 25. 3. Multiply/Divide L-to-R: 3 × 25 = 75, then 75 ÷ 5 = 15. 4. Subtract: 40 - 15 = 25.

COUNT-01 Way 1: Build a (2+3)=5 square (25 tiles), triple to 75, divide into 5 rows (15), and subtract from 40 -> 25.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Inside-Out Chunk Simplification

Start inside the parentheses (2+3 = 5), square it (25), divide 25 ÷ 5 = 5, multiply by 3 (15), and subtract from 40: 40 - 15 = 25.

COUNT-01 Way 2: Take the 5×5 square (25), divide into 5 columns of 5, keep 3 columns (15), and remove 15 from 40 -> 25!

SC-6 Keystrokes: `40` `ENTER` `3` `ENTER` `2` `ENTER` `3` `+` `x²` `×` `5` `÷` `-`

Answer: 25

SECTION 1.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Movie Night Snack Combo: 4 friends each buy a $9 ticket and a $5 popcorn: 4 × (9 + 5). Evaluate the expression using the order of operations.**

Ans: _____

2 **Skateboard Build: You buy a $45 deck and 4 wheels at $7 each: 45 + 4 × 7. Why must you multiply 4 × 7 before adding 45? Find the total cost.**

Ans: _____

3 **Laser Tag Score: You start with 100 points, lose 3 penalties of 8 points, and win a 25-point bonus: 100 - 3 × 8 + 25. Find your final score.**

Ans: _____

4 **Tile Border Courtyard: A 10 × 10 courtyard has a 6 × 6 fountain in the center. Evaluate $10^2 - 6^2$ to find the area of the tile walkway.**

Ans: _____

5 **Aquarium Ticket Group: Admission is $14 for adults and $9 for kids. Evaluate 2 × 14 + 5 × 9 for 2 adults and 5 kids.**

Ans: _____

6 **Robotics Points Formula: A robot scores 6 × (3^2 + 5) - 14 points in the autonomous round. Evaluate the expression step by step.**

Ans: _____

7 **Craft Fair Booth: You make 72 bracelets, keep 12 for friends, and pack the rest into boxes of 6: (72 - 12) ÷ 6. How many boxes do you fill?**

Ans: _____

8 **Left-to-Right Rule: Evaluate 48 ÷ 6 × 4 + 10. Remember: division and multiplication have equal priority and go from left to right!**

Ans: _____

9 **Bakery Muffin Boxes: A baker makes 5 trays of 12 blueberry muffins and 3 trays of 8 banana muffins, then packs them into 7 gift baskets: (5×12 + 3×8) ÷ 7.**

Ans: _____

10 **Escape Room Final Lock: Evaluate (4^3 - 14) ÷ (2 + 3) + 8^2 to unlock the final puzzle door.**

Ans: _____

BLOCK KIT: PRIME-01 • Prime Factor Studio

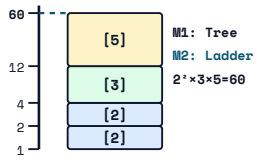
Section 1.4 — Practice A: Prime Factorization

Key Vocabulary: factor pair • prime number • composite number • prime factorization • factor tree

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (PRIME-01: PRIME FACTOR STUDIO)





DEMONSTRATE BOTH METHODS WITH PRIME-01 BLOCKS:

Block Way 1 (+10 pts): Split 60 into factor pairs (6 × 10), then trade 6 for [2][3] and 10 for [2][5] PRIME-01 blocks.

Block Way 2 (+20 pts): Stack [2] + [2] + [3] + [5] vertically in one PRIME-01 socket—any order reaches the 60 ruler mark!

Both Ways Proof: Branching a factor tree and stacking a division ladder produce the exact same [2][2][3][5] = 60 tower!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Write the prime factorization of 60 using a factor tree, Prime Blocks, and [g] [FACT].

METHOD 1 (+10 pts — 1st Token): Use a Factor Tree

Branch 60 into 6 × 10. Split 6 into 2 × 3 and 10 into 2 × 5. Collect the prime leaves in order: 60 = 2² × 3 × 5.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Use a Division Ladder

Divide by smallest primes: 60 ÷ 2 = 30 -> 30 ÷ 2 = 15 -> 15 ÷ 3 = 5 (prime!). Product of divisors: 2² × 3 × 5 = 60.

SC-6 Keystrokes: `60` `g` `FACT`

Answer: 60 = 2² × 3 × 5

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 1.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Factor Tree Diagram | Method 2: Prime Division La..

1 Use a factor tree or division ladder to write the prime factorization of 12.

Ans: _____

2 Write the prime factorization of 36 using exponents.

Ans: _____

3 Determine whether 29 and 57 are prime or composite. If composite, write the prime factorization.

Ans: _____

4 Find the prime factorization of 45.

Ans: _____

5 Build the Prime Tower for 72. Show how 8 × 9 and 6 × 12 use the exact same blocks.

Ans: _____

6 Find the prime factorization of 84.

Ans: _____

7 Find the prime factorization of 100.

Ans: _____

8 Find the whole number whose prime factorization is 2² × 3 × 7.

Ans: _____

9 Write the prime factorization of 180 using two different starting factor pairs (such as 18 × 10 and 12 × 15).

Ans: _____

10 Modeling Real Life: A vault code is the number with prime factorization 2³ × 3² × 5. Find the 3-digit code.

Ans: _____

BLOCK KIT: PRIME-01 • Prime Factor Studio

Section 1.4 — Practice B: Modeling Real Life (Prime Factorization)

Textbook Methods: Method 1: Factor Tree Diagram | Method 2: Prime Division Ladder & Logarithmic Prime Tower

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: MARCHING BAND FORMATIONS (PRIME FACTORIZATION OF 120) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A marching band has $2^3 \times 3 \times 5$ musicians. Find the total number of musicians and two different factor pairs for rectangular rows.

METHOD 1 (+10 pts — 1st Token): Expand the Prime Powers

Evaluate $2^3 = 2 \times 2 \times 2 = 8$. Multiply by the remaining primes: $8 \times 3 \times 5 = 24 \times 5 = 120$ musicians.

PRIME-01 Way 1: Stack three [2] blocks (=8), one [3] block, and one [5] block to reach the 120 mark.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Regroup Prime Factors into Factor Pairs

Pair $(2 \times 5) = 10$ first: $(2^2 \times 3) \times (2 \times 5) = 12 \times 10 = 120$. Two formations: 8 rows of 15, or 10 rows of 12!

PRIME-01 Way 2: Group [2][5]=10 and [2][2][3]=12 (10×12), or [2][2][2]=8 and [3][5]=15 (8×15)!

SC-6 Keystrokes: 2 g x³ 3 × 5 ×

Answer: 120 musicians (e.g., 10×12 or 8×15)

SECTION 1.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Solar Panel Grid: A satellite wing has 48 solar cells. Write all factor pairs of 48 and write its prime factorization using exponents.

Ans: _____

2 Drumline Formation: A drumline of 54 musicians wants to march in equal rows. Write the prime factorization of 54 using a factor tree or division ladder.

Ans: _____

3 Prime Jersey Check: Jersey numbers 37, 51, and 61 are on the bench. One of them is composite! Find which one is composite and write its prime factors.

Ans: _____

4 Collector Display Case: You have 80 mineral specimens. Write the prime factorization of 80 using a factor tree starting with 8×10.

Ans: _____

5 Mystery Prime Tower: A student stacks one [2] block, two [3] blocks, and one [7] block ($2 \times 3^2 \times 7$). What composite number does this tower represent?

Ans: _____

6 Century Mark: Write the prime factorization of 100 by branching 10×10 on a factor tree.

Ans: _____

7 Science Fair Tables: 140 projects are arranged in equal rows. Find the prime factorization of 140.

Ans: _____

8 Gross of Pencils: A 'gross' is a dozen dozens (144 pencils). Write the prime factorization of 144 using exponents.

Ans: _____

9 Two Factor Trees, Same Leaves: Factor 90 first by starting with 9×10, and second by starting with 6×15. Verify both trees give the same prime factorization!

Ans: _____

10 Secret Locker Combination: The locker number is the product of the first four prime numbers ($2 \times 3 \times 5 \times 7$). What is the locker number?

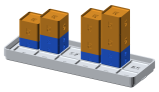
Ans: _____

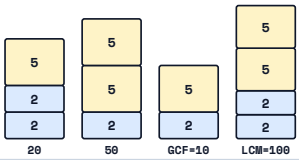
Section 1.5 — Practice A: Greatest Common Factor (GCF) & Least Common Multiple (LCM)

Key Vocabulary: common factor • greatest common factor (GCF) • common multiple • least common multiple (LCM) • Venn di..

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (PRIME-01: PRIME FACTOR STUDIO)





DEMONSTRATE BOTH METHODS WITH PRIME-01 BLOCKS:

Block Way 1 (+10 pts): Stand PRIME-01 Tower 20 ([2][2][5]) beside Tower 50 ([2][5][5]); pull out shared blocks [2][5] = 10 (GCF).

Block Way 2 (+20 pts): Build the Union Tower using the tallest count of each prime: two [2]s and two [5]s -> [2][2][5][5] = 100 (LCM)!

Both Ways Proof: Shared overlap blocks give GCF = 10, while the tallest-count union tower gives LCM = 100 (and 10×100 = 20×50)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find both the GCF and LCM of 20 and 50 using Prime Towers and the SC-6.

METHOD 1 (+10 pts — 1st Token): List Factors & Multiples

Factors of 20: 1, 2, 4, 5, 10, 20; of 50: 1, 2, 5, 10, 25, 50 -> GCF = 10. Multiples: 20, 40, 60, 80, 100 and 50, 100 -> LCM = 100.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Prime Factor Venn Diagram

$20 = 2^2 \cdot 5$ and $50 = 2 \cdot 5^2$. Shared overlap = $2 \cdot 5 = 10$ (GCF). Union of tallest prime powers = $2^2 \cdot 5^2 = 100$ (LCM).

SC-6 Keystrokes: `20` `ENTER` `50` `GCF` `20` `ENTER` `50` `f` `LCM`

Answer: GCF = 10, LCM = 100

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 1.5 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Lists of Factors & Multiples | Method 2: Prime Fa..

1 Find the greatest common factor (GCF) and least common multiple (LCM) of 12 and 18 using a Venn diagram or factor lists.

Ans: _____

2 Find the GCF and LCM of 8 and 12.

Ans: _____

3 Find the GCF and LCM of 15 and 25.

Ans: _____

4 Find the GCF and LCM of 14 and 35.

Ans: _____

5 Find the GCF and LCM of 16 and 24.

Ans: _____

6 Find the GCF and LCM of 28 and 42.

Ans: _____

7 Two prime numbers are 5 and 7. What is their GCF and what is their LCM?

Ans: _____

8 Verify that GCF(15, 20) × LCM(15, 20) equals 15 × 20.

Ans: _____

9 Hot dogs come in packs of 12 and buns in packs of 15. What is the smallest number of each to buy with 0 leftovers?

Ans: _____

10 You have 36 red tiles and 48 blue tiles to pack into identical kit bags with none left over. What is the max number of bags?

Ans: _____

BLOCK KIT: PRIME-01 • Prime Factor Studio

Section 1.5 — Practice B: Modeling Real Life (Greatest Common Factor (GCF) & Least Common Multiple..

Textbook Methods: Method 1: Lists of Factors & Multiples | Method 2: Prime Factorization Venn Diagram

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: IDENTICAL SNACK PACKS (18 PRETZELS & 24 FRUIT BARS) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: You have 18 pretzel bags and 24 fruit bars to make identical snack packs with none left over. What is the greatest number of packs (GCF)?

METHOD 1 (+10 pts — 1st Token): List Factor Pairs of 18 and 24

Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Common factors: 1, 2, 3, 6 -> GCF = 6 packs.

PRIME-01 Way 1: Compare Tower 18 ([2][3][3]) and Tower 24 ([2][2][2][3]) -> shared blocks [2][3] = 6 packs.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Prime Factorization Overlap

18 = 2 × 3² and 24 = 2³ × 3. Multiply shared prime factors: 2 × 3 = 6 packs (each with 18÷6 = 3 pretzels and 24÷6 = 4 bars).

COUNT-01 Way 2: Deal 18 pretzels and 24 bars into 6 equal rows -> each row gets 3 pretzels and 4 bars!

SC-6 Keystrokes: 18 ENTER 24 GCF 18 ENTER 24 f LCM

Answer: GCF = 6 packs (3 pretzels, 4 bars each)

SECTION 1.5 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 First-Aid Kits: A school nurse has 28 bandages and 42 antiseptic wipes to make identical first-aid kits with none left over. What is the greatest number of kits (GCF)?

Ans: _____

2 Lighthouse Flash Cycle: Red Beacon flashes every 12 seconds and Blue Beacon flashes every 15 seconds. They just flashed together. In how many seconds will they flash together again (LCM)?

Ans: _____

3 Hot Dogs & Buns: Hot dogs come in packs of 8 and buns come in packs of 12. What is the least number of hot dogs and buns you can buy to have an equal number of each (LCM)?

Ans: _____

4 Choir Riser Rows: The 6th-grade choir has 32 altos and 48 sopranos. Each row must have the same number of singers and only one voice part. What is the greatest number of singers per row (GCF)?

Ans: _____

5 Bus Station Loop: Route A returns to the station every 18 minutes and Route B returns every 24 minutes. Both leave at 9:00 AM. How many minutes until both are at the station together again?

Ans: _____

6 Square Quilt Patches: You have a sheet of fabric 36 inches wide by 60 inches long. You want to cut it into the largest possible equal squares with no waste. What is the side length of each square (GCF)?

Ans: _____

7 Gear Mesh Teeth: A small gear with 14 teeth meshes with a large gear with 21 teeth. Find the GCF and LCM of 14 and 21.

Ans: _____

8 Triple Track Laps: Three go-karts complete a lap in 4, 6, and 10 minutes. If they start together, after how many minutes will all three cross the start line together?

Ans: _____

9 Balloon Bouquets: You have 45 gold balloons and 75 silver balloons to tie into identical bouquets. Find the greatest number of bouquets (GCF) and how many gold balloons go in each.

Ans: _____

10 GCF × LCM Discovery: Find the GCF and LCM of 15 and 20. Multiply GCF × LCM and compare it to 15 × 20. What do you notice?

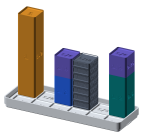
Ans: _____

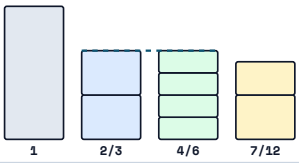
Section 2.1 — Practice A: Equivalent Fractions, Mixed Numbers & LCD Addition

Key Vocabulary: equivalent fractions • simplest form • least common denominator (LCD) • mixed number • improper fraction

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (FRAC-01: 84MM FRACTION STUDIO)





DEMONSTRATE BOTH METHODS WITH FRAC-01 BLOCKS:

Block Way 1 (+10 pts): Swap FRAC-01 1/3 (28mm) for four 1/12 blocks (28mm) and 1/4 (21mm) for three 1/12 blocks -> 7/12.

Block Way 2 (+20 pts): Stack the 1/3 block (28mm) directly on the 1/4 block (21mm) against the 84mm Whole (49mm / 84mm = 7/12)!

Both Ways Proof: Trading for common 1/12 blocks (4/12 + 3/12) matches stacking 28mm + 21mm = 49mm against the 84mm Whole!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Simplify 12/18 step-by-step and compare tower heights.

METHOD 1 (+10 pts — 1st Token): Step-by-Step Prime Factor Reduction

Divide numerator and denominator of 12/18 by 2 first: 12/18 = 6/9. Then divide by 3: 6/9 = 2/3.

METHOD 2 (+20 pts DOUBLE — 2nd Token): One-Step GCF Division

Find GCF(12, 18) = 6. Divide top and bottom by 6 in one step: (12 ÷ 6) / (18 ÷ 6) = 2/3.

SC-6 Keystrokes: `.` `12` `.` `18` `ENTER` `g` `SIMP` `g` `SIMP`

Answer: 12/18 -> 6/9 -> 2/3

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 2.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Divide Step-by-Step by Common Factors | Method 2:..

1 Write the fraction 8/12 in simplest form by dividing by common factors.

Ans: _____

2 Write 24/36 in simplest form, and write its equivalent repeating decimal.

Ans: _____

3 Write the mixed number 2 3/5 as an improper fraction and as a decimal.

Ans: _____

4 Write the improper fraction 19/8 as a mixed number and as a decimal.

Ans: _____

5 Add 1/2 + 1/3 using a common denominator (sixths) and check with 84mm Fraction Blocks.

Ans: _____

6 Add 3/4 + 5/6 using the LCD (12). Write both the mixed number and improper fraction.

Ans: _____

7 Subtract 7/8 - 1/4 using a common denominator, and check your answer by adding.

Ans: _____

8 Subtract mixed numbers: 4 1/3 - 1 5/6.

Ans: _____

9 Build 1/3 + 1/4 with 84mm towers (28mm + 21mm) AND solve using the LCD (twelfths).

Ans: _____

10 A trail is 3 3/4 miles long. You hiked 1 7/8 miles. How much trail is left? Solve 2 ways!

Ans: _____

BLOCK KIT: FRAC-01 • 84mm Fraction Studio

Section 2.1 — Practice B: Modeling Real Life (Equivalent Fractions, Mixed Numbers & LCD Addition)

Textbook Methods: Method 1: Divide Step-by-Step by Common Factors | Method 2: Divide Once by the GCF / Use LCD

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: COMBINING TRAIL LENGTHS (1/3 MI + 1/4 MI & 1 3/4 MI) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Convert 1 3/4 miles to an improper fraction and decimal, then find the total length of a 1/3-mile loop plus a 1/4-mile spur.

METHOD 1 (+10 pts — 1st Token): Least Common Denominator (LCD) & Quarter Trade

1 3/4 = 4/4 + 3/4 = 7/4 = 1.75. For 1/3 + 1/4, use LCD(3, 4) = 12: 4/12 + 3/12 = 7/12 mile.

FRAC-01 Way 1: Stack 1/3 (28mm) + 1/4 (21mm) = 49mm, which matches seven 1/12 blocks (7 × 7mm = 49mm = 7/12).

METHOD 2 (+20 pts DOUBLE — 2nd Token): 84mm Fraction Tower Height Sum

1/3 block = 28 mm and 1/4 block = 21 mm. Total height = 28 + 21 = 49 mm. Since 1/12 = 7 mm, 49 ÷ 7 = 7 twelfths (7/12)!

FRAC-01 Way 2: Stack seven 1/4 blocks (7/4) beside one 84mm Whole + three 1/4 blocks (1 3/4 = 1.75)!

SC-6 Keystrokes: `1` `.` `3` `.` `4` `ENTER` `f` `d/c` `FDISP`

Answer: 7/4 = 1.75; and 1/3 + 1/4 = 7/12 mi

SECTION 2.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Smoothie Recipe: You blend 3/8 cup of blueberries and 1/4 cup of strawberries. What fraction of a cup of berries did you use in all?

Ans: _____

2 3D Print Progress: A print is 18/24 complete. Simplify 18/24 to simplest form, and find what fraction of the print remains.

Ans: _____

3 Skateboard Hardware: A bolt is 1 5/8 inches long. Write 1 5/8 as an improper fraction and as a decimal.

Ans: _____

4 Hiking Trail Gap: The Summit Trail is 5/6 mile long and the Creek Trail is 3/8 mile long. Using LCD = 24, how much longer is the Summit Trail?

Ans: _____

5 Pizza Leftovers: After a party, there are 2 1/3 cheese pizzas and 1 1/2 pepperoni pizzas left. How many pizzas are left in all?

Ans: _____

6 Carpentry Trim: You cut 1 3/4 feet off a 4 1/2-foot board. How many feet of board are left?

Ans: _____

7 Free-Throw Practice: Maya made 15 out of 25 shots. Write 15/25 in simplest form and as a decimal.

Ans: _____

8 Science Beaker Mix: A beaker holds 2/5 liter of water. You pour in 3/10 liter more, then pour out 1/2 liter. How much water is left?

Ans: _____

9 Ribbon Spool: You need 2 2/3 yards of blue ribbon and 1 3/4 yards of gold ribbon. Do you have enough if a spool holds 5 yards? Find the total used and how much is left.

Ans: _____

10 Triathlon Training: You swim 3/8 mile, bike 2 1/2 miles, and run 1 3/4 miles. What is the total distance of your workout?

Ans: _____

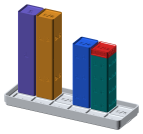
BLOCK KIT: FRAC-01 • 84mm Fraction Studio

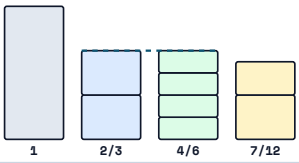
Section 2.2 — Practice A: Multiplying Fractions and Mixed Numbers

Key Vocabulary: product of fractions • area model • cross-canceling (dividing out common factors) • improper fraction

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (FRAC-01: 84MM FRACTION STUDIO)





DEMONSTRATE BOTH METHODS WITH FRAC-01 BLOCKS:

Block Way 1 (+10 pts): Take three 1/4 FRAC-01 blocks (3/4 = 63mm) and split that height in half (31.5mm = three 1/8 blocks = 3/8).

Block Way 2 (+20 pts): Build a 2 × 4 COUNT-01 grid (8 tiles); shade 1 of 2 rows and 3 of 4 columns -> overlap is 3 of 8 tiles (3/8)!

Both Ways Proof: Halving a 3/4 FRAC-01 tower (31.5mm) and overlapping a 1×3 corner inside a 2×4 array both yield 3/8!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find 1/2 × 3/4 using 84mm Fraction Towers and the SC-6.

METHOD 1 (+10 pts — 1st Token): Area / Fraction Tower Partition

Take half of each of the three 1/4 blocks: half of 1/4 is 1/8, so half of three 1/4 blocks is three 1/8 blocks (3/8).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Multiply Numerators & Denominators

Multiply numerators (1 × 3 = 3) and multiply denominators (2 × 4 = 8) -> (1 × 3)/(2 × 4) = 3/8.

SC-6 Keystrokes: [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [×]

Answer: 3/8

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 2.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Multiply Straight Across & Simplify | Method 2: C..

1 Find 1/2 × 2/3. (Hold two 1/3 blocks and take half of them!)

Ans: _____

2 Multiply 2/3 × 3/4. Divide out common factors before multiplying, and write the product in simplest form.

Ans: _____

3 Multiply 3/5 × 5/9.

Ans: _____

4 Multiply a whole number by a fraction: 12 × 3/4.

Ans: _____

5 Multiply mixed numbers: 1 3/4 × 2 2/7.

Ans: _____

6 Multiply 2 1/4 × 1 1/3.

Ans: _____

7 Multiply 3 1/2 × 2 1/2. Write the answer as a mixed number and decimal.

Ans: _____

8 Prediction Check: Without calculating, is 5 × 2/3 greater than 5 or less than 5? Now compute it!

Ans: _____

9 A slime recipe uses 1 3/4 cups of glue per batch. How much glue is needed for 2 1/2 batches?

Ans: _____

10 Evaluate the three-fraction product 4/5 × 5/6 × 3/4 by dividing out common factors.

Ans: _____

BLOCK KIT: FRAC-01 • 84mm Fraction Studio

Section 2.2 — Practice B: Modeling Real Life (Multiplying Fractions and Mixed Numbers)

Textbook Methods: Method 1: Multiply Straight Across & Simplify | Method 2: Cross-Cancel Common Factors First

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: SCALING A RECIPE (1 1/2 × 2 2/3 CUPS) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A batch of granola uses 1 1/2 cups of oats. You make 2 2/3 batches. How many cups of oats do you need?

METHOD 1 (+10 pts — 1st Token): Convert to Improper Fractions & Multiply Across

1 1/2 = 3/2 and 2 2/3 = 8/3. Multiply straight across: (3 × 8) / (2 × 3) = 24 / 6 = 4 cups.

FRAC-01 Way 1: Build two sets of (2 Whole + two 1/3 towers) + two-thirds of a set -> four 84mm Wholes (4 cups).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Cross-Cancel Common Factors Before Multiplying

In (3/2) × (8/3), divide out the common factor 3 (3÷3 = 1) and divide 8 by 2 (8÷2 = 4): (1/1) × (4/1) = 4 cups!

FRAC-01 Way 2: Convert 2 2/3 to eight 1/3 blocks (8/3) and 1 1/2 to three 1/2 blocks (3/2); cross-cancel to 4 Wholes!

SC-6 Keystrokes: `1` `.` `1` `.` `2` `ENTER` `2` `.` `2` `.` `3` `×`

Answer: 4 cups

SECTION 2.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Garden Herb Plot: 2/3 of a garden is planted with vegetables, and 3/5 of the vegetables are tomatoes. What fraction of the whole garden is tomatoes?

Ans: _____

2 Phone Battery Drain: Your phone battery is at 4/5 charge. A video game uses 1/2 of that charge. What fraction of a full battery did the game use?

Ans: _____

3 Pancake Recipe Doubler: A pancake recipe calls for 1 3/4 cups of flour. You make 2 1/2 times the recipe for a team breakfast. How much flour do you need?

Ans: _____

4 Miniature Sticker Area: A rectangular laptop sticker measures 2 1/4 inches wide by 1 1/3 inches tall. Find its area in square inches.

Ans: _____

5 Trail Hike Distance: You hike at 3 1/5 miles per hour for 1 1/4 hours. How many miles do you hike?

Ans: _____

6 Art Canvas Frame: A painting is 3/4 yard wide and 8/9 yard tall. Cross-cancel common factors to find its area in square yards.

Ans: _____

7 Rain Barrel Water: A 24-gallon rain barrel is 5/8 full. You use 2/3 of the water in the barrel for the garden. How many gallons did you use?

Ans: _____

8 Cubic Fractional Block: Find the volume of a cube whose edges are each 1/2 inch long: $(1/2)^3 = 1/2 \times 1/2 \times 1/2$.

Ans: _____

9 Skate Ramp Plywood: A plywood sheet is 3 1/2 feet by 2 2/7 feet. Convert to improper fractions and cross-cancel to find the area.

Ans: _____

10 Sleep Schedule: You sleep 1/3 of each 24-hour day, and 1/4 of your sleep is REM dream sleep. What fraction of the day (and how many hours) is REM sleep?

Ans: _____

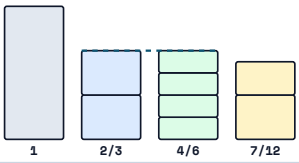
Section 2.3 — Practice A: Dividing Fractions and Mixed Numbers

Key Vocabulary: reciprocals • multiplicative inverses • common-denominator division • measurement division

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (FRAC-01: 84MM FRACTION STUDIO)





DEMONSTRATE BOTH METHODS WITH FRAC-01 BLOCKS:

Block Way 1 (+10 pts): Stand the 3/4 FRAC-01 tower (63mm) beside 1/8 blocks (10.5mm each) -> exactly 6 eighth-blocks match 63mm!

Block Way 2 (+20 pts): Count that 8 eighth-blocks fit in 1 Whole (reciprocal = 8/1), then take 3/4 of those 8 blocks (3/4 × 8 = 6)!

Both Ways Proof: Measuring how many 1/8 blocks fit in 3/4 (Method 1) is the same as multiplying 3/4 by 8 per Whole (Method 2)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Divide 3/4 ÷ 1/8 by comparing tower heights and by multiplying by the reciprocal with [f] [1/x].

METHOD 1 (+10 pts — 1st Token): Common Denominator Method

Rewrite 3/4 with denominator 8: 6/8 ÷ 1/8. Divide the numerators: 6 eighths ÷ 1 eighth = 6 ÷ 1 = 6.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Multiply by the Reciprocal

The reciprocal of 1/8 is 8/1 (since 1/8 × 8/1 = 1). Multiply: 3/4 × 8/1 = 24/4 = 6.

SC-6 Keystrokes: [.] [3] [.] [4] [ENTER] [.] [1] [.] [8] [f] [1/x] [×]

Answer: 6

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 2.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Common-Denominator Measurement ('How Many Fit?') |..

1 How many 1/4 blocks fit in a 1/2 tower? Compute 1/2 ÷ 1/4.

Ans: _____

2 How many 1/6 blocks (14mm) fit in a 2/3 tower (56mm)? Compute 2/3 ÷ 1/6.

Ans: _____

3 Divide 5/6 ÷ 5/12 by multiplying by the reciprocal. Write the quotient in simplest form.

Ans: _____

4 Divide a whole number by a fraction: 6 ÷ 3/4.

Ans: _____

5 Divide 3/4 ÷ 2/5. Write the answer as a mixed number.

Ans: _____

6 Divide a fraction by a whole number: 4/5 ÷ 2.

Ans: _____

7 Divide mixed numbers: 4 1/2 ÷ 3/4.

Ans: _____

8 Divide the mixed numbers 5 1/4 ÷ 1 3/4. Check your quotient by multiplying.

Ans: _____

9 A board is 6 3/4 ft long. You cut it into 3/4-ft shelves. How many shelves can you cut? Solve 2 ways!

Ans: _____

10 A 3 3/4-cup pitcher of smoothie is poured into 5/8-cup glasses. How many glasses does it fill?

Ans: _____

BLOCK KIT: FRAC-01 • 84mm Fraction Studio

Section 2.3 — Practice B: Modeling Real Life (Dividing Fractions and Mixed Numbers)

Textbook Methods: Method 1: Common-Denominator Measurement ('How Many Fit?') | Method 2: Multiply by the Reciprocal

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: CUTTING WOOD PLANKS (6 3/4 FT ÷ 1 1/8 FT) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A carpenter cuts a 6 3/4-foot cedar board into shelves that are each 1 1/8 feet long. How many shelves can be cut?

METHOD 1 (+10 pts — 1st Token): Improper Fractions & Reciprocal Flip

6 3/4 = 27/4 and 1 1/8 = 9/8. Multiply by reciprocal of 9/8: (27/4) × (8/9) = (3/1) × (2/1) = 6 shelves.

FRAC-01 Way 1: Build 4 1/2 Wholes (9 half-blocks = 18 quarter-blocks) and group into threes (3/4) -> 6 groups.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Common Denominator Eighths

Rewrite both in eighths: 6 3/4 = 54/8 and 1 1/8 = 9/8. Divide numerators: 54 eighths ÷ 9 eighths = 54 ÷ 9 = 6 shelves!

FRAC-01 Way 2: Since four 1/4-blocks fit in 1 Whole, multiply 9/2 × 4/3 = 36/6 = 6 shelves!

SC-6 Keystrokes: 6 . 3 . 4 ENTER 1 . 1 . 8 ÷

Answer: 6 shelves

SECTION 2.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Pizza Slice Sharing: How many 1/8-size slices are in 3/4 of a pizza? Solve by multiplying 3/4 by the reciprocal of 1/8.

Ans: _____

2 Trail Mix Scoops: A bin holds 4 1/2 pounds of almonds. Each snack bag holds 3/8 pound. How many snack bags can you fill?

Ans: _____

3 Relay Race Legs: A 3 3/4-mile charity relay is split into equal legs of 5/8 mile each. How many runners are on each relay team?

Ans: _____

4 LED Strip Cutting: You have a 5 1/4-meter LED light strip and cut it into 3/4-meter segments for desk shelves. How many segments do you get?

Ans: _____

5 3D Print Nozzle Layers: A 6-millimeter bracket is printed in layers that are 3/20 millimeter thick. How many layers tall is the bracket?

Ans: _____

6 Garden Stepping Stones: A 7 1/2-foot garden path has stepping stones spaced every 1 1/4 feet. How many spaces are along the path?

Ans: _____

7 Missing Width of a Rectangle: A rectangular solar panel has an area of 4 1/2 square meters and a length of 2 1/4 meters. Divide Area ÷ Length to find its width.

Ans: _____

8 Juice Pitcher Servings: A pitcher holds 2 2/3 quarts of lemonade. Each cup holds 1/3 quart. How many cups can be poured?

Ans: _____

9 Reciprocal Proof: Find the reciprocal of 2 2/5 (12/5), and verify that multiplying the number by its reciprocal equals 1.

Ans: _____

10 Lumber Yard Cuts: You have a 10 1/2-foot board. You cut off 1 1/2 feet first, then divide the remaining board into 3/4-foot pieces. How many 3/4-foot pieces do you get?

Ans: _____

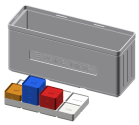
BLOCK KIT: PLACE-01 • Place Value Studio

Section 2.4 — Practice A: Adding, Subtracting, and Multiplying Decimals

Key Vocabulary: tenths • hundredths • thousandths • place-value alignment • leading/trailing zeros • estimation

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (PLACE-01: PLACE VALUE STUDIO)



Input: 14.85

M1: Whole	M2: Decimal
14	0.85
[f] [IP]	[g] [FP]

DEMONSTRATE BOTH METHODS WITH PLACE-01 BLOCKS:

Block Way 1 (+10 pts): Align PLACE-01 Tens, Ones, Tenths (0.1), and Hundredths (0.01) in vertical tray slots; regroup 10 tenths -> 1 One.

Block Way 2 (+20 pts): Split Whole Part ([f][IP]: 14 + 3 = 17) and Decimal Part ([g][FP]: 0.60 + 0.85 = 1.45) -> 17 + 1.45 = 18.45!

Both Ways Proof: Column regrouping in PLACE-01 and splitting Whole + Fractional parts both produce 18.45!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Split 14.85 into its whole and decimal parts, then compute 14.60 + 3.85.

METHOD 1 (+10 pts — 1st Token): Line Up Decimal Points Vertically

Annex a trailing zero to write 14.6 as 14.60. Add hundredths (0+5=5), tenths (6+8=14, carry 1), ones (4+3+1=8) -> 18.45.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Place-Value Split & Inverse Check

Add wholes (14 + 3 = 17) and decimals (0.60 + 0.85 = 1.45) -> 18.45. Check by subtracting: 18.45 - 3.85 = 14.60.

SC-6 Keystrokes: `14.6` `ENTER` `3.85` `+` `f` `LASTX` `-`

Answer: 18.45 (Checked: 18.45 - 3.85 = 14.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 2.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Vertical Decimal Alignment / Place Value | Method..

1 Decompose 27.64 into its whole-number part and its decimal part (in hundredths).

Ans: _____

2 Line up the decimal points vertically to add 12.8 + 4.75. Check by subtracting.

Ans: _____

3 Annex zeros to subtract 20.00 - 8.65.

Ans: _____

4 Multiply 4.5 × 0.6.

Ans: _____

5 Evaluate 25.4 - 9.86 + 3.2.

Ans: _____

6 Multiply 3.6 × 2.25 and check the whole-number ballpark.

Ans: _____

7 Multiply 0.15 × 0.4. Write the answer both as a decimal and as a fraction in simplest form.

Ans: _____

8 Find the area of a rectangular desk top measuring 1.8 m by 0.75 m.

Ans: _____

9 You pay with a $20 bill for a $8.75 burger and $4.60 smoothie. How much change do you get?

Ans: _____

10 Multiply 4.75 × 1.2 both as decimals and as mixed numbers (4 3/4 × 1 1/5).

Ans: _____

BLOCK KIT: PLACE-01 • Place Value Studio

Section 2.4 — Practice B: Modeling Real Life (Adding, Subtracting, and Multiplying Decimals)

Textbook Methods: Method 1: Vertical Decimal Alignment / Place Value | Method 2: Inverse Check & Fraction Form

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: BUYING BULK TRAIL MIX (3.6 LB × $2.25/LB) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: You buy 3.6 pounds of trail mix at $2.25 per pound. Find the total cost in two different ways.

METHOD 1 (+10 pts — 1st Token): Count Decimal Places in the Product

Multiply as whole numbers: 225 × 36 = 8,100. Count decimal places (1 place in 3.6 + 2 places in 2.25 = 3 places) -> 8.100 = $8.10.

PLACE-01 Way 1: Build 4 groups of $1.80 ($7.20) plus half of $1.80 ($0.90) -> $7.20 + $0.90 = $8.10.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Fraction / Mixed-Number Multiplication

Write 3.6 = 3 3/5 = 18/5 and 2.25 = 2 1/4 = 9/4. Multiply: (18/5) × (9/4) = 162/20 = 81/10 = 8.1 = $8.10!

FRAC-01 Way 2: Multiply 4 1/2 (9/2) × 1 4/5 (9/5) = 81/10 = 8 1/10 = $8.10!

SC-6 Keystrokes: `3.6` `ENTER` `2.25` `×` `FDISP`

Answer: $8.10 (8 1/10)

SECTION 2.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 3D Filament Order: You buy a spool of Matte Navy PLA for $18.75 and a spool of Silk Gold PLA for $22.49. What is the total cost?

Ans: _____

2 Change at the Book Fair: You buy a graphic novel for $13.65 and pay with a $20 bill. How much change should you receive?

Ans: _____

3 Sprint Stopwatch: In the 100-meter dash, Maya ran 13.42 seconds and Leo ran 12.87 seconds. How much faster was Leo's time?

Ans: _____

4 Bulk Honeycrisp Apples: Apples cost $2.40 per pound. You pick a bag weighing 3.25 pounds. What is the total price?

Ans: _____

5 Microchip Dimensions: A silicon processor die measures 1.4 cm by 0.85 cm. Find its area in square centimeters.

Ans: _____

6 Road Trip Fuel: Your family car buys 11.5 gallons of gas at $3.60 per gallon. How much does the fill-up cost?

Ans: _____

7 Drone Flight Altitude: A drone rises 24.8 meters, drops 6.35 meters, and rises 11.9 meters. What is its final altitude?

Ans: _____

8 Skate Shop Receipt: You buy 2 sets of bearings at $14.50 each and grip tape for $6.75. How much do you spend in all?

Ans: _____

9 Gridfinity Baseplate Tolerance: Four 42.0 mm Gridfinity squares sit side-by-side in a drawer that is 170.5 mm wide. How much clearance gap remains?

Ans: _____

10 Snack Bar Profit: You buy 25 smoothies for $1.80 each and sell all 25 for $3.20 each. Find your total profit in two ways!

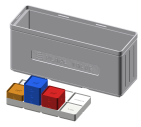
Ans: _____

Section 2.5 — Practice A: Dividing Whole Numbers and Decimals

Key Vocabulary: terminating decimal • power-of-10 shift • equivalent division expression • mixed number

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (FRAC-01: FRACTION & PLACE VALUE STUDIO)



Input: 15 ÷ 4

M1: Whole	M2: Decimal
3 R 3	3.75 = 3 3/4
[f] [IP]	[g] [FP]

DEMONSTRATE BOTH METHODS WITH FRAC-01 BLOCKS:

Block Way 1 (+10 pts): Share 23 PLACE-01 Ones into 4 groups (5 each), then trade 3 leftover Ones for 30 Tenths and 20 Hundredths -> 5.75.

Block Way 2 (+20 pts): Use FRAC-01 to model 23 ÷ 4 as 5 Whole towers + three 1/4 blocks (3/4 = 0.75) -> 5 3/4 = 5.75!

Both Ways Proof: Regrouping Ones into Tenths/Hundredths (PLACE-01) and converting R=3 into 3/4 = 0.75 (FRAC-01) both give 5.75!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Write 23 ÷ 4 as a decimal, mixed number, improper fraction, and quotient with remainder.

METHOD 1 (+10 pts — 1st Token): Annex Zeros & Divide

Write 23 as 23.00 and divide by 4: 23 ÷ 4 = 5 with 3 left; 30 tenths ÷ 4 = 7 tenths; 20 hundredths ÷ 4 = 5 hundredths -> 5.75.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Mixed Number to Decimal

23 ÷ 4 = 23/4 = 5 3/4. Since 3/4 = 75/100 = 0.75, 5 3/4 = 5.75 (and 5 R 3).

SC-6 Keystrokes: `23` `ENTER` `4` `÷` `FDISP` `f` `d/c`

Answer: 5.75 = 5 3/4 = 23/4 = 5 R 3

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 2.5 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Power-of-10 Decimal Shift | Method 2: Fraction /..

1 Write 39 ÷ 8 as (a) Quotient + Remainder, (b) Decimal, and (c) Mixed Number.

Ans: _____

2 Divide a decimal by a whole number: 18.4 ÷ 8.

Ans: _____

3 How many quarters ($0.25) are in $11.25? Compute 11.25 ÷ 0.25.

Ans: _____

4 Divide 7.2 ÷ 0.9 by shifting the decimal point in the divisor and dividend. Check by multiplying.

Ans: _____

5 Divide 15.6 ÷ 0.12.

Ans: _____

6 Divide 9.45 ÷ 1.5. Write the quotient both as a decimal and as a mixed number in simplest form.

Ans: _____

7 Divide 0.84 ÷ 0.07.

Ans: _____

8 Error Analysis: Leo says 4.8 ÷ 0.2 = 0.24. Use estimation or multiplication to explain and correct his error.

Ans: _____

9 A runner completes a 4.5-mile race in 0.75-mile laps. How many laps is the race? Solve 2 ways!

Ans: _____

10 A 6-pack of sports drinks costs $7.50. What is the cost of 1 bottle, and how many bottles can $20 buy?

Ans: _____

Section 2.5 — Practice B: Modeling Real Life (Dividing Whole Numbers and Decimals)

Textbook Methods: Method 1: Power-of-10 Decimal Shift | Method 2: Fraction / Mixed-Number Conversion

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: ARCADE TOKEN MACHINE ($6.75 ÷ $0.25) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: An arcade token machine takes quarters ($0.25). How many quarters are in $6.75, and why is the quotient larger than 6.75?

METHOD 1 (+10 pts — 1st Token): Multiply Dividend & Divisor by 100

Shift both decimal points 2 places right (× 100) so the divisor is a whole number: 6.75 ÷ 0.25 = 675 ÷ 25 = 27 quarters.

PLACE-01 Way 1: Convert $6.75 into 675 Hundredths (cents) and group into stacks of 25 Hundredths -> 27 stacks.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Divide Using Fourths (Reciprocal of 1/4)

Since $0.25 = 1/4 dollar, every $1 holds 4 quarters: 6.75 ÷ (1/4) = 6.75 × 4 = (6 × 4) + (0.75 × 4) = 24 + 3 = 27 quarters!

FRAC-01 Way 2: Since 4 quarters ($0.25) fit in every $1.00 Whole, multiply 6 × 4 = 24 quarters + 3 quarters = 27!

SC-6 Keystrokes: `6.75` `ENTER` `0.25` `÷` `f` `LASTX` `×`

Answer: 27 quarters

SECTION 2.5 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **4-Way Translator: Divide 35 ÷ 8. Write the result as (a) quotient with remainder, (b) mixed number, (c) improper fraction, and (d) exact decimal.**

Ans: _____

2 **Quarter Stack Height: A stack of quarters is 28.0 mm tall. Each quarter is 1.75 mm thick. How many quarters are in the stack?**

Ans: _____

3 **Splitting the Pizza Bill: Four friends split a $34.60 pizza order equally. How much does each friend pay?**

Ans: _____

4 **3D Print Layer Slicer: A 14.4 mm model is sliced into 0.20 mm layers. Shift both decimals 2 places right (1,440 ÷ 20) to find the number of layers.**

Ans: _____

5 **Bulk Granola Scoops: How many 0.35-kilogram bags can be filled from a 5.25-kilogram bin of granola?**

Ans: _____

6 **Track Lap Pace: A runner completes 4.5 miles in 0.75 hours. Divide 4.5 ÷ 0.75 to find her speed in miles per hour.**

Ans: _____

7 **Garden Fence Posts: A 22.5-meter garden wall has posts spaced every 1.25 meters. How many 1.25-meter sections are there?**

Ans: _____

8 **Lemonade Stand Cups: A cooler holds 13.5 liters of lemonade. Each cup holds 0.45 liter. How many cups can you fill?**

Ans: _____

9 **Movie Ticket Group: A group of friends spent $87.50 on movie tickets that cost $12.50 each. How many friends went to the movie?**

Ans: _____

10 **Two-Way Decimal & Fraction Check: Evaluate 10.5 ÷ 1.75 both by shifting decimals (1,050 ÷ 175) and by dividing mixed numbers (10 1/2 ÷ 1 3/4).**

Ans: _____

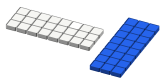
BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 3.1 — Practice A: Ratios and Tape Diagrams

Key Vocabulary: ratio • part-to-part ratio • part-to-whole ratio • equivalent ratios • tape diagram

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)



Part A (3 boxes):

4	4	4

Part B (2 boxes):

4	4

M1: 3:2 Tape | M2: 5 Boxes

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Lay 3 Blue COUNT-01 tiles above 2 Yellow tiles (Part-to-Part 3:2) and place 4 counters on each tile (12 and 8).

Block Way 2 (+20 pts): Snap all 3 + 2 = 5 tiles into one joined Part-to-Whole bar of 20 counters (20 ÷ 5 = 4 per box)!

Both Ways Proof: Two separate rows (3:2) and one joined 5-box bar (3:5 of 20) both put 4 counters in each box (12 Blue, 8 Yellow)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: A gaming clan has 18 builders and 12 speedrunners. Simplify the Part-to-Part and Part-to-Whole ratios.

METHOD 1 (+10 pts — 1st Token): Divide Both Terms by GCF

Part-to-Part = 18:12. Divide by GCF(18, 12) = 6 -> 3:2. Whole = 18 + 12 = 30 -> Part-to-Whole = 18:30 = 3:5.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Use a Tape Diagram

Draw 3 boxes for Builders and 2 boxes for Speedrunners (5 boxes total, each box = 6 players). Builders : Total = 3 boxes : 5 boxes = 3:5.

SC-6 Keystrokes: `18` `ENTER` `12` `g` `a:b`

Answer: Part-to-Part = 3:2; Part-to-Whole.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 3.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Tape Diagram Model (Equal Boxes) | Method 2: GCF..

1 Write the ratio 12 : 18 in simplest form by dividing both terms by their GCF.

Ans: _____

2 Write the ratio 24 : 16 in simplest form.

Ans: _____

3 A bag has 4 red marbles and 6 blue marbles. Find the simplified ratio of red to TOTAL marbles.

Ans: _____

4 In a 3:4 tape diagram, each tile is worth 6. Find the two parts and the total.

Ans: _____

5 Green paint mixes blue and yellow in a 2:3 ratio. If you use 25 cups total, how many cups of each?

Ans: _____

6 The ratio of cats to dogs at a shelter is 4:5. There are 36 animals total. How many are dogs?

Ans: _____

7 The ratio of boys to girls in a club is 3:5. If there are 15 boys, how many girls are there?

Ans: _____

8 Difference Tape Model: The ratio of acoustic to electric guitars is 2:5. There are 18 MORE electric guitars. Find how many electric guitars!

Ans: _____

9 Two robot teams split 84 gears in a 4:3 ratio. How many gears does each team get? Solve 2 ways!

Ans: _____

10 Ratio of gold to silver coins is 7:4. There are 21 more gold coins than silver. How many coins in all?

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 3.1 — Practice B: Modeling Real Life (Ratios and Tape Diagrams)

Textbook Methods: Method 1: Tape Diagram Model (Equal Boxes) | Method 2: GCF Simplification & Total Parts

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: ESPORTS TEAM RECORD (RATIO 5:3, 32 GAMES TOTAL) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: An esports team's ratio of wins to losses is 5:3. If the team played 32 games in all, how many games did they win?

METHOD 1 (+10 pts — 1st Token): Tape Diagram (Count Total Boxes)

Draw 5 Win boxes and 3 Loss boxes -> 5 + 3 = 8 equal boxes total. Each box = 32 ÷ 8 = 4 games. Wins = 5 boxes × 4 = 20 wins.

COUNT-01 Way 1: Snap 5 Win boxes + 3 Loss boxes into 8 boxes total; deal 32 games equally (4 per box -> 20 Wins).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Part-to-Whole Equivalent Ratio

Wins to Total Games is 5 to (5+3) = 5/8. Scale 5/8 to a denominator of 32 (multiply by 4): (5 × 4)/(8 × 4) = 20/32 -> 20 wins!

COUNT-01 Way 2: Scale the 8-box row by ×4 (8 × 4 = 32), so 5 × 4 = 20 Wins and 3 × 4 = 12 Losses!

SC-6 Keystrokes: 32 ENTER 5 ENTER 3 + ÷ 5 ×

Answer: 20 wins (and 12 losses)

SECTION 3.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Robotics Club Roles: A club has 15 coders and 10 builders. Write the simplified Part-to-Part ratio (coders to builders) and Part-to-Whole ratio (coders to total).

Ans: _____

2 Smoothie Berry Mix: The ratio of strawberries to bananas in a smoothie is 4:3. If you use 28 pieces of fruit in all, use a 7-box tape diagram to find how many are strawberries.

Ans: _____

3 Paint Color Mixing: Green forest paint uses 5 parts blue to 2 parts yellow. If you use 15 cups of blue paint, how many cups of yellow paint do you need?

Ans: _____

4 Animal Shelter Dogs & Cats: The ratio of dogs to cats is 7:5. There are 60 animals in all. How many dogs and how many cats are at the shelter?

Ans: _____

5 Tape Diagram Difference: The ratio of gold coins to silver coins in a treasure chest is 7:3. There are 24 MORE gold coins than silver coins (4 extra boxes = 24). How many gold coins are there?

Ans: _____

6 Bike Gear Sprockets: The front chainring has 48 teeth and the rear cog has 16 teeth. Simplify the gear ratio 48 : 16.

Ans: _____

7 School Band Instruments: A band has 16 flutes, 12 clarinets, and 8 trumpets. Find the simplified ratio of flutes to the total number of instruments.

Ans: _____

8 Screen Aspect Ratio: A monitor is 1,920 pixels wide by 1,080 pixels tall. Divide both by GCF(1920, 1080) = 120 to find the classic widescreen aspect ratio!

Ans: _____

9 Trail Mix Ratio: A trail mix uses almonds, cashews, and raisins in a 3 : 2 : 5 ratio (10 parts total). How many grams of each are in a 500-gram bag?

Ans: _____

10 Chess Tournament Record: You won 24 games and lost 8 games. Find the ratio of wins to total games in simplest form.

Ans: _____

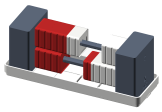
BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 3.2 — Practice A: Using Ratio Tables and Double Number Lines

Key Vocabulary: ratio table • equivalent ratios • double number line • scaling (multiplying or dividing)

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)



Part A (3 boxes):

1	1	1

Part B (5 boxes):

1	1	1	1	1

M1: 3:5 Tape | M2: 8 Boxes

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Add repeated COUNT-01 jumps of (+8 m, +3 s) five times across the baseplate: 8 -> 16 -> 24 -> 32 -> 40 m.

Block Way 2 (+20 pts): Build a 5-row array of 8 tiles (8 × 5 = 40 m) paired with a 5-row array of 3 tiles (3 × 5 = 15 s)!

Both Ways Proof: Five additive jumps of +8 (Double Number Line) equals a 5 × 8 rectangular array (Scale Factor ×5)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: A drone flies 8 meters every 3 seconds. Build a 5-step table to find how far it flies in 15 seconds.

METHOD 1 (+10 pts — 1st Token): Repeated Addition in a Ratio Table

Add 8 m for every 3 s: (3 s, 8 m) -> (6 s, 16 m) -> (9 s, 24 m) -> (12 s, 32 m) -> (15 s, 40 m).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Multiply by the Scale Factor

To go from 3 s to 15 s, multiply by 15 ÷ 3 = 5. Multiply distance by the same factor: 8 m × 5 = 40 m.

SC-6 Keystrokes: `8` `g` `REP` `+` `+` `+` `+`

Answer: 40 meters in 15 seconds

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 3.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Repeated Addition / Double Number Line | Method 2..

1 Complete a ratio table to write the first 4 equivalent ratios of 4 : 9.

Ans: _____

2 In a ratio table where A : B = 4 : 9, find B when A = 28.

Ans: _____

3 In a ratio table where X : Y = 5 : 12, find X when Y = 60.

Ans: _____

4 A printer prints 15 pages every 2 minutes. How many pages in 8 minutes?

Ans: _____

5 Hot cocoa uses 3 tbsp cocoa for every 8 oz milk. How much milk for 18 tbsp cocoa?

Ans: _____

6 8 markers cost $12. Scale down to 2 markers, then scale up to find the cost of 10 markers.

Ans: _____

7 A heart beats 21 times in 15 seconds. How many beats in 60 seconds (1 minute)?

Ans: _____

8 Error Analysis: To find a ratio equivalent to 3:5, Sam added 4 to both terms and wrote 7:9. Explain and fix his error.

Ans: _____

9 9 smoothies take 12 cups of berries. How many cups of berries for 15 smoothies? Solve 2 ways!

Ans: _____

10 An e-bike travels 14 miles on 35% battery. How far can it travel on 100% battery?

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 3.2 — Practice B: Modeling Real Life (Using Ratio Tables and Double Number Lines)

Textbook Methods: Method 1: Repeated Addition / Double Number Line | Method 2: Multiplicative Scale Factor

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: SKATE PARK PASSES (6 TICKETS FOR $45 -> 10 TICKETS) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Six skate park tickets cost $45. Use a ratio table to find the cost of 10 tickets.

METHOD 1 (+10 pts — 1st Token): Scale Down then Scale Up in a Ratio Table
Divide (6 tickets, $45) by 3 to get (2 tickets, $15). Then multiply (2 tickets, $15) by 5 to get (10 tickets, $75)!

COUNT-01 Way 1: Partition the 6-ticket bar ($45) into three 2-ticket chunks ($15 each), then lay 5 chunks ($75).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Unit Price per 1 Ticket
Find 1 ticket first: $45 ÷ 6 = $7.50 per ticket. Multiply by 10 tickets: $7.50 × 10 = $75.

COUNT-01 Way 2: Divide $45 into 6 single-ticket boxes ($7.50 each), then line up 10 boxes -> $75!

SC-6 Keystrokes: 45 ENTER 3 ÷ 5 ×

Answer: $75 for 10 tickets

SECTION 3.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 3D Printing Speed: A printer extrudes 15 grams of filament every 20 minutes. Use a ratio table to find how many grams it extrudes in 60 minutes (1 hour).

Ans: _____

2 Aquarium Salt Mix: A marine tank needs 7 scoops of sea salt for every 2 gallons of water. How many scoops are needed for a 16-gallon tank?

Ans: _____

3 Scale Down & Scale Up: 8 notebooks cost $12. Divide by 4 to find the cost of 2 notebooks, then multiply by 5 to find the cost of 10 notebooks!

Ans: _____

4 Heartbeat Counter: A runner counts 28 heartbeats in 15 seconds. At this rate, how many times does her heart beat in 60 seconds (1 minute)?

Ans: _____

5 Maple Syrup Sap: It takes 40 gallons of tree sap to boil down 1 gallon of pure maple syrup. How many gallons of sap are needed for 6.5 gallons of syrup?

Ans: _____

6 Moon Weight Ratio: An object weighing 60 pounds on Earth weighs 10 pounds on the Moon. How much would a 150-pound astronaut weigh on the Moon?

Ans: _____

7 Pizza Dough Flour: 6 cups of flour make 4 pizza crusts. Use a ratio table to find how many cups of flour are needed for 10 pizza crusts.

Ans: _____

8 Solar Charger: A portable solar panel charges 18% of a battery every 15 minutes. How long does it take to charge 90% of the battery?

Ans: _____

9 Model Train Scale: A scale model uses 2 inches to represent 15 feet of real train. If a real boxcar is 60 feet long, how many inches long is the model?

Ans: _____

10 Smoothie Stand Lemonade: 9 lemons make 6 cups of lemonade. How many lemons do you need to make 20 cups of lemonade?

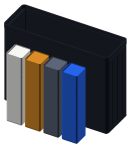
Ans: _____

Section 3.3 — Practice A: Rates, Unit Rates, and Comparing Rates

Key Vocabulary: rate • unit rate • unit cost (unit price) • equivalent rates

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)



M1 ZONE 1 TILES:
Bag A: $4.20 ÷ 12 = $0.35/oz

M2 ZONE 2 / SPLIT:
Bag B: $5.12 ÷ 16 = $0.32/oz

M1 = M2 ==> Bag B Wins!

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Deal 315 into 5 equal COUNT-01 boxes so 1 single unit box holds 315 ÷ 5 = 63 words per 1 minute.

Block Way 2 (+20 pts): To compare two rates, either compare 1-box unit heights OR scale both bars to a common number of boxes!

Both Ways Proof: Finding the 1-box Unit Rate and scaling to a Common Multiple both reveal the exact same winner!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the unit rate in words per minute for 315 words in 5 minutes (even if you typed 5 first!).

METHOD 1 (+10 pts — 1st Token): Divide by the Denominator for 1 Unit

Write 315 words / 5 minutes. Divide top and bottom by 5: (315 ÷ 5) / (5 ÷ 5) = 63 words per 1 minute.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Double Number Line / Multiply Check

Split 315 into 300 + 15: 300÷5 = 60 and 15÷5 = 3 -> 63 wpm. Check: 63 × 5 = 315 words.

SC-6 Keystrokes: `5` `ENTER` `315` `x⇔y` `÷`

Answer: 63 words per minute

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 3.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Unit Rate (Divide to Get 1 Unit) | Method 2: Equi..

1 Find the unit speed in miles per hour: 195 miles in 3 hours.

Ans: _____

2 You earn $36 for 4 hours of tutoring. Find the unit rate in dollars per hour.

Ans: _____

3 Find the unit price per pound: $7.80 for 6 pounds of apples.

Ans: _____

4 A 3D printer uses 90 grams of filament in 6 hours. How many grams per hour?

Ans: _____

5 Best Buy: Cereal A is $4.50 for 15 oz. Cereal B is $5.04 for 18 oz. Find both unit prices!

Ans: _____

6 Best Buy: 6-pack of pens for $5.10 vs. 10-pack of pens for $8.20.

Ans: _____

7 A train travels 210 miles in 3.5 hours. At this rate, how far does it travel in 5 hours?

Ans: _____

8 Pump A fills 45 gallons in 5 min. Pump B fills 64 gallons in 8 min. Which pump is faster?

Ans: _____

9 An extruder pushes 45 mm of filament in 15 seconds. How much in 60 seconds (1 min)? Solve 2 ways!

Ans: _____

10 Car A drives 168 miles on 6 gallons. Car B drives 232 miles on 8 gallons. Which car gets more mpg?

Ans: _____

Section 3.3 — Practice B: Modeling Real Life (Rates, Unit Rates, and Comparing Rates)

Textbook Methods: Method 1: Unit Rate (Divide to Get 1 Unit) | Method 2: Equivalent Rate / Common Multiple Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: BEST-BUY BATTERY SHOWDOWN (PACK A VS. PACK B) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Pack A has 8 AA batteries for $11.60. Pack B has 12 AA batteries for $16.80. Which pack is the better buy?

METHOD 1 (+10 pts — 1st Token): Compare Unit Prices (Cost per 1 Battery)
Pack A: $11.60 ÷ 8 = $1.45 per battery. Pack B: $16.80 ÷ 12 = $1.40 per battery. Pack B saves $0.05 per battery!

PLACE-01 Way 1: Divide Pack A ($5.80 ÷ 4 = $1.45/box) vs. Pack B ($8.40 ÷ 6 = $1.40/box) -> Pack B is cheaper.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Scale Both Packs to LCM(8, 12) = 24 Batteries
For 24 batteries: 3 of Pack A cost 3 × $11.60 = $34.80; 2 of Pack B cost 2 × $16.80 = $33.60. Pack B is $1.20 cheaper!

COUNT-01 Way 2: Scale both to a 12-battery bar: 3 Pack As = $17.40 vs. 2 Pack Bs = $16.80 (saves $0.60)!

SC-6 Keystrokes: `11.60` `ENTER` `8` `÷` `16.80` `ENTER` `12` `÷`

Answer: Pack B ($1.40/battery vs. $1.45/battery)

SECTION 3.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 High-Speed Train: A bullet train travels 450 miles in 3 hours. Find its unit rate (average speed) in miles per hour.

Ans: _____

2 Best-Buy Granola Bars: Box A has 6 bars for $4.50. Box B has 10 bars for $7.00. Find the unit cost per bar for each box. Which is the better buy?

Ans: _____

3 3D Filament Comparison: Spool A is 500 grams for $11.00. Spool B is 1,000 grams for $19.50. Find the price per 100 grams for each. Which is better?

Ans: _____

4 Swimming Pool Pump: A pump fills 180 gallons in 12 minutes. Find the unit rate in gallons per minute, and find how much it pumps in 25 minutes.

Ans: _____

5 Typing Sprint Showdown: Maya types 220 words in 4 minutes. Leo types 285 words in 5 minutes. Find each student's unit rate (wpm). Who types faster?

Ans: _____

6 Fuel Economy Test: Car A drives 336 miles on 12 gallons of gas. Car B drives 405 miles on 15 gallons. Which car gets more miles per gallon (mpg)?

Ans: _____

7 Dog Walking Job: You earn $54 for walking 6 dogs. At this unit rate, how many dogs must you walk to earn $99?

Ans: _____

8 Juice Box Bulk Pack: An 8-pack of juice boxes costs $3.60. A 12-pack costs $5.04. Find the unit price per juice box for each pack.

Ans: _____

9 Snowfall Rate: During a blizzard, 14 inches of snow fall in 3.5 hours. What is the unit rate in inches per hour?

Ans: _____

10 Unit Rate Distance Prediction: A rover drives 27 meters in 6 seconds. Find its speed in meters per second, and find how far it travels in 20 seconds.

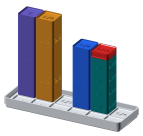
Ans: _____

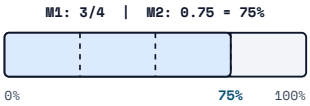
Section 3.4 — Practice A: Percents, Fractions, and Decimals

Key Vocabulary: percent ('per hundred') • hundredths grid • benchmark percent • part • whole

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (FRAC-01: 84MM FRACTION STUDIO)





DEMONSTRATE BOTH METHODS WITH FRAC-01 BLOCKS:

Block Way 1 (+10 pts): On the 10×10 COUNT-01 grid (100 squares), scale 7/20 by ×5 to cover 35 squares -> 35/100 = 35% = 0.35.

Block Way 2 (+20 pts): Compare FRAC-01 towers against 1/10 (10%) and 1/20 (5%) blocks: seven 5% blocks = 35% = 0.35!

Both Ways Proof: Covering 35 of 100 grid tiles (COUNT-01) and stacking seven 1/20 blocks (FRAC-01) both equal 0.35 = 35%!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Convert 7/20 into a decimal and a percent using [FDISP] and [f] [%].

METHOD 1 (+10 pts — 1st Token): Write an Equivalent Fraction with 100

Since 20 × 5 = 100, multiply numerator and denominator of 7/20 by 5: (7 × 5)/(20 × 5) = 35/100 = 0.35 = 35%.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Divide Numerator by Denominator

Divide 7 ÷ 20 = 0.35. Multiply by 100 (shift decimal point 2 places right) to get 35%.

SC-6 Keystrokes: `.` `7` `.` `20` `ENTER` `FDISP` `100` `×`

Answer: 0.35 and 35%

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 3.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Scale Denominator to 100 | Method 2: Divide Numer..

1 Write 3/5 as a decimal and as a percent.

Ans: _____

2 Write 9/25 as a decimal and as a percent.

Ans: _____

3 Write 45% as a decimal and as a fraction in simplest form.

Ans: _____

4 Write 0.125 as a percent and as a fraction in simplest form.

Ans: _____

5 Find 30% of 80 using a benchmark percent (3 × 10%) or decimal multiplication.

Ans: _____

6 Find 15% of 60.

Ans: _____

7 You answered 21 out of 25 questions right. What is your score as a percent?

Ans: _____

8 Convert 150% into a decimal and a mixed fraction.

Ans: _____

9 Order 5/8, 60%, and 0.61 from least to greatest by converting all 3 to decimals (and percents!).

Ans: _____

10 A phone battery is at 35% of its 4,000 mAh capacity. How many mAh remain? Solve 2 ways!

Ans: _____

BLOCK KIT: FRAC-01 • 84mm Fraction Studio

Section 3.4 — Practice B: Modeling Real Life (Percents, Fractions, and Decimals)

Textbook Methods: Method 1: Scale Denominator to 100 | Method 2: Divide Numerator by Denominator

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: BASKETBALL FREE-THROW PERCENTAGE (18 OUT OF 24) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: You make 18 out of 24 free throws. Find your shooting rate as a simplified fraction, decimal, and percent.

METHOD 1 (+10 pts — 1st Token): Simplify by GCF First (Benchmark Fraction)
Divide 18 and 24 by GCF(18, 24) = 6: 18/24 = 3/4. Recognize the benchmark fraction 3/4 = 75/100 = 0.75 = 75%!

FRAC-01 Way 1: Simplify 18/24 to three 1/4 blocks (3/4); since each 1/4 block is 25%, three blocks = 75%.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Direct Division (Part ÷ Whole)
Divide 18 ÷ 24 = 0.75. Shift the decimal 2 places right -> 75%, and press [FDISP] to confirm 3/4.

COUNT-01 Way 2: Divide 18 ÷ 24 = 0.75 (75 squares on the 10×10 hundred-grid = 75%)!

SC-6 Keystrokes: 18 ENTER 24 ÷ FDISP

Answer: 3/4 = 0.75 = 75%

SECTION 3.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Phone Storage Bar: You have used 17 out of 20 gigabytes on your tablet. Write 17/20 as a decimal and as a percent.

Ans: _____

2 Soccer Goal Rate: A striker scores on 9 out of 25 penalty kicks. Write 9/25 as a decimal and as a percent.

Ans: _____

3 Quiz Mastery Score: You earn 40% of the points on Part A (out of 50 points) and 80% on Part B (out of 50 points). How many total points did you score?

Ans: _____

4 School Survey: In a poll of 160 middle schoolers, 65% voted for a robotics elective. How many students voted for robotics?

Ans: _____

5 Battery Charge Left: A laptop shows 15% battery remaining. Write 15% as a decimal and as a fraction in simplest form.

Ans: _____

6 Finding the Whole from a Part: 18 students in the science club are 6th graders. If 6th graders make up 40% of the club, how many total students are in the club?

Ans: _____

7 Benchmark Percent Mental Math: Find 15% of $60 by finding 10% of $60 ($6) plus half of that for 5% ($3).

Ans: _____

8 Archery Target Bullseyes: An archer hits the bullseye 21 times out of 24 arrows. Write her accuracy as a simplified fraction, decimal, and percent.

Ans: _____

9 Fundraiser Goal: A class has raised $360, which is 75% (3/4) of their goal. What is the total fundraiser goal?

Ans: _____

10 Three-Costume Ordering: Order 3/5, 58%, and 0.62 from least to greatest by converting all three to percents.

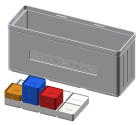
Ans: _____

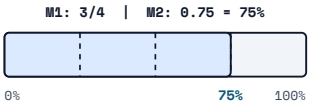
Section 3.5 — Practice A: Solving Percent Problems, Discounts & Conversions

Key Vocabulary: discount • sale price • markup • sales tax • percent of change • conversion factor

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (PLACE-01: PLACE VALUE & FRACTION STUDIO)





DEMONSTRATE BOTH METHODS WITH PLACE-01 BLOCKS:

Block Way 1 (+10 pts): Build a 4-block COUNT-01 bar for $60 (each 25% block = $15) and remove 1 block: $60 - $15 = $45.

Block Way 2 (+20 pts): Keep the 3 remaining 25% blocks directly (100% - 25% = 75%): 3 × $15 = $45 in one step!

Both Ways Proof: Subtracting 1 removed 25% block ($60 - $15) leaves the exact same 3 blocks as finding 75% of $60 ($45)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: A $60 video game is 25% off. Find the discount and the sale price.

METHOD 1 (+10 pts — 1st Token): Find Discount First, Then Subtract

Discount = 25% of $60 = 0.25 × 60 = $15. Subtract from original price: $60 - $15 = $45.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Multiply by Remaining Percent

If 25% is taken off, you pay 100% - 25% = 75% of the original price: 0.75 × $60 = (3/4) × 60 = $45!

SC-6 Keystrokes: `60` `ENTER` `25` `f` `%` `-`

Answer: Discount = $15; Sale Price = $45 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 3.5 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Subtract the Discount from Original | Method 2: M..

1 An $80 pair of sneakers is on sale for 30% off. Find the discount and the sale price.

Ans: _____

2 A $40 restaurant meal includes a 15% tip. Find the tip amount and the total bill.

Ans: _____

3 Find the percent of change from 50 to 65 using (Change ÷ Original) × 100%.

Ans: _____

4 Find the percent of decrease from 80 down to 60.

Ans: _____

5 Use the conversion factor 2.54 cm = 1 in to convert 10 inches to centimeters.

Ans: _____

6 Use the conversion factor 1 in = 2.54 cm to convert 50.8 centimeters to inches.

Ans: _____

7 Convert 11 pounds to kilograms (use 1 lb ≈ 0.4536 kg or 1 kg ≈ 2.2 lb).

Ans: _____

8 Reverse Percent: 18 is 25% of what number? (Divide 18 by 0.25!)

Ans: _____

9 Modeling Real Life: A $120 skateboard is 20% off. Find the sale price in two ways (subtracting the 20% discount vs. finding 80% of $120).

Ans: _____

10 Modeling Real Life: A 32-inch skateboard deck weighs 5 pounds. Convert its length to centimeters and its weight to kilograms.

Ans: _____

Section 3.5 — Practice B: Modeling Real Life (Solving Percent Problems, Discounts & Conversions)

Textbook Methods: Method 1: Subtract the Discount from Original | Method 2: Multiply by Remaining Percent (100% - p%)

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: CHANNEL GROWTH (250 TO 325) & UNIT FACTOR (32 IN -> CM) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A maker channel grows from 250 to 325 subscribers. Find the percent increase, and convert a 32-inch build plate to centimeters.

METHOD 1 (+10 pts — 1st Token): Change ÷ Original Formula

Increase = 325 - 250 = 75. Divide by ORIGINAL: 75 ÷ 250 = 3/10 = 0.30 = +30%. For 32 in: 32 × 2.54 cm/in = 81.28 cm.

COUNT-01 Way 1: Compare the 6-block growth (26 - 20 = 6) to the original 20-block bar: 6/20 = 30/100 = +30%.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Ratio of New-to-Old & SC-6 [%CHG]

New ÷ Old = 325 ÷ 250 = 1.30 = 130% of original (a +30% increase!). Verify with 250 [ENTER] 325 [g] [%CHG] and 32 [f] [->cm].

PLACE-01 Way 2: Scale 32 inches by 2.54 cm per 1-inch tile -> 32 × 2.54 = 81.28 cm!

SC-6 Keystrokes: 250 ENTER 325 g %CHG 32 f ->cm

Answer: +30% increase; 81.28 cm

SECTION 3.5 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Gaming Headset Deal: A $75 wireless headset is on sale for 20% off. Find the discount and the sale price in two ways.

Ans: _____

2 Pizza Team Dinner Tip: A team dinner costs $64.00 before tip. You leave a 15% tip. What is the tip amount and the total bill?

Ans: _____

3 Sales Tax Checkout: You buy a $48.00 3D build plate plus 7.5% sales tax. Find the tax amount and the final total.

Ans: _____

4 Ticket Price Percent Increase: Concert tickets rose from $40 last year to $52 this year. Find the percent of increase.

Ans: _____

5 Winter Jacket Clearance: A jacket was marked down from $90 to $63. Find the percent of decrease.

Ans: _____

6 5K Race Distance Conversion: A race is 5 kilometers long. Since 1 mile ≈ 1.609 km (or convert 5000 m), convert 10 miles into kilometers.

Ans: _____

7 customs Luggage Weight: Your suitcase weighs 20 kilograms. Convert 20 kg into pounds (1 kg ≈ 2.2046 lb).

Ans: _____

8 Water Bottle Capacity: A sports jug holds 1 gallon of water. Convert 1 gallon to liters ([f] [->L]) and fluid ounces (128 fl oz).

Ans: _____

9 Double Coupon Trap: A $100 bike has a 20% off store sale, and you have an extra 10% off coupon applied to the sale price. Why is the final price $72 instead of $70?

Ans: _____

10 Temperature Lab Conversion: Room temperature is 20°C. Convert 20°C to degrees Fahrenheit using F = 1.8C + 32 or [g] [->°F].

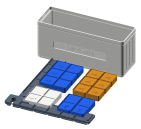
Ans: _____

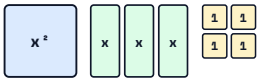
Section 4.1 — Practice A: Writing and Evaluating Algebraic Expressions

Key Vocabulary: algebraic expression • variable • term • coefficient • constant • evaluate • substitute

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (ALGB-01: ALGEBRA BOXES & TILES)





M1: Sort Shapes | M2: Sub x

x²-Flats | x-Rods | 1-Cubes

DEMONSTRATE BOTH METHODS WITH ALGB-01 BLOCKS:

Block Way 1 (+10 pts): Load 4 unit cubes into each of 3 x-boxes (12) and 5 cubes into each of 3 y-boxes (15) on ALGB-01 -> 12 + 15 = 27.

Block Way 2 (+20 pts): Pair 1 x-box + 1 y-box (4 + 5 = 9 cubes) and build 3 identical groups on ALGB-01 -> 3 × 9 = 27!

Both Ways Proof: Filling 3x and 3y separately (12 + 15) holds the exact same cubes as 3 combined (x + y) groups (3 × 9 = 27)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Evaluate 4x + 7 when x = 5 using Algebra Tiles and the SC-6.

METHOD 1 (+10 pts — 1st Token): Substitute with Parentheses & PEMDAS

Replace x with (5) in 4x + 7: 4(5) + 7. Multiply first (4 × 5 = 20), then add 7: 20 + 7 = 27.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Physical Tile Replacement

Trade each of the 4 green x-rods for 5 unit cubes (5 + 5 + 5 + 5 = 20 cubes) plus 7 cubes = 27 cubes.

SC-6 Keystrokes: `4` `ENTER` `5` `×` `7` `+`

Answer: 27

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 4.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Algebra Tile Replacement Model | Method 2: Parent..

1 Evaluate 6x - 9 when x = 7.

Ans: _____

2 Evaluate 5n + 14 when n = 8.

Ans: _____

3 Evaluate 40 - 3m when m = 9. (Park 40 on Y first!)

Ans: _____

4 Evaluate $x^2 + 11$ when x = 6.

Ans: _____

5 Evaluate $2y^2 - 5$ when y = 4. (Square 4 before multiplying by 2!)

Ans: _____

6 Evaluate 7a + 3b when a = 4 and b = 9.

Ans: _____

7 Evaluate (x + y) ÷ 4 when x = 19 and y = 13.

Ans: _____

8 Evaluate 5k - 2.4 when k = 3.5.

Ans: _____

9 Skate park rental costs $8 flat plus $4.50 per hour (8 + 4.50h). Find the cost for 4 hours 2 ways!

Ans: _____

10 Surface area of a cube is $6s^2$. Find the surface area when edge s = 5 cm.

Ans: _____

Section 4.1 — Practice B: Modeling Real Life (Writing and Evaluating Algebraic Expressions)

Textbook Methods: Method 1: Algebra Tile Replacement Model | Method 2: Parentheses Substitution & PEMDAS

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: TWO-VARIABLE SCORE FORMULA (3A² + 4B FOR A = 5, B = 6) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: In a game, each boss trophy (a) is worth 3a² points and each gem (b) is worth 4b points. Find your score for a = 5 and b = 6.

METHOD 1 (+10 pts — 1st Token): Substitute & Evaluate Powers First

Substitute a = 5 and b = 6: 3(5)² + 4(6). Evaluate the power first (5² = 25), multiply (3×25 = 75 and 4×6 = 24), then add: 75 + 24 = 99.

ALGB-01 Way 1: Build a 5×5×5 cube (125) on the left and remove (2 × 8 + 10 = 26) cubes -> 99 cubes remain.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Separate Term Chunks on the SC-6 Stack

Chunk 1 (Bosses): 5 [x²] 3 [×] = 75. Chunk 2 (Gems): 4 [ENTER] 6 [×] = 24. Combine both chunks with [+] -> 99 points!

ALGB-01 Way 2: Evaluate the 5³ cube (125), subtract two 8-rods (109), then subtract the 10-tray -> 99!

SC-6 Keystrokes: `5` `x²` `3` `×` `4` `ENTER` `6` `×` `+`

Answer: 99 points

SECTION 4.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Bowling Shoe Rental: Bowling costs $6 for shoe rental plus $5 per game g: 5g + 6. Evaluate the cost for g = 4 games.

Ans: _____

2 Carnival Ride Wristband: You start with 50 tickets and spend 4 tickets per ride r: 50 - 4r. How many tickets are left after r = 9 rides?

Ans: _____

3 Square Tile Patio Cost: A square patio of side s feet costs 8s² + 25 dollars to pave. Find the cost when s = 5 feet.

Ans: _____

4 Esports Tournament Points: A team scores 10w + 3d points for w wins and d draws. Find their score for w = 7 wins and d = 4 draws.

Ans: _____

5 Falling Object Physics: A ball dropped from a tower falls 16t² feet in t seconds. How far does it fall in t = 3 seconds?

Ans: _____

6 Surface Area of a Cube Formula: The surface area of a cube is 6s². Evaluate 6s² when the edge length is s = 4 inches.

Ans: _____

7 Movie Theater Concession: Popcorn costs $7 (p) and drinks cost $4 (d): 7p + 4d. Evaluate for p = 3 popcorns and d = 5 drinks.

Ans: _____

8 Trapezoid Bridge Truss: Evaluate ½h(a + b) when h = 6, a = 11, and b = 15.

Ans: _____

9 3D Print Time Formula: A batch of k keycaps takes 12k + 15 minutes (including 15 min warm-up). How long does a batch of k = 8 keycaps take?

Ans: _____

10 Two-Variable Exponent Challenge: Evaluate 2x³ - 3y² when x = 4 and y = 5.

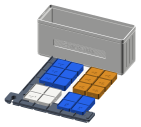
Ans: _____

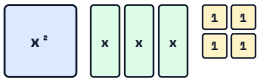
Section 4.2 — Practice A: Simplifying Expressions by Combining Like Terms

Key Vocabulary: like terms • simplest form • Commutative Property • Associative Property • equivalent expressions

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (ALGB-01: ALGEBRA BOXES & TILES)





M1: Sort Shapes | M2: Sub x

x²-Flats | x-Rods | 1-Cubes

DEMONSTRATE BOTH METHODS WITH ALGB-01 BLOCKS:

Block Way 1 (+10 pts): Sort ALGB-01 tiles by physical shape: combine 4 x-rods + 2 x-rods = 6 x-rods, and 5 + 3 = 8 unit cubes (6x + 8).

Block Way 2 (+20 pts): Test x = 2 cubes per x-rod in both layouts: (8 + 5 + 4 + 3 = 20 cubes) matches (6×2 + 8 = 20 cubes)!

Both Ways Proof: Only matching tile shapes (x² flats, x rods, 1 cubes) can stack together, and substituting x = 2 proves equivalence!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Simplify 4x + 5 + 2x + 3 and prove it by testing x = 10 on your SC-6.

METHOD 1 (+10 pts — 1st Token): Group Like Terms & Add Coefficients

Reorder using Commutative/Associative Properties: (4x + 2x) + (5 + 3) = (4 + 2)x + 8 = 6x + 8.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Substitute x = 10 to Prove Equivalence

Original at x = 10: 4(10)+5+2(10)+3 = 40+5+20+3 = 68. Simplified at x = 10: 6(10)+8 = 68 (Exact match!).

SC-6 Keystrokes: 4 ENTER 2 + 5 ENTER 3 +

Answer: 6x + 8

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 4.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Sort & Group Algebra Tiles by Shape | Method 2: A..

1 Sort and simplify: 7x + 4 + 2x + 9.

Ans: _____

2 Simplify: 8x + 11 - 3x - 4.

Ans: _____

3 Simplify: 9y - y + 6. (Remember: -y means -1y!)

Ans: _____

4 Simplify: 12a + 5 - 4a + 8.

Ans: _____

5 Sort all 3 tile shapes: $4x^2 + 3x + 2x^2 + 5x + 7$.

Ans: _____

6 Simplify: $6n^2 + 9n - 2n^2 - 4n + 10$.

Ans: _____

7 Simplify with decimals: 3.5x + 4.2 + 1.8x - 1.5.

Ans: _____

8 Error Analysis: Leo simplified 5x + 3 to 8x. Substitute x = 10 into both expressions to explain why they are not equivalent.

Ans: _____

9 A rectangle has two sides of (2x + 3) and two sides of (x + 5). Write its simplified perimeter & test x=10!

Ans: _____

10 A triangle has sides (3x + 2), (2x + 7), and (4x - 1). Write the simplified perimeter & test x=5!

Ans: _____

Section 4.2 — Practice B: Modeling Real Life (Simplifying Expressions by Combining Like Terms)

Textbook Methods: Method 1: Sort & Group Algebra Tiles by Shape | Method 2: Add/Subtract Coefficients & Test a Value

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: COMBINING X²-FLATS, X-RODS & CONSTANTS (3X² + 5X + 2X² - X + 6) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Simplify the expression $3x^2 + 5x + 2x^2 - x + 6$ by combining like terms, and check with x = 2.

METHOD 1 (+10 pts — 1st Token): Sort by Variable & Exponent Power

x^2-terms: $3x^2 + 2x^2 = 5x^2$. x-terms: 5x - 1x = 4x (remember -x means -1x!). Constant: +6 -> $5x^2 + 4x + 6$.

ALGB-01 Way 1: Group three x^2-flats + two x^2-flats ($5x^2$), seven x-rods minus three x-rods (4x), and 6 unit cubes.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Test x = 2 in Both Expressions

At x = 2: Original = 3(4) + 5(2) + 2(4) - 2 + 6 = 12 + 10 + 8 - 2 + 6 = 34. Simplified = 5(4) + 4(2) + 6 = 20 + 8 + 6 = 34!

ALGB-01 Way 2: Substitute x = 2 into each tile ($x^2 = 4$, x = 2): both arrangements hold 5(4) + 4(2) + 6 = 34 cubes!

SC-6 Keystrokes: 3 ENTER 2 + 5 ENTER 1 -

Answer: $5x^2 + 4x + 6$

SECTION 4.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Triangle Perimeter Expression: A triangle has side lengths (2x + 3), (3x + 1), and (4x + 5). Combine like terms to write its perimeter in simplest form.**

Ans: _____

2 **Rectangle Perimeter: A garden is 3x + 4 feet long and x + 2 feet wide. Add all 4 sides (or 2 lengths + 2 widths) and simplify.**

Ans: _____

3 **Arcade Token Pouches: You have 6 pouches of x tokens plus 9 loose tokens, give away 2 pouches, and find 5 more loose tokens: 6x + 9 - 2x + 5. Simplify.**

Ans: _____

4 **Algebra Tile Sort: Simplify $4x^2 + 7x + 3 + 2x^2 - 3x + 8$ by grouping x^2-flats, x-rods, and unit cubes.**

Ans: _____

5 **Bake Sale Revenue: On Saturday you sell 8c + 12 items and on Sunday you sell 5c + 19 items. Write a simplified expression for the weekend total.**

Ans: _____

6 **Identify Parts of an Expression: In 9x + 14 - 2x + 5, simplify first, then state the coefficient of x and the constant term.**

Ans: _____

7 **Fraction Coefficients: Combine like terms to simplify (3/8)n + 4 + (1/8)n + 9.**

Ans: _____

8 **Decimal Coefficients: Combine like terms to simplify 2.4y + 5.8 + 1.6y - 2.3.**

Ans: _____

9 **Equivalence Test: Simplify 10m - 4m + 15 - 6, and verify your answer by substituting m = 5 into both the original and simplified expressions.**

Ans: _____

10 **Square Frame Tiles: Combine $5x^2 + 6x + 10 - 2x^2 - 6x - 3$. Notice what happens to the x-terms!**

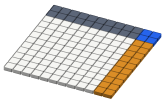
Ans: _____

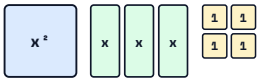
Section 4.3 — Practice A: The Distributive Property

Key Vocabulary: Distributive Property • area model • partial products • equivalent expressions

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (ALGB-01: ALGEBRA BOXES & TILES)





M1: Sort Shapes | M2: Sub x

x²-Flats | x-Rods | 1-Cubes

DEMONSTRATE BOTH METHODS WITH ALGB-01 BLOCKS:

Block Way 1 (+10 pts): Join widths 10 + 4 = 14 first on COUNT-01 / ALGB-01 and build one 6 × 14 rectangle (84 tiles).

Block Way 2 (+20 pts): Split the 6 × 14 rectangle at the partition line into a 6 × 10 array (60) plus a 6 × 4 array (24) -> 84!

Both Ways Proof: One joined 6×(10+4) rectangle covers the exact same 84 baseplate sockets as two split 6×10 + 6×4 rectangles!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Compute 6 × (10 + 4) both ways on your SC-6 to see why distribution works.

METHOD 1 (+10 pts — 1st Token): Add Inside Parentheses First

Combine the total width: 6 × (10 + 4) = 6 × 14 = 84.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Use the Distributive Property

Multiply 6 by each term inside the parentheses: 6(10) + 6(4) = 60 + 24 = 84.

SC-6 Keystrokes: 6 ENTER 10 × 6 ENTER 4 × +

Answer: 84 (Matches 6 × 14 = 84)

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 4.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Total Width First a(b + c) | Method 2: Split Area..

1 Evaluate 7 × (20 + 3) in two ways: using the order of operations and using the Distributive Property.

Ans: _____

2 Build 4 rows of (x + 5) and write the distributed expression for 4(x + 5).

Ans: _____

3 Use the Distributive Property to expand 6(3x + 4).

Ans: _____

4 Expand 8(2y - 5). Watch the subtraction sign!

Ans: _____

5 Mental Math Split: Compute 9 × 43 by splitting 43 into (40 + 3).

Ans: _____

6 Distribute and combine like terms: 3(4x + 2) + 5x.

Ans: _____

7 Distribute and combine: 4(3n + 5) - 7.

Ans: _____

8 Error Analysis: Mia wrote 5(x + 6) = 5x + 6. Use an area model to show what she forgot to multiply by 5.

Ans: _____

9 Simplify 5(2x + 3) + 4x AND prove your answer matches the original when x = 2!

Ans: _____

10 6 friends each buy a burger ($x) and $3.50 fries: 6(x + 3.50). Expand it and evaluate for x = $6.50!

Ans: _____

BLOCK KIT: ALGB-01 • Algebra Boxes & Tiles

Section 4.3 — Practice B: Modeling Real Life (The Distributive Property)

Textbook Methods: Method 1: Total Width First a(b + c) | Method 2: Split Area Model ab + ac

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: COMBO MEAL ORDER (SIMPLIFY 5(2X + 3) + 4X) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Five friends each buy 2 tacos (x) and a $3 drink, plus 4 extra tacos: 5(2x + 3) + 4x. Simplify and check for x = $2.

METHOD 1 (+10 pts — 1st Token): Distribute First, Then Combine Like Terms
Distribute 5 across (2x + 3): 5(2x) + 5(3) + 4x = 10x + 15 + 4x. Combine x-terms: (10x + 4x) + 15 = 14x + 15.

ALGB-01 Way 1: Build a 4 × (2x + 5) rectangle (8x + 20) and a 3 × (2x - 5) rectangle (6x - 5) -> 14x + 15.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Check Both Forms at x = $2
Original at x = 2: 5(2·2 + 3) + 4(2) = 5(7) + 8 = $43. Simplified at x = 2: 14(2) + 15 = 28 + 15 = $43!

ALGB-01 Way 2: Put x = 2 cubes per x-rod: 4(9) + 3(4) - 5 = 36 + 7 = 43 cubes, matching 14(2) + 15 = 43!

SC-6 Keystrokes: 5 ENTER 2 × 4 + 5 ENTER 3 ×

Answer: 14x + 15 ($43 at x = $2)

SECTION 4.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Mental Math Ticket Split: Use the Distributive Property to compute 8 × 53 as 8(50 + 3) in your head.**

Ans: _____

2 **99-Cent Store Trick: Use the Distributive Property Over Subtraction to compute 6 × $9.90 as 6(10 - 0.10).**

Ans: _____

3 **Split Rectangle Garden: A garden is 7 feet wide. Its length is split into an x-foot herb bed and an 8-foot flower bed: 7(x + 8). Write the expanded area.**

Ans: _____

4 **Soccer Uniform Order: 9 players each need a jersey (2x dollars) and shorts (7 dollars): 9(2x + 7). Use the Distributive Property to simplify.**

Ans: _____

5 **Coupon Discount Expression: Four friends each buy a ticket (t) with a $3 coupon: 4(t - 3). Expand the expression and evaluate for t = $15.**

Ans: _____

6 **Distribute & Combine: Simplify 3(4x + 5) + 2x by distributing 3 first and then combining the x-terms.**

Ans: _____

7 **Two-Group Order: Simplify 4(2x + 3) + 5(x + 2) by distributing both groups and combining like terms.**

Ans: _____

8 **Fractional Distribution: Use the Distributive Property to simplify (1/2)(6x + 10).**

Ans: _____

9 **Rectangle Perimeter Formula: The perimeter of a rectangle with length (2x + 5) and width (x + 3) is 2(2x + 5) + 2(x + 3). Simplify completely.**

Ans: _____

10 **Two-Way Proof: Evaluate 12 × (10 + 5) both by adding inside parentheses first (12 × 15) and by distributing (120 + 60).**

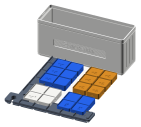
Ans: _____

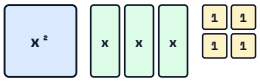
Section 4.4 — Practice A: Factoring Expressions Using the GCF

Key Vocabulary: factoring an expression • greatest common factor (GCF) • factored form • expanded form

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (ALGB-01: ALGEBRA & PRIME STUDIO)





M1: Sort Shapes | M2: Sub x

x²-Flats | x-Rods | 1-Cubes

DEMONSTRATE BOTH METHODS WITH ALGB-01 BLOCKS:

Block Way 1 (+10 pts): Arrange 18 x-rods and 24 unit cubes on ALGB-01 into 6 equal rows -> each row has 3 x-rods + 4 cubes: 6(3x + 4).

Block Way 2 (+20 pts): Use PRIME-01 towers to find GCF(18, 24) = 6 as the shared row count, then distribute 6(3x + 4) to check!

Both Ways Proof: Factoring with GCF = 6 finds the outside height of the ALGB-01 rectangle whose inside width is (3x + 4)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Factor 18x + 24 completely using the [GCF] key.

METHOD 1 (+10 pts — 1st Token): Factor Out GCF(18, 24) = 6

Find GCF(18, 24) = 6. Divide each term by 6: 18x ÷ 6 = 3x and 24 ÷ 6 = 4 -> Factored form: 6(3x + 4).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Distribute Back to Verify

Multiply 6(3x + 4) using the Distributive Property: 6(3x) + 6(4) = 18x + 24 (Matches original expression!).

SC-6 Keystrokes: `18` `ENTER` `24` `GCF`

Answer: 6(3x + 4)

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 4.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Divide Out GCF Using Prime Factors | Method 2: Re..

1 Factor the numerical expression 12 + 20 in the form a(b + c) by dividing out the GCF.

Ans: _____

2 Factor the algebraic expression 8x + 12 using the GCF.

Ans: _____

3 Factor the algebraic expression 15y - 25 using the GCF.

Ans: _____

4 Factor 24x + 36 completely using the greatest common factor (GCF).

Ans: _____

5 Factor 32n - 48 completely using the GCF.

Ans: _____

6 Factor 42a + 14 completely. (Careful: what is 14 ÷ 14?)

Ans: _____

7 Factor 36m - 60 completely using the GCF.

Ans: _____

8 Error Analysis: Sam factored 24x + 36 as 6(4x + 6). Explain why it is not factored completely and write the complete factorization.

Ans: _____

9 Modeling Real Life: A rectangular banner has area (28x + 42) square inches and height 14 inches. Factor to find its width.

Ans: _____

10 A square garden has perimeter (36x + 48) ft. Factor out 4 to find the length of 1 side!

Ans: _____

Section 4.4 — Practice B: Modeling Real Life (Factoring Expressions Using the GCF)

Textbook Methods: Method 1: Divide Out GCF Using Prime Factors | Method 2: Reverse Area Model & Distribute to Check

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: GARDEN BED DIMENSIONS (FACTOR 45X - 30) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A rectangular garden has area (45x - 30) square feet. Factor the expression using the GCF to find its width and length.

METHOD 1 (+10 pts — 1st Token): Find GCF of 45 and 30 & Divide

GCF(45, 30) = 15. Divide both terms by 15: 45x ÷ 15 = 3x and 30 ÷ 15 = 2 -> Factored form: 15(3x - 2).

PRIME-01 Way 1: Build Tower 45 ([3][3][5]) and Tower 30 ([2][3][5]) -> shared GCF tower is [3][5] = 15.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Check by Distributing & Testing x = 4

Distribute: 15(3x) - 15(2) = 45x - 30. At x = 4: 45(4) - 30 = 150, and 15(3·4 - 2) = 15(10) = 150!

ALGB-01 Way 2: Arrange 45x - 30 into 15 equal rows of (3x - 2) -> 15(3x - 2)!

SC-6 Keystrokes: 45 ENTER 30 GCF 45 x⇔y ÷

Answer: 15(3x - 2)

SECTION 4.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Identical Gift Bags: You have 16x + 24 stickers to divide equally into the greatest possible number of gift bags. Factor out the GCF.**

Ans: _____

2 **Rectangular Solar Array: A solar panel array has area (21x + 35) square meters and width 7 meters. Factor out 7 to find the length of the array.**

Ans: _____

3 **Square Perimeter Unwrapping: A square has a total perimeter of (20x + 36) inches. Factor out 4 to find the length of 1 side of the square!**

Ans: _____

4 **Equilateral Triangle Side: An equilateral triangle (3 equal sides) has perimeter (18y - 27) cm. Factor out 3 to find the length of 1 side.**

Ans: _____

5 **Complete GCF Factoring: Factor 40x + 64 completely by dividing out the greatest common factor.**

Ans: _____

6 **Club Shirt Order: The total cost of shirts and hats is 36s + 54. Factor out the GCF to find the largest number of identical player kits.**

Ans: _____

7 **Three-Term Factoring: Find the GCF of 12, 18, and 30, and factor the expression 12x + 18y + 30.**

Ans: _____

8 **Classroom Desk Rows: A classroom has (35d - 14) square feet of desk space arranged in 7 equal rows. Factor to find the area per row.**

Ans: _____

9 **Simplify First, Then Factor: Combine like terms in 10x + 18 + 14x + 18, and then factor out the GCF completely!**

Ans: _____

10 **Factor & Evaluate Both Ways: Factor 27x - 18 completely, then evaluate both expressions when x = 2.**

Ans: _____

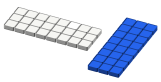
BLOCK KIT: COUNT-01 • Counting & Array Tiles

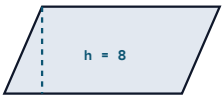
Section 5.1 — Practice A: Areas of Parallelograms and Rectangles

Key Vocabulary: polygon • parallelogram • base (b) • perpendicular height (h) • slant length • perimeter

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)





M1: Cut-Slide | M2: b×h=12×8

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Cut the overhanging right triangle of COUNT-01 tiles from the left of the parallelogram and slide it to the right -> 12×8 = 96.

Block Way 2 (+20 pts): Measure the 12-tile base and 8-tile perpendicular height (90° corner, NOT the 10-tile slant) -> 12 × 8 = 96!

Both Ways Proof: Sliding the triangle turns the slanted parallelogram into a straight 12 × 8 rectangle with the exact same 96 tiles!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the area of a parallelogram with base b = 12 cm, vertical height h = 8 cm, and slanted side 10 cm.

METHOD 1 (+10 pts — 1st Token): Cut-and-Slide Rectangle Model

Slide the left right-triangle overhang to the right side to form a 12 cm × 8 cm rectangle -> Area = 96 cm².

METHOD 2 (+20 pts DOUBLE — 2nd Token): Formula A = bh (Ignore Slant Side)

Use perpendicular base b = 12 and height h = 8: A = 12 × 8 = 96 cm² (the 10 cm slant is only for Perimeter = 2(12+10) = 44 cm).

SC-6 Keystrokes: 12 ENTER 8 f RECT

Answer: 96 cm²

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 5.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Cut-and-Slide into a Rectangle | Method 2: Area F..

1 Find the area of a parallelogram with base b = 15 m, vertical height h = 9 m, and slanted side 11 m.

Ans: _____

2 Find the area and perimeter of a rectangle measuring 18 feet long by 12 feet wide.

Ans: _____

3 Find the area of a parallelogram with b = 8.5 m and vertical height h = 6.4 m.

Ans: _____

4 Find the area of a square tile with side 13 cm.

Ans: _____

5 A parallelogram has base b = 6 1/2 in and height h = 4 in. Find its area.

Ans: _____

6 Missing Height: A parallelogram has Area = 108 cm² and base b = 12 cm. Find h.

Ans: _____

7 Missing Base: A parallelogram has Area = 135 ft² and vertical height h = 7.5 ft. Find b.

Ans: _____

8 Error Analysis: A parallelogram has b = 10, vertical h = 6, slanted side = 7.5. Sam got 75. Fix it!

Ans: _____

9 A slanted parking space has base 9 ft and straight length 18.5 ft. How much asphalt covers 4 spaces?

Ans: _____

10 A rectangular patio (15 ft × 12 ft) has a parallelogram planter (b=4 ft, h=3 ft) inside. Find the remaining patio area!

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 5.1 — Practice B: Modeling Real Life (Areas of Parallelograms and Rectangles)

Textbook Methods: Method 1: Cut-and-Slide into a Rectangle | Method 2: Area Formula A = bh & Inverse Division

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: FINDING MISSING HEIGHT OF A BANNER (A = 168 IN², B = 14 IN) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A parallelogram-shaped team banner has an area of 168 square inches and a base of 14 inches. Find its vertical height h.

METHOD 1 (+10 pts — 1st Token): Substitute into A = bh & Solve by Division
Write 168 = 14h. Divide both sides by 14: h = 168 ÷ 14 = 12 inches.

COUNT-01 Way 1: Arrange 180 area tiles into columns of 15 tiles (base = 15) -> forms 12 rows high (h = 12).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Split Division & Multiply Back to Check
Split 168 = 140 + 28: (140 ÷ 14) + (28 ÷ 14) = 10 + 2 = 12 in. Check with [f] [RECT]: 14 × 12 = 168 in²!

ALGB-01 Way 2: Balance 15h = 180 and divide by 15 -> h = 12 in (check: 15 × 12 = 180)!

SC-6 Keystrokes: 168 ENTER 14 ÷ f LASTX f RECT

Answer: h = 12 in

SECTION 5.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Diagonal Parking Space: A parallelogram parking space has a base of 10 feet, a perpendicular height of 18 feet, and a slanted side of 20 feet. Find its area.

Ans: _____

2 Parking Space Stripes: For the same parking space (base 10 ft, slanted side 20 ft), what is the perimeter around all 4 outer edges?

Ans: _____

3 Staircase Railing Glass: A parallelogram glass panel has base b = 4.5 feet and vertical height h = 3.2 feet. Find its area.

Ans: _____

4 Missing Base of a Garden Tile: A parallelogram patio stone has an area of 135 square inches and a height of 9 inches. Find its base length b.

Ans: _____

5 Quilt Pattern Rhombus: A quilt uses 8 identical parallelogram patches, each with base 6 cm and height 4.5 cm. What is the total area of all 8 patches?

Ans: _____

6 Basketball Key Paint: The rectangular free-throw lane is 19 feet long by 12 feet wide. How many square feet of paint are needed to cover it?

Ans: _____

7 Lawn Mower Coverage: A rectangular yard is 65 feet by 40 feet. A shed covers a 10 ft by 8 ft corner. How many square feet of grass remain?

Ans: _____

8 Fractional Parallelogram: Find the area of a parallelogram with base b = 5 1/2 inches and perpendicular height h = 4 inches.

Ans: _____

9 Same Base & Height Discovery: Rectangle A and Parallelogram B both have base = 16 cm and perpendicular height = 9 cm. Compare their areas!

Ans: _____

10 Banner Cost by Area: A parallelogram banner has base 8 ft and height 3.5 ft. Vinyl printing costs $4.00 per square foot. Find the total cost.

Ans: _____

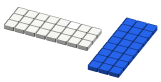
BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 5.2 — Practice A: Areas of Triangles

Key Vocabulary: triangle • congruent triangles • perpendicular base & height • Area Formula A = ½bh

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)





M1: ½(b×h) | M2: (b÷2)×h

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Build the full 10 × 6 = 60 COUNT-01 rectangle around the triangle, then split along the diagonal (60 ÷ 2 = 30).

Block Way 2 (+20 pts): Fold the top half of the triangle down into a 5 × 6 rectangle (half-base × height = 5 × 6 = 30)!

Both Ways Proof: Two congruent triangles always lock together to form one full b × h rectangle on the COUNT-01 baseplate!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the area of a triangle with base b = 10 cm and height h = 6 cm.

METHOD 1 (+10 pts — 1st Token): Half of the Enclosing Rectangle

Find the area of the 10 cm × 6 cm rectangle (10 × 6 = 60 cm²), then divide by 2 -> 60 ÷ 2 = 30 cm².

METHOD 2 (+20 pts DOUBLE — 2nd Token): Halve the Even Base or Height First

Take half of b = 10 first (½ × 10 = 5), then multiply by h = 6 -> 5 × 6 = 30 cm² (faster mental math!).

SC-6 Keystrokes: `10` `ENTER` `6` `g` `TRI`

Answer: 30 cm²

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 5.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Half of a Parallelogram/Rectangle | Method 2: For..

1 Find the area of a triangle with base b = 14 ft and perpendicular height h = 18 ft.

Ans: _____

2 Find the area of a right triangle with base 9 ft, vertical height 4 ft, and slanted hypotenuse 9.85 ft.

Ans: _____

3 Find the area of a triangle with b = 16 cm and h = 11 cm.

Ans: _____

4 Find the area of a triangle with decimal dimensions: b = 12.5 m, h = 8.4 m.

Ans: _____

5 Find the area of a triangle with fractional dimensions: b = 4 1/2 in, h = 6 in.

Ans: _____

6 Missing Height: A triangle has Area = 54 cm² and base b = 12 cm. (Double 54 first, then ÷ 12!)

Ans: _____

7 Missing Base: A triangle has Area = 84 m² and height h = 14 m. Find base b.

Ans: _____

8 Error Analysis: Leo found the area of a b=8, h=5 triangle as 40. What did he calculate instead?

Ans: _____

9 A sailboat has a main sail (b=12 ft, h=25 ft) and jib sail (b=8 ft, h=15 ft). Find total sail area 2 ways!

Ans: _____

10 Modeling Real Life: A triangular gable window has an area of 36 ft² and a base of 9 ft. Find its height h.

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 5.2 — Practice B: Modeling Real Life (Areas of Triangles)

Textbook Methods: Method 1: Half of a Parallelogram/Rectangle | Method 2: Formula A = ½bh (or ½b × h)

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: CHAMPIONSHIP PENNANT HEIGHT (AREA = 60 IN², BASE = 15 IN) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A triangular sports pennant has an area of 60 square inches and a base of 15 inches. Find its height h.

METHOD 1 (+10 pts — 1st Token): Double the Triangle into a Rectangle First
Two identical pennants form a rectangle of area 2 × 60 = 120 in². Divide by base 15: h = 120 ÷ 15 = 8 inches.

COUNT-01 Way 1: Double the 56 in² triangle into a full 112 in² rectangle of base 14 -> height = 112 ÷ 14 = 8 in.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Substitute into A = ½bh & Solve
60 = ½(15)h -> 60 = 7.5h -> h = 60 ÷ 7.5 = 8 inches. Check: 15 [ENTER] 8 [g] [TRI] = 60 in²!

ALGB-01 Way 2: Take half of base 14 first (7 tiles wide), then divide 56 ÷ 7 = 8 rows high!

SC-6 Keystrokes: 60 ENTER 2 × 15 ÷

Answer: h = 8 in

SECTION 5.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Sailboat Mainsail: A right-triangular sail has a base of 12 feet and a height of 25 feet. Find the area of the sail (A = ½bh).

Ans: _____

2 Skateboard Launch Ramp: The triangular side panel of a launch ramp has base 8 feet and height 3.5 feet. Find the area of ONE side panel, and of BOTH side panels.

Ans: _____

3 Hang Glider Wing: A triangular hang glider wing has a wingspan (base) of 30 feet and a center chord (height) of 9.4 feet. Find its wing area.

Ans: _____

4 Missing Base of a Yield Sign: A triangular road sign has an area of 450 square inches and a height of 25 inches. Double the area and divide by 25 to find the base!

Ans: _____

5 Missing Height of a Tent Door: A triangular tent door has an area of 18 square feet and a base of 6 feet. Find the height of the tent door.

Ans: _____

6 Sandwich Diagonal Cut: A square piece of bread is 4.5 inches on each side and is cut diagonally into 2 congruent triangles. What is the area of 1 triangle?

Ans: _____

7 Corner Garden Bed: A right-triangular flower bed in the corner of a yard has perpendicular legs of 9 feet and 12 feet (and a 15-foot front edge). Find its area.

Ans: _____

8 Paper Airplane Wing: You fold a rectangular 8.5 in × 11 in sheet. Two right triangles of base 4 in and height 8 in form the wings. What is their combined area?

Ans: _____

9 Fractional Triangle: Find the area of a triangle with base b = 6 1/2 inches and height h = 4 inches.

Ans: _____

10 Pyramid Glass Panels: A skylight has 4 congruent triangular glass panes, each with base 5 ft and slant height 6.4 ft. Find the total glass area.

Ans: _____

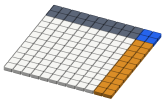
BLOCK KIT: COUNT-01 • Counting & Array Tiles

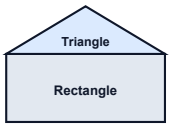
Section 5.3 — Practice A: Areas of Trapezoids and Composite Figures

Key Vocabulary: trapezoid • parallel bases (b1, b2) • height (h) • composite figure • decompose

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)





M1: Split Sum | M2: Box-Cut

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Split the trapezoid on COUNT-01 into an 8 × 9 rectangle (72) plus a 12 × 9 ÷ 2 triangle (54) -> 72 + 54 = 126.

Block Way 2 (+20 pts): Place two identical (8, 20, h=9) trapezoids together (one flipped) to form a 28 × 9 = 252 parallelogram, then halve -> 126!

Both Ways Proof: Splitting into Rectangle + Triangle (72 + 54) equals halving the doubled (b1 + b2) × h parallelogram (252 ÷ 2 = 126)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the total area of a 12 ft × 8 ft rectangular wall topped by a roof triangle (b = 12 ft, h = 5 ft).

METHOD 1 (+10 pts — 1st Token): Decompose into Rectangle + Triangle

Bottom rectangle: 12 × 8 = 96 ft². Top roof triangle: ½ × 12 × 5 = 30 ft². Total area = 96 + 30 = 126 ft².

METHOD 2 (+20 pts DOUBLE — 2nd Token): Accumulate Chunks with [Σ+]

Clear stats [g][CLΣ], compute 12 [ENTER] 8 [f][RECT] [Σ+] (96), then 12 [ENTER] 5 [g][TRI] [Σ+] (30), and read [f][Σx] = 126 ft².

SC-6 Keystrokes: g CLΣ 12 ENTER 8 f RECT Σ+ 12 ENTER 5 g TRI Σ+ f Σx

Answer: 126 ft²

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 5.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Decompose into Rectangles & Triangles | Method 2:..

1 Find the area of a trapezoid with parallel bases b1 = 10 cm and b2 = 14 cm, and height h = 6 cm.

Ans: _____

2 Find the area of a trapezoid with b1 = 18 ft, b2 = 12 ft, and h = 9 ft.

Ans: _____

3 Decompose an L-shaped room into Rectangle A (10 ft × 6 ft) and Rectangle B (8 ft × 4 ft). Find the total floor area.

Ans: _____

4 Banner Arrow: A 15 × 6 rectangle attached to a triangle (b = 10, h = 6). Find total area.

Ans: _____

5 Find the area of a trapezoid with decimal bases b1 = 8.5 m, b2 = 5.5 m, and h = 4.2 m.

Ans: _____

6 A 10 in × 8 in picture frame has a 6 in × 4 in center opening cut out. Find the area of the wooden border.

Ans: _____

7 Missing Trapezoid Height: A trapezoid has Area = 90 cm² and bases b1 = 11 cm, b2 = 7 cm. Find h.

Ans: _____

8 A mini-golf hole is a 12 × 5 ft rectangle plus a trapezoid (b1=5, b2=9, h=4). Find total area.

Ans: _____

9 Find the area of a trapezoid (b1=10, b2=16, h=8) both with the trapezoid formula AND by splitting into a 10×8 rect + 6×8 triangle!

Ans: _____

10 A 20 ft × 15 ft lawn has a triangular flowerbed (b=8 ft, h=6 ft) and square shed (5 ft × 5 ft). Find grass area!

Ans: _____

Section 5.3 — Practice B: Modeling Real Life (Areas of Trapezoids and Composite Figures)

Textbook Methods: Method 1: Decompose into Rectangles & Triangles | Method 2: Trapezoid Formula A = ½h(b1 + b2)

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: TRAPEZOID MAKER DESK (B1 = 48 IN, B2 = 36 IN, H = 24 IN) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A trapezoid classroom desk has parallel bases of 48 inches and 36 inches, and a depth (height) of 24 inches. Find its area.

METHOD 1 (+10 pts — 1st Token): Trapezoid Area Formula A = ½h(b1 + b2)

Add the parallel bases (48 + 36 = 84 in), multiply by ½ of height (½ × 24 = 12 in): 84 × 12 = 1,008 in².

COUNT-01 Way 1: Build the 36 × 20 rectangle (720) and stack the 16 × 36 ÷ 2 trapezoid roof (288) -> 1,008.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Split into 36×24 Rectangle + Triangle Overhangs

Center rectangle = 36 × 24 = 864 in². Extra base width = 48 - 36 = 12 in -> Triangle area = ½ × 12 × 24 = 144 in². Sum: 864 + 144 = 1,008 in²!

COUNT-01 Way 2: Frame a 36 × 36 square (1,296) and remove the two top corner triangles (288) -> 1,008!

SC-6 Keystrokes: 48 ENTER 36 + 24 g TRI

Answer: 1,008 in²

SECTION 5.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Vaulting Box Side: The trapezoid side of a gym vaulting box has top base b1 = 16 in, bottom base b2 = 28 in, and height h = 20 in. Find its area.**

Ans: _____

2 **Popcorn Bucket Panel: Each of the 4 sides of a popcorn tub is a trapezoid with b1 = 4 in, b2 = 6 in, and h = 8 in. Find the area of 1 side and all 4 sides.**

Ans: _____

3 **Mini-Golf Hole Green: An L-shaped putting green splits into a 14 ft × 6 ft rectangle and a 10 ft × 5 ft rectangle. Find the total turf area.**

Ans: _____

4 **Rocket Fin Blueprint: A model rocket fin is a trapezoid with parallel edges of 5 cm and 9 cm, and width (height) of 6 cm. Find the area of 1 fin and 3 fins.**

Ans: _____

5 **Arrow Sign Composite: An arrow sign is made of a 15 in × 6 in rectangle attached to a triangle of base 12 in and height 8 in. Find the total area.**

Ans: _____

6 **Cornhole Board Hole Cutout: A rectangular board is 48 in × 24 in. A square cutout of 6 in × 6 in is removed. Find the remaining area.**

Ans: _____

7 **Missing Height of a Trapezoid: A trapezoid patio has parallel bases of 10 ft and 16 ft, and an area of 104 ft². Since ½(10+16) = 13, divide 104 ÷ 13 to find h!**

Ans: _____

8 **Barn Wall Paint: A barn wall is a 20 ft × 14 ft rectangle topped by a gable roof triangle of base 20 ft and height 6 ft. Find the total area to paint.**

Ans: _____

9 **Swimming Pool Deck Border: A 20 m × 10 m pool is surrounded by a concrete deck whose outer boundary is 26 m × 16 m. Subtract to find the deck area!**

Ans: _____

10 **Hexagon Nut Face: A regular hexagon splits into 2 identical trapezoids, each with b1 = 4 cm, b2 = 8 cm, and h = 3.5 cm. Find the total hexagon area.**

Ans: _____

BLOCK KIT: TIME-01 • Planetary Geared Clock

Section 5.4 — Practice A: Polygons in the Coordinate Plane & Elapsed Time

Key Vocabulary: coordinate grid • vertices • horizontal side • vertical side • elapsed time • base-60 regrouping

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (TIME-01: PLANETARY GEARED CLOCK)





M1: Base-60 | M2: [f][CLK]

DEMONSTRATE BOTH METHODS WITH TIME-01 BLOCKS:

Block Way 1 (+10 pts): Count grid squares between vertex pegs on COUNT-01 (W=7, H=5); rotate TIME-01 gears forward across 12:00.

Block Way 2 (+20 pts): Subtract coordinates (|5 - (-2)| = 7, |4 - (-1)| = 5); regroup 1 hr = 60 min on the TIME-01 60-tooth ring!

Both Ways Proof: Counting physical grid squares / gear teeth matches subtracting coordinates and base-60 time!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: A rectangle has vertices (2, 3), (9, 3), (9, 8), and (2, 8). Find its width, height, perimeter, and area.

METHOD 1 (+10 pts — 1st Token): Subtract x- and y-Coordinates

Width = 9 - 2 = 7 units; Height = 8 - 3 = 5 units. Perimeter = 2(7 + 5) = 24 units; Area = 7 × 5 = 35 units².

METHOD 2 (+20 pts DOUBLE — 2nd Token): Count Grid Squares & Opposite Sides

Count 7 columns across and 5 rows up on the baseplate: 7 + 5 + 7 + 5 = 24 around, and 5 rows of 7 = 35 squares inside.

SC-6 Keystrokes: 9 ENTER 2 - 8 ENTER 3 - f RECT

Answer: Width = 7, Height = 5, Perimeter.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 5.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Grid Counting / Clock Gear Dialing | Method 2: Co..

1 Find the distance between (3, 4) and (11, 4) on a coordinate grid.

Ans: _____

2 A rectangle has corners (1, 2), (10, 2), (10, 8), (1, 8). Find its area.

Ans: _____

3 Find the total elapsed time: 1 hour 45 minutes + 1 hour 35 minutes (regroup 60 minutes into 1 hour).

Ans: _____

4 Convert 2 hours 30 minutes into decimal hours.

Ans: _____

5 Convert 4 hours 45 minutes into decimal hours.

Ans: _____

6 Convert 1.75 decimal hours into hours and minutes.

Ans: _____

7 Find the elapsed time from 8:50 AM to 11:25 AM.

Ans: _____

8 A right triangle on a grid has vertices (2, 1), (10, 1), and (2, 7). Find its area!

Ans: _____

9 Modeling Real Life: You tutor for 3 hours 15 minutes at $16 per hour. Find your total pay in two ways.

Ans: _____

10 Coordinate Park: Vertices (2, 2), (14, 2), (14, 11), (2, 11) where 1 unit = 10 yd. Find perimeter and area!

Ans: _____

BLOCK KIT: TIME-01 • Planetary Geared Clock

Section 5.4 — Practice B: Modeling Real Life (Polygons in the Coordinate Plane & Elapsed Time)

Textbook Methods: Method 1: Grid Counting / Clock Gear Dialing | Method 2: Coordinate Subtraction & [CLK] Arithmetic

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: 3D PRINT FINISH TIME (START 9:40 + 2 HR 50 MIN) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A 3D print starts at 9:40 AM and takes 2 hours 50 minutes. When does it finish? Also write 3 hr 15 min as decimal hours.

METHOD 1 (+10 pts — 1st Token): Base-60 Regrouping (60 min = 1 hr)

Add hours (9 + 2 = 11 hr) and minutes (40 + 50 = 90 min = 1 hr 30 min) -> 12:30 PM. For 3 hr 15 min: 15/60 = 1/4 = 0.25 -> 3.25 hr.

TIME-01 Way 1: Turn the geared hands 2 hr 15 min from 8:40 AM to 11:55 AM, then 35 min more to 12:30 PM.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Bridge to the Next Whole Hour + SC-6 [CLK]

From 9:40, add 20 min to reach 10:00 AM, then add the remaining 2 hr 30 min -> 12:30 PM! Verify with [CLK] 9.40 [ENTER] 2.50 [+].

FRAC-01 Way 2: Model 15 min as a 1/4-hour block (0.25 hr) -> 3 hr 15 min = 3 1/4 = 3.25 hours!

SC-6 Keystrokes: [CLK] [9.40] [ENTER] [2.50] [+] [3.15] [f] [->HR]

Answer: 12:30 PM; and 3.25 hours

SECTION 5.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Coordinate Soccer Field: A field on a map has corners at (1, 2), (11, 2), (11, 8), and (1, 8). Find its width, height, perimeter, and area.

Ans: _____

2 Coordinate Right Triangle: A triangle has vertices at (2, 1), (10, 1), and (2, 7). Find the horizontal base, vertical height, and area.

Ans: _____

3 Flight Duration: A flight departs at 8:35 AM and lands 3 hours 40 minutes later. Regroup 60 minutes into 1 hour to find the landing time.

Ans: _____

4 Science Lab Timer: An experiment starts at 10:45 AM and ends at 2:20 PM (14:20 in 24-hour time). How many hours and minutes did it run?

Ans: _____

5 Marathon Split Sum: A runner completes Half 1 in 1 hr 52 min and Half 2 in 1 hr 58 min. What is her total marathon time?

Ans: _____

6 Lifeguard Shift Pay: You work a shift from 1:15 PM to 4:45 PM (3 hr 30 min) at $14 per hour. Convert 3 hr 30 min to 3.5 hours and find your pay!

Ans: _____

7 Decimal Hours to H:MM: A battery test lasts 2.75 hours. Convert 2.75 hours into hours and minutes (0.75 × 60 min).

Ans: _____

8 3D Print Queue: Print A takes 1 hr 25 min, Print B takes 2 hr 15 min, and Print C takes 50 min. What is the total print time?

Ans: _____

9 Missing Vertex of a Rectangle: Three corners of a rectangle are (3, 4), (9, 4), and (9, 10). What are the coordinates of the 4th corner, and what is the area?

Ans: _____

10 Speed & Elapsed Time: A train travels for 2 hours 15 minutes (2.25 hours) at 80 miles per hour. How many miles does it travel?

Ans: _____

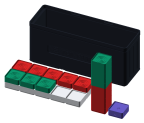
BLOCK KIT: BASE-10 • Ten-Frame Zero-Pair Counters

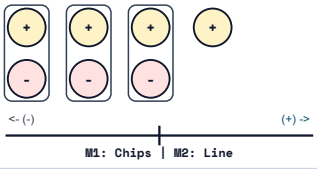
Section 6.1 — Practice A: Positive and Negative Integers & Opposites

Key Vocabulary: integer • positive number • negative number • opposites • additive inverse • zero pair

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (BASE-10: TEN-FRAME ZERO-PAIR COUNTERS)





DEMONSTRATE BOTH METHODS WITH BASE-10 BLOCKS:

Block Way 1 (+10 pts): Place pegs at -15 and +15 on RACK-01 at equal 15-unit distances on opposite sides of 0.

Block Way 2 (+20 pts): Pair 15 Red (-) counters with 15 Yellow (+) counters in RACK-01—all 15 zero pairs cancel to 0!

Both Ways Proof: Opposites sit at mirror positions across 0 (Method 1) and form 100% cancelling Zero Pairs (Method 2)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Enter -15 on the SC-6, find its opposite, and verify that -15 + 15 = 0.

METHOD 1 (+10 pts — 1st Token): Opposite Sides of Zero on a Number Line

-15 is 15 units left of 0; its opposite is 15 units right of 0 (+15). Moving 15 left then 15 right lands back at 0.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Zero Pairs & Sign Flip [+/-]

15 negative counters + 15 positive counters form 15 zero pairs = 0. On SC-6: 15 [+/-] [ENTER] 15 [+] = 0.

SC-6 Keystrokes: `15` `+/-` `ENTER` `15` `+`

Answer: Opposite of -15 is +15; -15 + 15.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 6.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Number Line Distance from 0 | Method 2: Zero-Pair..

1 Write the integer that represents '25 feet below sea level' and state its opposite.

Ans: _____

2 Find the opposite of -42, and then evaluate the opposite of the opposite: -(-(-42)).

Ans: _____

3 Zero-Pair Cancellation: A submarine is at -18 m and rises +18 m. Where is it now?

Ans: _____

4 Combine 6 Yellow (+1) counters and 6 Red (-1) counters. What is the value?

Ans: _____

5 A hiker starts at -120 ft in a canyon and climbs +120 ft. What is her elevation?

Ans: _____

6 Which integer is its own opposite? Explain using a number line.

Ans: _____

7 The temperature starts at 0°F, drops 12°F, and then rises 19°F. Find the final temperature.

Ans: _____

8 Golf Par: -3 means 3 strokes under par. If you shoot -3 on Saturday and +3 on Sunday, what is your total?

Ans: _____

9 Modeling Real Life: An elevator starts at Floor 0, goes up 14 floors, down 20 floors, and up 9 floors. Find the final floor.

Ans: _____

10 Game XP: You gain 40 XP, lose 65 XP, and gain 25 XP. Did you break even? Prove 2 ways!

Ans: _____

BLOCK KIT: BASE-10 • Ten-Frame Zero-Pair Counters

Section 6.1 — Practice B: Modeling Real Life (Positive and Negative Integers & Opposites)

Textbook Methods: Method 1: Number Line Distance from 0 | Method 2: Zero-Pair Cancellation & [+/-] Flip

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: BANK ACCOUNT LEDGER (+$35, -$50, +$20) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: You start with $0, deposit $35 (+35), spend $50 (-50), and deposit $20 (+20). What is your final balance?

METHOD 1 (+10 pts — 1st Token): Group All Positives & All Negatives First
Total deposits = +$35 + $20 = +$55. Subtract the $50 withdrawal: $55 - $50 = +$5 balance.

RACK-01 Way 1: Place 20 Red (-) and 25 Yellow (+) counters; 20 zero pairs cancel, leaving 5 Yellow (+$5).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Step-by-Step Running Balance on a Number Line
Start at 0 -> +35 -> subtract 50 (pass 0 to -15) -> add 20 (-15 + 20 = +5). Final balance = +$5!

RACK-01 Way 2: Start at 0 on the number rail, slide 20 steps Left to -20, then 25 steps Right to +5!

SC-6 Keystrokes: 35 ENTER 50 +/- + 20 +

Answer: +$5

SECTION 6.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS 1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Submarine Dive & Climb: A submarine starts at sea level (0 ft), dives 120 feet (-120), and then rises 120 feet (+120). Where is it now, and what integer pair is this?**

Ans: _____

2 **Death Valley vs. Mt. Whitney: Badwater Basin is 282 feet below sea level. Write this elevation as an integer and state its opposite.**

Ans: _____

3 **Football Yardage Drive: A team gains 12 yards (+12), loses 8 yards (-8), and gains 15 yards (+15). What is their net yardage change?**

Ans: _____

4 **Freezer Temperature Drop: A lab sample starts at 15°C and cools by 25°C. What is the new temperature, and how far below 0°C is it?**

Ans: _____

5 **Zero-Pair Counter Challenge: You place 18 red negative counters (-18) and 25 yellow positive counters (+25) on the desk. How many zero pairs cancel, and what remains?**

Ans: _____

6 **Golf Tournament Score: In golf, par is 0. A player shoots -4 on Friday, +2 on Saturday, and -3 on Sunday. What is her 3-day total score?**

Ans: _____

7 **Opposite of the Opposite: If a scuba diver is at x = -35 feet, what is -x (the opposite of x), and what is -(-x)?**

Ans: _____

8 **Gift Card Balance: You have a $25 gift card, spend $25 on a game, and earn a $10 reward credit. What is your balance?**

Ans: _____

9 **Mountain Trail Hiker: A hiker starts in a canyon at -40 meters, climbs 95 meters, and descends 20 meters. What is her final elevation?**

Ans: _____

10 **Zero-Sum Balancing: Three numbers add to 0. Two of the numbers are -14 and -9. What must the third number be?**

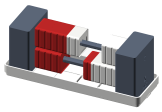
Ans: _____

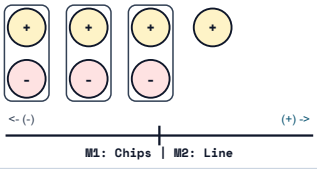
Section 6.2 — Practice A: Comparing and Ordering Integers & Rational Numbers

Key Vocabulary: rational number • inequality (<, >) • number line order (left = less, right = greater)

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (RACK-01: NUMBER RACK & NUMBER LINE)





DEMONSTRATE BOTH METHODS WITH RACK-01 BLOCKS:

Block Way 1 (+10 pts): Place pegs at -5 and -18 on the RACK-01 number rail; -5 sits to the Right of -18, so -5 > -18.

Block Way 2 (+20 pts): Compare Red (-) counter stacks on RACK-01: 5 Red counters (owing 5) is less debt than 18 Red counters!

Both Ways Proof: Further Right on the RACK-01 rail always corresponds to fewer Red negative counters (closer to 0)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Compare -5 and -18. Which one is greater?

METHOD 1 (+10 pts — 1st Token): Plot on a Horizontal Number Line

Plot -5 and -18. Since -5 is to the right of -18 (closer to 0), -5 is greater than -18 (-5 > -18).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Thermometer / Elevation Context & [f][>]

-5°F is warmer than -18°F, and -5 ft is higher than -18 ft -> -5 > -18. Verify: 5 [+/-] [ENTER] 18 [+/-] [f][>] -> 1 (TRUE).

SC-6 Keystrokes: `5` `+/-` `ENTER` `18` `+/-` `f` `>`

Answer: -5 > -18 (SC-6 outputs 1 = TRUE) 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 6.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Horizontal/Vertical Number Line Position | Method..

1 Compare -12°C and -5°C using < or >. Which temperature is warmer?

Ans: _____

2 Compare -9 and -2 using > or <.

Ans: _____

3 Compare -15 and 0 using > or <.

Ans: _____

4 Order the integers from least to greatest: 4, -7, 0, -2, -10.

Ans: _____

5 Compare -3/4 and -0.6. (Hint: -3/4 = -0.75. Which is further right?)

Ans: _____

6 Compare -2 1/2 and -2.3 using > or <.

Ans: _____

7 Compare -5/8 and -7/8.

Ans: _____

8 Error Analysis: Leo says -4.5 > -4.1 because 4.5 > 4.1. Explain why -4.1 is actually greater!

Ans: _____

9 Order the rational numbers -1.5, 1 1/4, -1 3/4, and -0.5 from least to greatest.

Ans: _____

10 Three divers are at -14.8 m, -14 1/2 m, and -15.2 m. Who is deepest (least) and who is shallowest (greatest)?

Ans: _____

BLOCK KIT: RACK-01 • Number Rack & Number Line

Section 6.2 — Practice B: Modeling Real Life (Comparing and Ordering Integers & Rational Numbers)

Textbook Methods: Method 1: Horizontal/Vertical Number Line Position | Method 2: Common Decimal or LCD Comparison

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: DIVING DEPTH COMPARISON (-4/5 M VS. -0.75 M) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Diver A is at -4/5 meter and Diver B is at -0.75 meter. Which diver is at a greater elevation (closer to the surface)?

METHOD 1 (+10 pts — 1st Token): Convert Both to Decimals & Compare on Number Line
-4/5 = -0.80 m. Compare -0.80 and -0.75: since -0.75 is closer to 0 (further right), -0.75 > -4/5 (Diver B is higher).

FRAC-01 Way 1: Compare 3/4 (63mm = 0.75) vs. 4/5 (67.2mm = 0.80) below 0 -> -0.75 sits above -0.80!

METHOD 2 (+20 pts DOUBLE — 2nd Token): Convert Both to Common Denominator Twentieths
-4/5 = -16/20 and -0.75 = -3/4 = -15/20. Because -15 > -16, -15/20 > -16/20, so -0.75 is greater!

FRAC-01 Way 2: Convert to twentieth-blocks: -15/20 is closer to 0 (warmer/higher) than -16/20!

SC-6 Keystrokes: [.] [4] [.] [5] [+/-] [FDISP] [0.75] [+/-] [g] [<]

Answer: -0.75 > -4/5 (Diver B is higher)

SECTION 6.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Arctic Weather Station: Monday was -18°C and Tuesday was -11°C. Write an inequality (< or >) to compare the temperatures. Which day was warmer?**

Ans: _____

2 **Ocean Trench Explorers: Rover Alpha is at -2.4 km and Rover Beta is at -2.15 km. Which rover is deeper, and which elevation is greater?**

Ans: _____

3 **Negative Fraction Showdown: Compare -2/3 and -5/6 by rewriting both with denominator 6. Which rational number is greater?**

Ans: _____

4 **Golf Leaderboard Order: Four players have scores of -6, +2, -9, and -1. Order the scores from least (best in golf!) to greatest.**

Ans: _____

5 **Freezing Point of Liquids: Water freezes at 0°C, olive oil at -6°C, mercury at -39°C, and rubbing alcohol at -89°C. Order from coldest (least) to warmest (greatest).**

Ans: _____

6 **Fraction vs. Decimal Depth: A coral reef is at -3 1/4 meters and a sea turtle is at -3.1 meters. Which one is closer to the surface (greater)?**

Ans: _____

7 **Stock Market Change: Tech stock changed by -1.8% and Retail stock changed by -1.08%. Which stock had the greater (less negative) change?**

Ans: _____

8 **Between Two Negatives: Find an integer and a decimal that lie between -4 and -3 on a number line.**

Ans: _____

9 **Four-Card Rational Sort: Order -7/4, -1.6, 3/2, and -2 from least to greatest by converting the fractions to decimals.**

Ans: _____

10 **True or False Inequality Check: If a < b and both a and b are negative (like -10 < -2), which one is further from 0?**

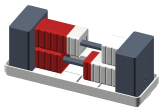
Ans: _____

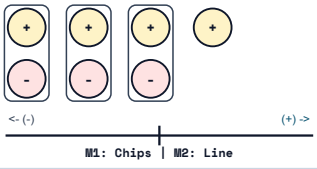
Section 6.3 — Practice A: Absolute Value and Magnitude

Key Vocabulary: absolute value (|x|) • magnitude • distance from zero • opposite vs. absolute value

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (RACK-01: NUMBER RACK & ZERO-PAIR STUDIO)





DEMONSTRATE BOTH METHODS WITH RACK-01 BLOCKS:

Block Way 1 (+10 pts): Count 14 un-signed steps from 0 to -14 and from 0 to +14 on the RACK-01 rail (|-14| = |14| = 14).

Block Way 2 (+20 pts): Flip 14 Red (-) counters to positive Yellow inside the |x| bars (+14), then apply any outside '-' sign last (-14)!

Both Ways Proof: Absolute value bars |x| measure un-signed counter count (distance from 0) before any outside sign!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find |-14|, |14|, and -|-14| on your SC-6.

METHOD 1 (+10 pts — 1st Token): Measure Distance from 0

Both -14 and 14 are 14 units away from 0, so |-14| = 14 and |14| = 14.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Inside Bars First, Outside Sign Second

In -|-14|, evaluate inside the bars first (|-14| = 14), then attach the outside negative sign -> -14.

SC-6 Keystrokes: `14` `+/-` `g` `|x|`

Answer: |-14| = 14, |14| = 14, -|-14| = -14 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 6.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Count Steps from Zero on Number Line | Method 2:..

1 Find the absolute value |-27|. Explain what it represents on a number line.

Ans: _____

2 Evaluate |-27| + |13|.

Ans: _____

3 Evaluate |-4.8| + |-3.2|.

Ans: _____

4 Evaluate |-35| - |-19|.

Ans: _____

5 Compare |-12| and |9| using > or <. (Which is further from 0?)

Ans: _____

6 Evaluate -|-24| by finding the absolute value inside the bars first.

Ans: _____

7 Evaluate 50 - |-18|. Find |-18| first, then subtract from 50.

Ans: _____

8 Order from least to greatest: |-8|, -6, |-3|, 0, -|-9|.

Ans: _____

9 A diver is at -32 ft and a seagull is at +25 ft. Which is closer to sea level (0), and by how many feet?

Ans: _____

10 Two accounts have balances of -$45 and -$28. Which account has the greater debt (|-45| vs |-28|), and which has the greater balance?

Ans: _____

Section 6.3 — Practice B: Modeling Real Life (Absolute Value and Magnitude)

Textbook Methods: Method 1: Count Steps from Zero on Number Line | Method 2: Evaluate |x| Bars First (Like Grouping Sy..

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: SUBMARINE DEPTH EXPRESSION (EVALUATE 50 - |-18|) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Evaluate 50 - |-18| and explain why it does NOT equal 68.

METHOD 1 (+10 pts — 1st Token): Treat Absolute Value Bars as Grouping Symbols

Step 1: Find |-18| = 18 first (absolute value bars are NOT parentheses!). Step 2: Subtract 50 - 18 = 32.

RACK-01 Way 1: Count 18 Red counters for the -18 drop (|-18| = 18), then subtract 18 from 50 -> 32.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Physical Distance Comparison

A rope is 50 ft long; a diver is 18 ft below sea level (depth = |-18| = 18 ft). Rope left = 50 - 18 = 32 ft!

RACK-01 Way 2: Start at +50 on the number rail and walk 18 steps toward 0 -> lands on 32!

SC-6 Keystrokes: `50` `ENTER` `18` `+/-` `g` `|x|` `-`

Answer: 32 (since |-18| = 18, and 50 - 18 = 32)

SECTION 6.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Scuba Depth vs. Seagull Height: A diver is at -28 feet and a seagull is at +28 feet. Find |-28| and |28|. Who is further from sea level?**

Ans: _____

2 **Credit Card Debt Magnitude: Account A has a balance of -$65 and Account B has -$40. Which account has the greater balance, and which has the greater debt (|-x|)?**

Ans: _____

3 **Absolute Value Sum: A drone flies 18 meters west (-18) and then 14 meters east (+14). Evaluate |-18| + |14| to find the total distance flown.**

Ans: _____

4 **Outside Negative Sign Trap: Evaluate both |-35| and -|-35|. Explain why the second answer is negative.**

Ans: _____

5 **Target Precision Error: Three darts land at -2.4 cm, +1.8 cm, and -1.5 cm from the bullseye (0). Use absolute value to find which dart is closest to the center!**

Ans: _____

6 **Evaluate the Expression: Compute | -42 | ÷ | -6 | + | -11 |.**

Ans: _____

7 **Submarine Cable Length: A ship is at +25 feet above the water and lowers a camera to -140 feet. Evaluate |25| + |-140| to find the cable length.**

Ans: _____

8 **Two Solutions Puzzle: Find BOTH numbers on the number line whose absolute value is 19 (|x| = 19).**

Ans: _____

9 **Order with Absolute Value Bars: Order |-8|, -6, |5|, and -|-9| from least to greatest. (Tip: evaluate each absolute value first!)**

Ans: _____

10 **Fractional Absolute Value: Evaluate |-3 1/2| + |2 1/4| - |-1 1/8|.**

Ans: _____

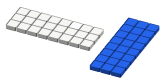
BLOCK KIT: COUNT-01 • 4x4 Gridfinity Baseplate & Tiles

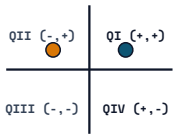
Section 6.4 — Practice A: The Four-Quadrant Coordinate Plane & Reflections

Key Vocabulary: coordinate plane • origin (0, 0) • x-axis • y-axis • quadrants (I, II, III, IV) • ordered pair (x, y) • reflection

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: 4X4 GRIDFINITY BASEPLATE & TILES)





DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): On the 4-Quadrant COUNT-01 grid, walk horizontally from (0,0) along x first, then vertically along y.

Block Way 2 (+20 pts): Fold/mirror across the x-axis by flipping the vertical y-coordinate, or across the y-axis by flipping x!

Both Ways Proof: Quadrant signs (+,-) and mirror reflections both follow horizontal-x and vertical-y grid steps!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Identify the quadrant for A(-6, 5), B(4, -7), and C(-3, -8).

METHOD 1 (+10 pts — 1st Token): Walk Horizontal (x) Then Vertical (y)

A(-6, 5): left 6, up 5 -> Quad II. B(4, -7): right 4, down 7 -> Quad IV. C(-3, -8): left 3, down 8 -> Quad III.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Use Quadrant Sign Chart

Quad I = (+, +), Quad II = (-, +), Quad III = (-, -), Quad IV = (+, -). Match each ordered pair's signs!

SC-6 Keystrokes: `6` `+/-` `ENTER` `5`

Answer: A: Quad II, B: Quad IV, C: Quad III 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 6.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Plot & Fold Across the Axis | Method 2: Coordinat..

1 Name the Quadrant (I, II, III, or IV) for point (7, -4).

Ans: _____

2 Name the Quadrant for point (-9, -2).

Ans: _____

3 Name the Quadrant for point (-5, 11).

Ans: _____

4 Reflect the point (6, 9) across the y-axis. Write the coordinates of the reflected point.

Ans: _____

5 Reflect the point (6, 9) across the x-axis. Write the coordinates of the reflected point.

Ans: _____

6 Reflect (-3, -7) across the x-axis.

Ans: _____

7 Reflect (-8, 4) across the y-axis, and then across the x-axis.

Ans: _____

8 On the Fence: Which axis does (0, -9) sit on, and which axis does (-7, 0) sit on?

Ans: _____

9 Start at (4, -5). Walk 7 units left and 8 units up. Where do you land, and in which quadrant?

Ans: _____

10 Points A(-4, 2), B(4, 2), and C(4, -5) are 3 corners of a rectangle. Find the 4th corner D!

Ans: _____

BLOCK KIT: COUNT-01 • 4x4 Gridfinity Baseplate & Tiles

Section 6.4 — Practice B: Modeling Real Life (The Four-Quadrant Coordinate Plane & Reflections)

Textbook Methods: Method 1: Plot & Fold Across the Axis | Method 2: Coordinate Sign Rules (Flip y for x-axis, Flip x for y-a..

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: DOUBLE MIRROR REFLECTION FROM P(-5, 8) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Start at P(-5, 8) in Quadrant II. Reflect across the x-axis, then reflect that new point across the y-axis. Where do you land?

METHOD 1 (+10 pts — 1st Token): Apply Axis Reflection Sign Rules

1. Reflect (-5, 8) across x-axis (keep x, flip y): (-5, -8). 2. Reflect (-5, -8) across y-axis (flip x, keep y): (5, -8) in Quad IV.

COUNT-01 Way 1: Place a peg at (-5, -8) in QIII, then mirror 5 units across the y-axis to (+5, -8) in QIV.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Count Equal Distances Across Each Axis

8 units above x-axis flips to 8 units below (-5, -8); 5 units left of y-axis flips to 5 units right -> (5, -8)!

COUNT-01 Way 2: Keep y = -8 fixed and flip x from -5 to +5 -> (5, -8), which is 5+5 = 10 units away!

SC-6 Keystrokes: `5` `+/-` `ENTER` `8` `+/-` `x⇔y` `+/-` `x⇔y`

Answer: (5, -8) in Quadrant IV

SECTION 6.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Treasure Map Quadrants: Tell which quadrant (I, II, III, or IV) contains each landmark: Cave (-8, 4), Lighthouse (7, 9), Shipwreck (-5, -6), and Pier (3, -10).

Ans: _____

2 On the Axis Check: A buoy is at (0, -7) and a dock is at (9, 0). Neither is in a quadrant! Tell which axis (x-axis or y-axis) each point lies on.

Ans: _____

3 Lake Reflection (x-Axis): A mountain peak is plotted at (8, 12). Its reflection in the lake flips across the x-axis. What are the coordinates of the reflection?

Ans: _____

4 Symmetry Wing Design (y-Axis): The left wingtip of a butterfly is at (-9, 5) in Quadrant II. Reflect it across the y-axis to place the right wingtip!

Ans: _____

5 Double Jump Reflection: Start at (-4, -9) in Quadrant III. Reflect across the x-axis first, then across the y-axis. Where do you land?

Ans: _____

6 Robot Navigation Directions: Starting from the origin (0, 0), a robot moves 6 units left and 8 units down, then 10 units right. What are its new coordinates and quadrant?

Ans: _____

7 Decimal Ordered Pairs: Plot mentally and identify the quadrant for P(-2.5, 4.5) and Q(-1.5, -3.5).

Ans: _____

8 Mystery Axis Reflection: Point M(7, -3) was reflected to M'(-7, -3). Which axis did it reflect across? How can you tell from the coordinates?

Ans: _____

9 Square Symmetry Around Origin: One corner of a square centered at (0, 0) is at (5, 5). Reflect across both axes to write the other 3 corners!

Ans: _____

10 Quadrant Sign Puzzle: If x is negative and y is positive, what quadrant is (x, y) in? What quadrant is (-x, -y) in?

Ans: _____

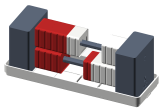
BLOCK KIT: COUNT-01 • Baseplate & Zero-Pair Studio

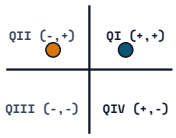
Section 6.5 — Practice A: Distances in the Coordinate Plane

Key Vocabulary: horizontal distance • vertical distance • same quadrant (subtract |x|) • different quadrants (add |x|)

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: BASEPLATE & ZERO-PAIR STUDIO)





DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): On COUNT-01, count 6 squares from (-6, 4) to the y-axis plus 9 squares from the y-axis to (9, 4) -> 6 + 9 = 15.

Block Way 2 (+20 pts): On RACK-01, subtract the coordinates and take absolute value: |9 - (-6)| = |9 + 6| = 15 units!

Both Ways Proof: Adding distances to the axis (|-6| + |9|) when crossing 0 equals subtracting coordinates (|9 - (-6)| = 15)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the distance between (-6, 4) and (9, 4), and between (-12, -3) and (-4, -3).

METHOD 1 (+10 pts — 1st Token): Add Across Axis / Subtract in Same Quadrant

(-6, 4) to (9, 4) crosses y-axis -> ADD |-6| + |9| = 15. (-12, -3) to (-4, -3) is in same quadrant -> SUBTRACT |-12| - |-4| = 8.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Absolute Difference |x1 - x2|

Subtract x-coordinates and take absolute value: |-6 - 9| = |-15| = 15, and |-12 - (-4)| = |-8| = 8.

SC-6 Keystrokes: `6` `+/-` `ENTER` `9` `-` `g` `|x|`

Answer: 15 units; and 8 units

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 6.5 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Add/Subtract Absolute Values to Axis | Method 2:..

1 Find the horizontal distance between (-8, 5) and (11, 5).

Ans: _____

2 Find the vertical distance between (-3, -4) and (-3, -15).

Ans: _____

3 Find the vertical distance between (7, 9) and (7, -6).

Ans: _____

4 Find the horizontal distance between (-14, 2) and (-5, 2).

Ans: _____

5 Find the distance between (-3.5, 4) and (6.5, 4).

Ans: _____

6 Find the distance between (2, -4 1/2) and (2, 3 1/2).

Ans: _____

7 A line segment starts at (-6, 3) and goes 10 units right. What are the coordinates of the right endpoint?

Ans: _____

8 Error Analysis: For (-5, 2) to (8, 2), Sam subtracted 8 - 5 = 3. Why is the distance 13, not 3?

Ans: _____

9 A treasure map rectangle has corners (-4, 6), (5, 6), (5, -2), (-4, -2). Find its perimeter & area 2 ways!

Ans: _____

10 On a city grid (1 unit = 0.5 mile), you walk from (-6, -2) to (-6, 8) to (4, 8). How many miles did you walk?

Ans: _____

BLOCK KIT: COUNT-01 • Baseplate & Zero-Pair Studio

Section 6.5 — Practice B: Modeling Real Life (Distances in the Coordinate Plane)

Textbook Methods: Method 1: Add/Subtract Absolute Values to Axis | Method 2: Universal Difference Formula |x1 - x2|

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: TOWN PARK BLUEPRINT ((-4, 6), (5, 6), (5, -2), (-4, -2)) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A rectangular park has vertices (-4, 6), (5, 6), (5, -2), and (-4, -2). Find its width, height, perimeter, and area.

METHOD 1 (+10 pts — 1st Token): Add Distances Across the x- and y-Axes

Width (crosses y-axis): |-4| + |5| = 4 + 5 = 9. Height (crosses x-axis): |6| + |-2| = 6 + 2 = 8. P = 2(9+8) = 34; A = 9×8 = 72.

COUNT-01 Way 1: Frame the rectangle on the 4-quadrant grid: width = 4 + 5 = 9 squares, height = 3 + 5 = 8 squares.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Coordinate Subtraction + [f] [RECT]

Width = |5 - (-4)| = 9; Height = |6 - (-2)| = 8. Area = 9 × 8 = 72 units²; Perimeter = 9 + 8 + 9 + 8 = 34 units.

COUNT-01 Way 2: Walk the 4 edges (9 + 8 + 9 + 8 = 34 perimeter) and fill the 9 × 8 interior (72 area squares)!

SC-6 Keystrokes: 4 +/- ENTER 5 - g |x| 6 ENTER 2 +/- - f RECT

Answer: W = 9, H = 8, Perimeter = 34, Area = 72

SECTION 6.5 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Bridge Across the River (y-Axis): West Tower is at (-8, 5) and East Tower is at (11, 5). Add their distances to the y-axis to find the bridge length!**

Ans: _____

2 **Same-Quadrant Sprint: On a campus map (1 unit = 10 meters), the gym is at (-14, 6) and the library is at (-5, 6). How many units (and meters) apart are they?**

Ans: _____

3 **Vertical Elevator Shaft: A mine shaft runs from a tower at (4, 15) straight down to a tunnel at (4, -35). What is the total vertical distance?**

Ans: _____

4 **Coordinate Dog Park Perimeter: A rectangular dog park has corners at (-3, 4), (7, 4), (7, -2), and (-3, -2). Find its width, height, and perimeter.**

Ans: _____

5 **Coordinate Dog Park Area: Find the area of the dog park from Problem 4 (Width = 10, Height = 6). Notice it spans across all 4 quadrants!**

Ans: _____

6 **Subway Line Stops: Stop A is at (-3.5, 2), Stop B is at (0, 2), and Stop C is at (4.25, 2). Find the total distance from Stop A to Stop C.**

Ans: _____

7 **Fractional Coordinate Distance: Find the vertical distance between (2, 3 1/2) and (2, -2 1/4).**

Ans: _____

8 **L-Shaped Drone Delivery: A drone flies from (-6, -2) to (-6, 5), and then turns east to (4, 5). What is the total two-segment flight distance?**

Ans: _____

9 **Two Possible Points: Point A is at (3, 4). Point B has the same y-coordinate (4) and is 8 units away horizontally. Find BOTH possible locations for Point B!**

Ans: _____

10 **Coordinate Triangle Area: A triangle has vertices at (-4, -1), (6, -1), and (-4, 7). Find its horizontal base, vertical height, and area.**

Ans: _____

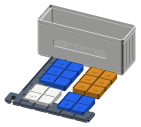
BLOCK KIT: ALGB-01 • Algebra Boxes & Stage

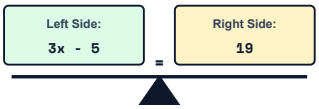
Section 7.1 — Practice A: Writing Equations in One Variable & Testing Solutions

Key Vocabulary: equation • solution of an equation • defining a variable • balance scale • turnaround words ('less than')

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (ALGB-01: ALGEBRA BOXES & STAGE)





M1/M2: x = 8

DEMONSTRATE BOTH METHODS WITH ALGB-01 BLOCKS:

Block Way 1 (+10 pts): On the ALGB-01 Balance Stage, load 9 cubes into each of the 4 x-boxes + 5 cubes -> 36 + 5 = 41 (Balances!).

Block Way 2 (+20 pts): Remove 5 cubes from both pans of ALGB-01 (4x = 36), then share 36 cubes equally into 4 x-boxes -> x = 9!

Both Ways Proof: Testing x = 9 by filling the boxes (41 = 41) and unwrapping the pans in reverse both prove x = 9!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Test whether x = 7 or x = 9 is the solution to 4x + 5 = 41.

METHOD 1 (+10 pts — 1st Token): Substitute Candidates to Test Equality

Test x = 7: 4(7) + 5 = 33 ≠ 41 (False). Test x = 9: 4(9) + 5 = 36 + 5 = 41 = 41 (True! x = 9 is the solution).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Balance Scale Unwrapping

Remove 5 cubes from both pans (41 - 5 = 36), then split 36 across 4 x-rods: 36 ÷ 4 = 9 -> x = 9.

SC-6 Keystrokes: `41` `ENTER` `4` `ENTER` `9` `×` `5` `+`

Answer: x = 9 is the solution (41 == 41) 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 7.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Substitute & Test Both Sides | Method 2: Mental /..

1 Test whether x = 6 or x = 8 solves 5x - 7 = 33.

Ans: _____

2 Test whether y = 12 or y = 15 solves y / 3 + 4 = 9.

Ans: _____

3 Translate into an equation: 'The sum of a number x and 14 is 52.'

Ans: _____

4 Translate into an equation: '6 times a number k equals 84.'

Ans: _____

5 Translate: '9 less than a number w is 17.' (Watch the order!)

Ans: _____

6 Translate: 'The quotient of a number m and 5 is 13.'

Ans: _____

7 4 pizza pies (p) plus a $6 delivery fee total $54. Write the equation and test p = 12!

Ans: _____

8 Error Analysis: For '12 less than x is 20', Sam wrote 12 - x = 20. Fix the equation and solve!

Ans: _____

9 A streaming pass costs $10 plus $3 per movie (10 + 3m = 28). Find m by testing AND by undoing!

Ans: _____

10 Three friends split a restaurant bill b equally and each pays $14.50. Write the equation and find b!

Ans: _____

BLOCK KIT: ALGB-01 • Algebra Boxes & Stage

Section 7.1 — Practice B: Modeling Real Life (Writing Equations in One Variable & Testing Solutions)

Textbook Methods: Method 1: Substitute & Test Both Sides | Method 2: Mental / Balance Scale Unwrapping

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: TURNAROUND PHRASE ('8 LESS THAN A NUMBER N IS 25') (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Write the word sentence '8 less than a number n is 25' as an equation, solve for n, and check your answer.

METHOD 1 (+10 pts — 1st Token): Translate Turnaround Phrase & Solve

'8 less than n' means start with n and subtract 8: n - 8 = 25. Add 8 to both sides: n = 25 + 8 = 33.

ALGB-01 Way 1: Place an n-box with 8 removed cubes balanced against 25 cubes; add 8 cubes to both pans -> n = 33.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Substitute n = 33 Back into the Word Sentence

Check: What is '8 less than 33'? 33 - 8 = 25! (Notice why 8 - n = 25 would be wrong.)

COUNT-01 Way 2: Join the 25 remaining drones and 8eliminated drones into one whole bar -> 25 + 8 = 33!

SC-6 Keystrokes: 25 ENTER 8 + 8 -

Answer: n - 8 = 25; n = 33

SECTION 7.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Escape Room Key Test: Which value of x from {6, 8, 10} is the solution of the equation 5x - 7 = 33? Substitute to test!

Ans: _____

2 Concert Ticket Bundle: Each ticket costs $15 plus a $5 order fee, for a total of $65. Write an equation in terms of t tickets and test t = 4.

Ans: _____

3 Turnaround Translation: Write the word sentence '12 less than a number k equals 29' as an equation, and find k.

Ans: _____

4 Quotient Translation: Write 'the quotient of a number n and 6 is 14' as an equation, and find n.

Ans: _____

5 Perimeter Equation Setup: An equilateral triangle has 3 equal sides of length s and a perimeter of 54 cm. Write an equation and find s.

Ans: _____

6 Battery Drain Test: Does m = 12 or m = 15 satisfy the equation 100 - 4m = 40? Substitute both values to check.

Ans: _____

7 Pizza Sharing Equation: Splitting p slices among 8 players gives each player 3 slices: p ÷ 8 = 3. Test p = 24.

Ans: _____

8 Savings Goal Equation: You have $45 saved and add d dollars to reach $120. Write an addition equation and find d.

Ans: _____

9 Two-Side Balance Check: Test whether x = 6 is a solution of 3x + 10 = 5x - 2 by evaluating both sides for x = 6.

Ans: _____

10 Age Puzzle Equation: Maya is 13 years old, which is 4 years younger than her brother Leo (b - 4 = 13). How old is Leo?

Ans: _____

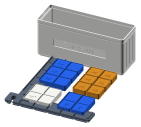
BLOCK KIT: ALGB-01 • Algebra Boxes & Stage

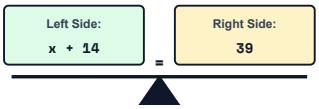
Section 7.2 — Practice A: Solving Equations Using Addition or Subtraction

Key Vocabulary: inverse operations • Addition Property of Equality • Subtraction Property of Equality • isolate the variable

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (ALGB-01: ALGEBRA BOXES & STAGE)





M1/M2: Remove 14 -> x = 25

DEMONSTRATE BOTH METHODS WITH ALGB-01 BLOCKS:

Block Way 1 (+10 pts): Remove 28 unit cubes from both pans of x + 28 = 73 on ALGB-01 so the x-box stands alone (x = 45).

Block Way 2 (+20 pts): Snap the x = 45 bar and 28 bar together on COUNT-01 to verify they match the 73 total bar (45 + 28 = 73)!

Both Ways Proof: Removing 28 from both balance pans (Inverse Op) is the exact reverse of joining 45 + 28 in a bar model!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Solve x + 28 = 73 using the Subtraction Property of Equality and check your solution.

METHOD 1 (+10 pts — 1st Token): Subtraction Property of Equality

Undo '+ 28' by subtracting 28 from both sides of x + 28 = 73: x = 73 - 28 = 45.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Substitution Check ([f] [LASTX] [+])

Substitute x = 45 into original equation: 45 + 28 = 73 (True!). On SC-6: 73 [ENTER] 28 [-] [f][LASTX] [+] -> 73.

SC-6 Keystrokes: 73 | ENTER | 28 | - | f | LASTX | +

Answer: x = 45 (Proven: 45 + 28 = 73)

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 7.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Inverse Operation (Properties of Equality) | Meth..

1 Solve the equation x + 28 = 73 using the Subtraction Property of Equality. Check your solution.

Ans: _____

2 Solve the equation y - 34 = 58 using the Addition Property of Equality. Check your solution.

Ans: _____

3 Solve 19 + n = 64.

Ans: _____

4 Solve k - 47 = 29.

Ans: _____

5 Solve w - 19.6 = 34.8.

Ans: _____

6 Solve p + 8.75 = 25.4.

Ans: _____

7 Solve the fraction equation m + 3/8 = 5/6. Write the solution in simplest form and check by adding.

Ans: _____

8 Solve a - 1 1/4 = 3 1/2.

Ans: _____

9 After spending $18.50 on a controller skin, you have $26.75 left (m - 18.50 = 26.75). Find starting money m!

Ans: _____

10 Modeling Real Life: A drone climbs 42.5 feet to reach an altitude of 110 feet (h + 42.5 = 110). Find its starting height h.

Ans: _____

BLOCK KIT: ALGB-01 • Algebra Boxes & Stage

Section 7.2 — Practice B: Modeling Real Life (Solving Equations Using Addition or Subtraction)

Textbook Methods: Method 1: Inverse Operation (Properties of Equality) | Method 2: Part-Part-Whole Bar Model & [LASTX]..

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: DECIMAL & FRACTION EQUATIONS (W - 19.6 = 34.8 & M + 3/8 = 5/6) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Solve w - 19.6 = 34.8 and m + 3/8 = 5/6 using inverse operations, and check each solution.

METHOD 1 (+10 pts — 1st Token): Apply Addition & Subtraction Properties of Equality

1. Add 19.6 to both sides: w = 34.8 + 19.6 = 54.4. 2. Subtract 3/8 from both sides using LCD = 24: m = 20/24 - 9/24 = 11/24.

PLACE-01 Way 1: Add 12.6 to both sides of w - 12.6 = 41.8 -> 41.8 + 12.6 = 54.4.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Check Both by Reversing the Operation

Check 1: 54.4 - 19.6 = 34.8. Check 2: 11/24 + 9/24 = 20/24 = 5/6 (Both verified!).

FRAC-01 Way 2: Subtract 3/8 (9/24) from 5/6 (20/24) using 1/24 towers -> 11/24!

SC-6 Keystrokes: `34.8` `ENTER` `19.6` `+` `.` `5` `.` `6` `ENTER` `.` `3` `.` `8` `-`

Answer: w = 54.4; m = 11/24

SECTION 7.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Video Game High Score: After scoring 45 bonus points, your total score is 210 (s + 45 = 210). Solve using the Subtraction Property of Equality and check.

Ans: _____

2 Skateboard Discount Equation: After a $18 coupon, a skateboard costs $67 (p - 18 = 67). Solve using the Addition Property of Equality to find the regular price p.

Ans: _____

3 Mountain Summit Climb: From Base Camp (h), a climber ascends 1,450 feet to reach a 5,200-foot peak (h + 1450 = 5200). Find the Base Camp elevation h.

Ans: _____

4 3D Filament Remaining: After a print uses 64.8 grams of PLA, 185.2 grams remain on the spool (f - 64.8 = 185.2). How much PLA was on the spool before the print?

Ans: _____

5 Stopwatch Improvement: Leo shaved 0.48 seconds off his 100m dash time t to run 12.65 seconds (t - 0.48 = 12.65). What was his previous time t?

Ans: _____

6 Fraction Cup Measuring: You have 3/8 cup of milk and need 7/8 cup in all (m + 3/8 = 7/8). Solve and simplify.

Ans: _____

7 Trail Hike Remaining: A trail is 4 1/2 miles long. You have hiked 2 3/4 miles (2 3/4 + d = 4 1/2). How many miles d remain?

Ans: _____

8 Triangle Angle Sum: The three angles of any triangle add to 180°. Two angles are 65° and 48° (65 + 48 + x = 180). Find the third angle x!

Ans: _____

9 Submarine Depth Change: A sub starts at -40 meters and dives d meters to reach -115 meters (-40 + d = -115). Solve for d.

Ans: _____

10 Two-Step Perimeter Reduction: A rectangle has perimeter 64 cm. Two widths total 20 cm (20 + L = 64, where L is the sum of the 2 lengths). Find L and 1 length!

Ans: _____

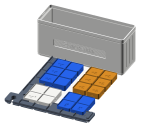
BLOCK KIT: ALGB-01 • Algebra & Fraction Studio

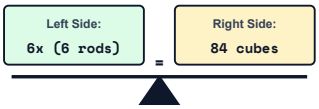
Section 7.3 — Practice A: Solving Equations Using Multiplication or Division

Key Vocabulary: Multiplication Property of Equality • Division Property of Equality • reciprocal coefficient

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (ALGB-01: ALGEBRA & FRACTION STUDIO)





M1/M2: 84 ÷ 6 -> x = 14

DEMONSTRATE BOTH METHODS WITH ALGB-01 BLOCKS:

Block Way 1 (+10 pts): Deal 136 cubes equally into the 8 x-boxes on ALGB-01 (136 ÷ 8 = 17 cubes in each x-box).

Block Way 2 (+20 pts): Combine all 8 boxes of 17 cubes (8 × 17 = 136); for k ÷ 7 = 14.5, join 7 boxes of 14.5 -> 101.5!

Both Ways Proof: Sharing 136 into 8 boxes (Division) and combining 8 boxes of 17 (Multiplication Check) are inverse partners!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Solve 8x = 136 and k / 7 = 14.5, and prove each with [f] [LASTX].

METHOD 1 (+10 pts — 1st Token): Division & Multiplication Properties

For 8x = 136, divide both sides by 8: x = 136 ÷ 8 = 17. For k/7 = 14.5, multiply both sides by 7: k = 14.5 × 7 = 101.5.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Inverse Substitution Check

Check x = 17: 8 × 17 = 136. Check k = 101.5: 101.5 ÷ 7 = 14.5.

SC-6 Keystrokes: 136 ENTER 8 ÷ 14.5 ENTER 7 ×

Answer: x = 17; k = 101.5

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 7.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Divide by the Coefficient | Method 2: Multiply by..

1 Solve 8x = 136 using the Division Property of Equality. Check your solution by multiplying.

Ans: _____

2 Solve 12y = 180 using the Division Property of Equality. Check your solution.

Ans: _____

3 Solve m ÷ 6 = 19 using the Multiplication Property of Equality. Check your solution.

Ans: _____

4 Solve k / 7 = 14.5.

Ans: _____

5 Solve 4.5w = 63.

Ans: _____

6 Solve 3/4 x = 18.

Ans: _____

7 Solve (2/5)n = 16 in two ways: by multiplying by the reciprocal (5/2) and by using a 5-box tape diagram.

Ans: _____

8 Solve 5/6 p = 2/3.

Ans: _____

9 A 3D printer prints at 1.5 cm/min. How long (t) to print a 27 cm tower (1.5t = 27)? Solve & prove!

Ans: _____

10 3/5 of the students in a club (3/5 s = 27) are 6th graders. How many total students s are in the club?

Ans: _____

Section 7.3 — Practice B: Modeling Real Life (Solving Equations Using Multiplication or Division)

Textbook Methods: Method 1: Divide by the Coefficient | Method 2: Multiply by the Reciprocal & Check

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: FRACTIONAL COEFFICIENT EQUATION ((2/5)N = 16) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Two-fifths of the songs on a playlist (2/5 n) are rock songs. There are 16 rock songs. Find the total songs n two ways.

METHOD 1 (+10 pts — 1st Token): Multiply Both Sides by the Reciprocal (5/2)
Multiply both sides of (2/5)n = 16 by 5/2: n = 16 × (5/2) = (16 ÷ 2) × 5 = 8 × 5 = 40 songs.

PLACE-01 Way 1: Divide $50.00 into $1.25 song stacks (5,000 ÷ 125 = 40 songs).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Tape Diagram (Find 1/5 First, Then 5/5)
If 2 fifth-boxes equal 16 songs, then 1 fifth-box = 16 ÷ 2 = 8 songs. All 5 fifth-boxes = 5 × 8 = 40 songs!

FRAC-01 Way 2: Since $1.25 = 5/4 dollar, multiply 50 by the reciprocal 4/5 -> 200 ÷ 5 = 40 songs!

SC-6 Keystrokes: 16 ENTER . 2 . 5 f 1/x ×

Answer: n = 40 songs

SECTION 7.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Robotics Club T-Shirts: A team paid $168 for 12 identical team shirts (12s = 168). Solve using the Division Property of Equality to find the price s per shirt.

Ans: _____

2 Pizza Slice Split: A giant party pizza is cut into equal slices for 9 friends, and each friend gets 3 slices (p ÷ 9 = 3). How many slices p were there in all?

Ans: _____

3 Distance-Rate-Time Equation: A train travels at 65 miles per hour for t hours to cover 325 miles (65t = 325). Solve for t.

Ans: _____

4 Bulk Almond Price: You buy 2.5 pounds of roasted almonds for $16.25 (2.5p = 16.25). Find the price p per pound.

Ans: _____

5 Comic Book Allowance: You spend 1/4 of your savings s on a $9 comic book (s ÷ 4 = 9). Solve using the Multiplication Property of Equality.

Ans: _____

6 Fractional Coefficient Reciprocal: Three-fourths of the students in a class (3/4 c = 21) brought a water bottle. Multiply by 4/3 to find the total students c!

Ans: _____

7 Discounted Sale Price Equation: A bike is on sale for 5/6 of its original price p, and the sale price is $150 ((5/6)p = 150). Find the original price p.

Ans: _____

8 Rectangular Banner Width: A banner has area 108 ft² and length 13.5 ft (13.5w = 108). Solve for the width w.

Ans: _____

9 Scale Model Rocket: A model rocket is 1/15 the height of the real rocket (h ÷ 15 = 3.2 ft). Find the height h of the real rocket.

Ans: _____

10 Percent as an Equation: 40% of a mystery number n is 56 (0.40n = 56). Solve for n both using decimals (÷ 0.40) and fractions (× 5/2)!

Ans: _____

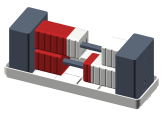
BLOCK KIT: RACK-01 • Number Rack & Number Line

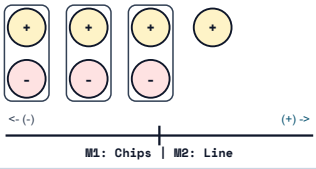
Section 7.4 — Practice A: Writing and Graphing Inequalities

Key Vocabulary: inequality • solution set • graph of an inequality • open circle (○ for <, >) • closed/solid circle (● for ≤, ≥)

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (RACK-01: NUMBER RACK & NUMBER LINE)





DEMONSTRATE BOTH METHODS WITH RACK-01 BLOCKS:

Block Way 1 (+10 pts): Add 6 cubes to both pans of x - 6 ≥ 13 on ALGB-01 to find the boundary peg x = 19 (solid peg on RACK-01).

Block Way 2 (+20 pts): Test a peg to the Right on RACK-01 (x = 20: 14 ≥ 13 TRUE) vs. Left (x = 18: 12 ≥ 13 FALSE) -> shade Right!

Both Ways Proof: Solving on the ALGB-01 stage finds the exact boundary peg (19), and testing pegs on RACK-01 proves the arrow direction!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Solve x - 7 ≥ 12, state whether the circle at 19 is Open or Solid, and test x = 20.

METHOD 1 (+10 pts — 1st Token): Addition Property of Inequality & Graph

Add 7 to both sides of x - 7 ≥ 12 -> x ≥ 19. Graph with a SOLID dot (●) at 19 and shade right.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Test a Shaded Point (x = 20)

Pick x = 20 from the shaded arrow: 20 - 7 = 13, and 13 ≥ 12 is TRUE (proving the arrow points in the right direction!).

SC-6 Keystrokes: `12` `ENTER` `7` `+` `20` `x⇄y` `f` `≥`

Answer: x ≥ 19 (Solid dot at 19, arrow Ri.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 7.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Inverse Operation + Boundary Circle Rule | Method..

1 Solve x + 8 > 21. Tell whether the circle at 13 is Open or Solid, and arrow direction.

Ans: _____

2 Solve y - 9 ≤ 15. Tell whether the circle at 24 is Open or Solid, and arrow direction.

Ans: _____

3 Solve 6n ≥ 78. Tell whether the circle at 13 is Open or Solid, and arrow direction.

Ans: _____

4 Solve m / 4 < 9. Tell whether the circle at 36 is Open or Solid, and arrow direction.

Ans: _____

5 Translate: 'You must be at least 48 inches tall (h) to ride.' Write the inequality!

Ans: _____

6 Translate: 'An elevator holds at most 2,000 pounds (w).' Write the inequality!

Ans: _____

7 Tell which values from the set {4, 5, 6} are solutions of the inequality 4n + 3 > 23.

Ans: _____

8 Which numbers from {4, 5, 6} are solutions to 4x - 3 ≥ 17?

Ans: _____

9 Modeling Real Life: You have $50 to buy $8 comic books (8b ≤ 50). Solve the inequality to find the maximum number of books you can buy.

Ans: _____

10 Modeling Real Life: A player needs at least 500 points to advance (p + 340 ≥ 500). Solve for p and graph the solution.

Ans: _____

BLOCK KIT: RACK-01 • Number Rack & Number Line

Section 7.4 — Practice B: Modeling Real Life (Writing and Graphing Inequalities)

Textbook Methods: Method 1: Inverse Operation + Boundary Circle Rule | Method 2: Shaded Test-Point Verification

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: BACKPACK WEIGHT LIMIT (3G + 15 ≤ 60) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Your adventure pack has a 15-lb shield plus gold bars weighing 3 lb each. Total weight can be at most 60 lb (3g + 15 ≤ 60). Find max bars g.

METHOD 1 (+10 pts — 1st Token): Subtract Fixed Weight, Then Divide

Subtract 15 lb shield from 60 lb limit: 3g ≤ 45. Divide by 3 lb per bar: g ≤ 15 (at most 15 gold bars).

ALGB-01 Way 1: Divide 180 lb capacity into 12 lb gold-bar boxes -> at most 15 bars (g ≤ 15).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Test Boundary g = 15 and g = 16

At g = 15: 3(15) + 15 = 60 ≤ 60 (True!). At g = 16: 3(16) + 15 = 63 lb > 60 lb (Too heavy!).

RACK-01 Way 2: Test g = 15 (180 ≤ 180 TRUE) vs. g = 16 (192 ≤ 180 FALSE) -> solid peg at 15, arrow Left!

SC-6 Keystrokes: 60 ENTER 15 - 3 ÷

Answer: g ≤ 15 (at most 15 gold bars)

SECTION 7.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Roller Coaster Height Rule: Riders must be at least 54 inches tall (h ≥ 54). Should the boundary circle at 54 be Open (○) or Solid (●)? If Leo is 49 inches tall, how many more inches must he grow?

Ans: _____

2 Elevator Capacity Limit: An elevator holds at most 12 passengers (p ≤ 12). There are already 5 passengers inside (p + 5 ≤ 12). How many more passengers can board?

Ans: _____

3 Free Shipping Threshold: Online orders over $35 get free shipping (22.50 + x ≥ 35). You have $22.50 in your cart. How much more x must you add?

Ans: _____

4 Arcade Game Budget: You have $28 to spend on $3.50 VR games (3.50g ≤ 28). Solve the inequality to find the maximum number of games g you can play.

Ans: _____

5 School Play Seating: A theater needs more than 240 chairs set up in rows of 15 (15r > 240). Solve for r. What is the smallest whole number of rows needed?

Ans: _____

6 Drone Battery Warning: A drone lands when battery b drops below 15% (b - 25 < 15 after a 25% flight). Solve for starting battery b and state Open vs. Solid circle.

Ans: _____

7 Solution Set Testing: Which values from {6, 8, 10} make the inequality 5x - 10 ≥ 30 true? Test each value!

Ans: _____

8 Bake Sale Profit Goal: Your club spends $30 on supplies and sells cupcakes for $2 each. To make a profit of at least $70, solve 2c - 30 ≥ 70.

Ans: _____

9 Speed Limit Sign: A school zone speed limit is 20 mph (s ≤ 20). Write two solutions and two non-solutions, and describe the number line graph.

Ans: _____

10 Cargo Van Crate Limit: A van can carry at most 800 lb. The driver weighs 170 lb and each crate weighs 45 lb (45c + 170 ≤ 800). How many crates c can fit?

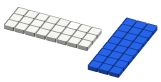
Ans: _____

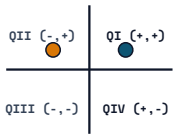
Section 7.5 — Practice A: Writing Equations in Two Variables (y = kx and y = x + b)

Key Vocabulary: independent variable (input x) • dependent variable (output y) • multiplicative (y = kx) • additive (y = x + b)

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)





DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): On COUNT-01, Table A triples/septuples each input tile (y = 7x); Table B removes a fixed 4-tile offset (y = x - 4).

Block Way 2 (+20 pts): Plot (x, y) pegs on the COUNT-01 grid—multiplicative y = 7x passes through (0,0), while y = x - 4 starts at (0, -4)!

Both Ways Proof: Equal-ratio tile stacks model y = kx (through origin), while fixed-offset bars model y = x + b!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Table A has (2, 14), (5, 35), (8, 56). Table B has (10, 6), (15, 11), (22, 18). Find both equations.

METHOD 1 (+10 pts — 1st Token): Check Ratio (y ÷ x) and Difference (y - x)

Table A: 14÷2 = 7, 35÷5 = 7, 56÷8 = 7 -> y = 7x. Table B: 6-10 = -4, 11-15 = -4, 18-22 = -4 -> y = x - 4.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Substitute Input x to Verify Output y

Test x = 8 in y = 7x -> y = 56 (Matches!). Test x = 22 in y = x - 4 -> y = 18 (Matches!).

SC-6 Keystrokes: `14` `ENTER` `2` `÷` `35` `ENTER` `5` `÷`

Answer: Table A: y = 7x; Table B: y = x - 4 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 7.5 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Test Constant Ratio y ÷ x vs. Difference y - x |..

1 Table has (1, 8), (2, 16), (3, 24). Write the equation and find y when x = 10.

Ans: _____

2 Table has (1, 9), (2, 10), (5, 13). Write the equation and find y when x = 20.

Ans: _____

3 Table has (2, 14), (5, 35), (8, 56). Write the equation and find y when x = 12.

Ans: _____

4 Table has (10, 6), (15, 11), (22, 18). Write the equation and find y when x = 30.

Ans: _____

5 Complete an input-output table for the multiplicative equation y = 6x using x = 1, 2, 3, and 4.

Ans: _____

6 In d = 65t (distance d in miles after t hours), which variable is Independent and which is Dependent?

Ans: _____

7 Using d = 65t, how far does the car travel in 4.5 hours?

Ans: _____

8 Does the graph of y = 4x pass through the origin (0, 0)? What about y = x + 4?

Ans: _____

9 You earn p = 12.50h. How many hours h must you work to earn $100? Solve by division AND ratio doubling!

Ans: _____

10 A gym charges a $25 signup fee plus $15 per month (m). Write the equation for total cost c and find c for 12 months!

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 7.5 — Practice B: Modeling Real Life (Writing Equations in Two Variables (y = kx and y = x + b))

Textbook Methods: Method 1: Test Constant Ratio y ÷ x vs. Difference y - x | Method 2: Input-Output Function Rule & Graph

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: LAWN MOWING EARNINGS (P = 12.50H FOR H = 6.5 HR) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: You earn $12.50 per hour mowing lawns (p = 12.50h). Identify the independent and dependent variables, and find your pay for 6.5 hours.

METHOD 1 (+10 pts — 1st Token): Substitute h = 6.5 into the Equation

Independent variable = hours worked (h); Dependent variable = total pay (p). Compute p = 12.50 × 6.5 = $81.25.

COUNT-01 Way 1: Build 6 full-hour rate blocks (6 × $12.50 = $75.00) plus a half-hour block ($6.25) -> $81.25.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Split into 6 Whole Hours + 1 Half-Hour

6 hours × $12.50 = $75.00. Half an hour (0.5 hr) = $6.25. Add them: $75.00 + $6.25 = $81.25!

PLACE-01 Way 2: Multiply 12.50 × 6.5 directly -> $81.25 (hours h = input, pay p = output)!

SC-6 Keystrokes: `12.50` `ENTER` `6.5` `×`

Answer: p = $81.25 (h = independent, p = dependent)

SECTION 7.5 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Streaming Subscription vs. Movie Rental: Plan A adds a $5 fee to your age-tier cost (y = x + 5). Plan B charges $8 per month m (y = 8m). Tell which is Additive and which is Multiplicative.

Ans: _____

2 3D Printer Filament Rate: A printer uses 18 grams of PLA per hour h (g = 18h). Identify the independent and dependent variables, and find g for h = 4.5 hours.

Ans: _____

3 Birthday Age Gap: Maya is always 4 years older than her sister Zoe (m = z + 4). Complete the table for z = 8, 12, and 15. Does the ratio m ÷ z stay constant?

Ans: _____

4 Custom Sticker Packs: Each sheet holds 15 stickers (s = 15p). How many sheets p are needed to print s = 180 stickers?

Ans: _____

5 Coupon Discount Rule: Every item on a table is $6 off its tag price x (y = x - 6). Find the sale price y when x = $24, $35, and $50.

Ans: _____

6 Bicycle Pedaling Gear: For every 1 pedal turn x, the rear wheel turns 3.5 times (y = 3.5x). How many times does the wheel turn in 40 pedal turns?

Ans: _____

7 Find the Rule from the Table: Input x = 3, 6, 9, 12 gives Output y = 27, 54, 81, 108. Write the equation relating x and y, and find y when x = 20.

Ans: _____

8 Two-Step Shipping Cost: Each board game costs $20 plus a flat $7 shipping fee for the box (c = 20g + 7). Find the total cost c for g = 6 games.

Ans: _____

9 Origin Graph Check: Which type of relationship ALWAYS passes through the origin (0, 0) on a coordinate graph: Multiplicative (y = kx) or Additive (y = x + b)? Explain why!

Ans: _____

10 Sound Wave Distance: Sound travels about 0.2 miles per second s (d = 0.2s). If you hear thunder 15 seconds after seeing lightning, how far away is the storm?

Ans: _____

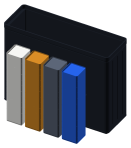
BLOCK KIT: COUNT-01 • 3x1 Gridfinity Box & Tiles

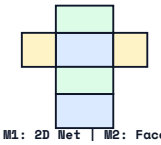
Section 8.1 — Practice A: Three-Dimensional Figures, Nets & Rectangular Prisms

Key Vocabulary: polyhedron • face • edge • vertex • rectangular prism • cube • net • surface area (SA)

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: 3X1 GRIDFINITY BOX & TILES)





M1: 2D Net | M2: Face Pairs

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Unfold the 7×5×3 prism into a flat 6-face net on COUNT-01 and add all 6 rectangles: 35+35+21+21+15+15 = 158 cm².

Block Way 2 (+20 pts): Stack matching opposite face pairs on COUNT-01: 2(35) + 2(21) + 2(15) = 70 + 42 + 30 = 158 cm²!

Both Ways Proof: Adding all 6 unfolded net faces (Method 1) and doubling the 3 unique face pairs (Method 2) both give 158 cm²!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the surface area of a rectangular prism measuring 8 cm by 5 cm by 3 cm.

METHOD 1 (+10 pts — 1st Token): Add All 6 Faces on a 2D Net

Top+Bottom = 40+40 = 80; Front+Back = 24+24 = 48; Left+Right = 15+15 = 30. Total SA = 80 + 48 + 30 = 158 cm².

METHOD 2 (+20 pts DOUBLE — 2nd Token): Sum 3 Distinct Pairs & Double

SA = 2(lw + lh + wh) = 2(8·5 + 8·3 + 5·3) = 2(40 + 24 + 15) = 2(79) = 158 cm².

SC-6 Keystrokes:

`g` `CLΣ` `8` `ENTER` `5` `f` `RECT` `2` `×` `Σ+` `8` `ENTER`
`3` `f` `RECT` `2` `×` `Σ+` `5` `ENTER` `3` `f` `RECT` `2` `×`
`Σ+` `f` `Σx`

Answer: 158 cm²

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 8.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Unfold into a 6-Face 2D Net | Method 2: Surface A..

1 Find the surface area of a cube with edge length s = 4 cm ($6s^2$).

Ans: _____

2 Find the surface area of a cube with edge length s = 9 in.

Ans: _____

3 Unfold a 6 cm by 4 cm by 3 cm rectangular prism into a 2D net and find its total surface area.

Ans: _____

4 Find the surface area of a 10 in by 5 in by 2 in rectangular box.

Ans: _____

5 Find the surface area of a prism with l = 12 ft, w = 8 ft, and h = 5 ft.

Ans: _____

6 Find the surface area of a prism with decimal edges: 5 m by 4 m by 2.5 m.

Ans: _____

7 A cube has a total surface area of 150 cm². What is the area of 1 face, and what is the edge length s?

Ans: _____

8 Error Analysis: For a 5×4×3 box, Leo added 20 + 15 + 12 = 47. What did he forget to do?

Ans: _____

9 An open-top 3D-printed bin is 10 cm long, 8 cm wide, and 5 cm tall (NO lid!). Find its outer surface area 2 ways!

Ans: _____

10 You wrap a gift box (15 in × 10 in × 4 in). If you have 550 in² of wrapping paper, is it enough?

Ans: _____

BLOCK KIT: COUNT-01 • 3x1 Gridfinity Box & Tiles

Section 8.1 — Practice B: Modeling Real Life (Three-Dimensional Figures, Nets & Rectangular Prisms)

Textbook Methods: Method 1: Unfold into a 6-Face 2D Net | Method 2: Surface Area Formula SA = 2(lw + lh + wh)

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: CUBE GIFT BOX (S = 7 IN) & OPEN-TOP ORGANIZER (10×8×5 CM) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Find the surface area of a cube with edge s = 7 in, and an open-top storage bin measuring 10 cm long, 8 cm wide, and 5 cm tall.

METHOD 1 (+10 pts — 1st Token): Use Cube Formula (6s²) & 5-Face Net

Cube: 6 identical 7×7 faces = 6 × 49 = 294 in². Open-Top Bin (only 1 bottom!): 10×8 + 2(10×5) + 2(8×5) = 80 + 100 + 80 = 260 cm².

COUNT-01 Way 1: Build six 7×7 square faces for the cube (6 × 49 = 294 in²).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Full 6-Face Prism Minus the Missing Top

Full 10×8×5 prism SA = 2(80 + 50 + 40) = 340 cm². Subtract the open 10×8 top (80 cm²): 340 - 80 = 260 cm²!

COUNT-01 Way 2: For the open-top bin, take the full 6-face area (340) and remove ONE 10×8 top lid (80) -> 260 cm²!

SC-6 Keystrokes: 7 x² 6 ×

Answer: Cube = 294 in²; Open-Top Bin = 260 cm²

SECTION 8.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS 1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Cereal Box Cardboard: A cereal box is a rectangular prism measuring 20 cm long, 8 cm wide, and 30 cm tall. Find its total surface area using SA = 2(lw + lh + wh).**

Ans: _____

2 **Collector Dice Cube: A giant foam die is a cube with edge length s = 9 inches. Find the area of 1 square face and the total surface area of all 6 faces (6s²).**

Ans: _____

3 **Glass Aquarium (Open Top!): A glass fish tank is 24 in long, 12 in wide, and 15 in tall, with NO glass lid on top. Find the area of the 5 glass panels!**

Ans: _____

4 **Wrapping Paper Roll: You wrap a game console box measuring 15 in × 10 in × 4 in. If your sheet of wrapping paper is 600 in², do you have enough paper?**

Ans: _____

5 **Room Painting (4 Walls + Ceiling): A bedroom is 12 ft long, 10 ft wide, and 8 ft high. You paint the 4 walls and the ceiling (not the floor!). Find the area painted.**

Ans: _____

6 **Find the Cube Edge from SA: A wooden cube has a total surface area of 150 cm². Divide by 6 to find the area of 1 face, then take the square root to find edge s!**

Ans: _____

7 **Shipping Trunk Plywood: A storage trunk is 4 ft × 2 ft × 2 ft. Find its total surface area across all 6 faces.**

Ans: _____

8 **Juice Box Foil: A mini juice box measures 6 cm by 4 cm by 10 cm. Find its surface area.**

Ans: _____

9 **Doubling Edge Lengths Effect: Cube A has edge 3 cm (SA = 6×3² = 54 cm²). Cube B has edge 6 cm (double the edge!). Find Cube B's SA. How many times larger is it?**

Ans: _____

10 **Two Cubes Glued Together: Two 4 cm × 4 cm × 4 cm cubes are glued face-to-face to form an 8 cm × 4 cm × 4 cm prism. Find the new surface area (10 exposed square faces).**

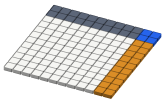
Ans: _____

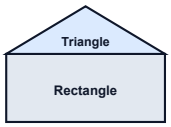
Section 8.2 — Practice A: Surface Areas of Triangular Prisms and Pyramids

Key Vocabulary: triangular prism • pyramid • lateral face • slant height • regular pyramid • 2D net

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)





M1: Split Sum | M2: Box-Cut

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Lay out the 10×10 square base (100) and 4 identical triangular faces (each ½×10×12 = 60) on COUNT-01 -> 340 m².

Block Way 2 (+20 pts): Line up the 4 triangles into one strip of base Perimeter P = 40 and height 12 (½ × 40 × 12 = 240) + 100 base = 340!

Both Ways Proof: Summing individual net faces and using the combined Lateral Perimeter strip both give 340 m²!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the surface area of a square pyramid with a 10 m × 10 m base and 4 triangles (b = 10 m, slant height = 12 m).

METHOD 1 (+10 pts — 1st Token): 2D Net (1 Square Base + 4 Triangles)

Square base = 10 × 10 = 100 m². Each triangle = ½ × 10 × 12 = 60 m². Total SA = 100 + 60 + 60 + 60 + 60 = 340 m².

METHOD 2 (+20 pts DOUBLE — 2nd Token): Base Area + 4 × Lateral Triangle

SA = B + 4(½bl) = 10² + 4(60) = 100 + 240 = 340 m².

SC-6 Keystrokes: g | CLΣ | 10 | x² | Σ+ | 10 | ENTER | 12 | g | TRI | 4 | × | Σ+ | f | Σx

Answer: 340 m²

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 8.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Sum Individual Faces on the 2D Net | Method 2: Ba..

1 Square Pyramid: Base is 6 cm × 6 cm; 4 triangles each have b = 6 cm, slant height = 5 cm. Find SA.

Ans: _____

2 Square Pyramid: Base is 8 in × 8 in; 4 triangles each have b = 8 in, slant height = 10 in. Find SA.

Ans: _____

3 Square Pyramid: Base is 12 ft × 12 ft; 4 triangles each have b = 12 ft, slant height = 9 ft. Find SA.

Ans: _____

4 Regular Tetrahedron: Made of 4 identical equilateral triangles (b = 6 cm, h = 5.2 cm). Find total SA.

Ans: _____

5 Triangular Prism: 2 triangle ends (b=4, h=3) + 3 rectangles of length 10 (widths 3, 4, 5). Find SA.

Ans: _____

6 Triangular Prism Tent: 2 triangle ends (b=8 ft, h=3 ft) + 3 rectangles of length 7 ft (widths 5, 5, 8 ft).

Ans: _____

7 A regular tetrahedron has 4 identical triangles with b = 8 cm and h = 6.93 cm. Find its surface area.

Ans: _____

8 How many faces (triangles and rectangles) does a Triangular Prism have vs. a Square Pyramid?

Ans: _____

9 Camping Tent (Triangular Prism): 2 ends (b=6 ft, h=4 ft), floor (6×8 ft), and 2 roof sides (5×8 ft each). Find total fabric 2 ways!

Ans: _____

10 A glass pyramid skylight has NO bottom floor—only 4 triangular glass panes (b = 9 ft, slant h = 6 ft). Find glass area!

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 8.2 — Practice B: Modeling Real Life (Surface Areas of Triangular Prisms and Pyramids)

Textbook Methods: Method 1: Sum Individual Faces on the 2D Net | Method 2: Base Area + Lateral Perimeter × Height

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: SKATEBOARD RAMP (TRIANGULAR PRISM: 6-8-10 RIGHT TRIANGLES, LENGTH 10 FT) (SOLVE 2 WAYS = +20 PTS DOUB..

Problem: A skate ramp has 2 right-triangle ends (base 6 ft, height 8 ft, ramp 10 ft) and 3 rectangular faces of length 10 ft. Find its surface area.

METHOD 1 (+10 pts — 1st Token): Add All 5 Faces on the Unfolded Net

2 Triangles = 2 × (½·6·8) = 48 ft². 3 Rectangles = (6×10) + (8×10) + (10×10) = 60 + 80 + 100 = 240 ft². Total = 48 + 240 = 288 ft².

COUNT-01 Way 1: Sum the two 6×8÷2 triangle ends (24 + 24 = 48) plus three 10-ft rectangular panels (80+80+80 = 240) -> 288 ft².

METHOD 2 (+20 pts DOUBLE — 2nd Token): Fold the 3 Rectangles into 1 Giant Strip

Triangle perimeter = 6 + 8 + 10 = 24 ft. Giant lateral rectangle = 24 × 10 = 240 ft². Add the 2 triangles (48 ft²): 240 + 48 = 288 ft²!

COUNT-01 Way 2: Unfold the 3 rectangles into one (8+8+8) × 10 = 24 × 10 = 240 ft² sheet + 48 ft² ends = 288 ft²!

SC-6 Keystrokes: 6 ENTER 8 g TRI 2 × 6 ENTER 8 + 10 + 10 f RECT +

Answer: 288 ft²

SECTION 8.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Camping Tent (Triangular Prism):** A tent has 2 triangular doors (base 6 ft, height 4 ft) and 3 rectangular panels (two 5 ft × 8 ft roof sides + one 6 ft × 8 ft floor). Find total fabric area.

Ans: _____

2 **Louvre Glass Pyramid Model:** A scale model of a square pyramid has a 12 cm × 12 cm base and 4 triangular glass sides with slant height 10 cm. Find its total surface area.

Ans: _____

3 **Lateral Area Only (Roof Shingles):** A gazebo roof is a square pyramid with base edge 8 ft and slant height 6 ft (no bottom base!). Find the area of the 4 triangular roof sides.

Ans: _____

4 **Toblerone Chocolate Box:** A triangular prism box has 2 equilateral triangle ends (each area = 6.9 cm²) and 3 identical rectangular sides (each 4 cm × 20 cm). Find total SA.

Ans: _____

5 **Wheel Chock Wedge:** A wooden wheel wedge is a right triangular prism with 2 triangle ends (b = 8 in, h = 6 in, slant = 10 in) and width 5 in. Find its total surface area.

Ans: _____

6 **Regular Tetrahedron (4 Triangles):** A 4-sided game die (D4) is made of 4 congruent equilateral triangles, each with base 3 cm and height 2.6 cm. Find the total surface area.

Ans: _____

7 **Attic Roof Prism:** An attic is a triangular prism 30 ft long. Its front and back gables have base 16 ft and height 6 ft, and the two roof slopes are 10 ft wide. Find the area of the 2 gables + 2 roof slopes.

Ans: _____

8 **Pyramid Paperweight:** A square pyramid has base 6 in × 6 in and total surface area 96 in². Subtract the base (36 in²) and divide by 4 to find the area of 1 triangular face!

Ans: _____

9 **Triangular Prism Gift Box:** Two triangle ends have b = 12 cm, h = 8 cm. Three rectangular sides are each 10 cm, 10 cm, and 12 cm wide by 15 cm long. Find total SA.

Ans: _____

10 **Compare Two Pyramids:** Pyramid A has base 10×10 and slant height 8. Pyramid B has base 8×8 and slant height 10. Find both surface areas—which is larger?

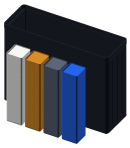
Ans: _____

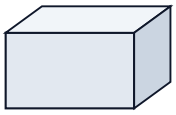
Section 8.3 — Practice A: Volumes of Rectangular Prisms

Key Vocabulary: volume (V) • cubic units • base area (B = lw) • V = Bh = lwh • fractional edge length

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & FRACTION STUDIO)





M1: Layer×h | M2: V = l×w×h

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Build one 10 × 8 = 80-cube floor layer on COUNT-01, then stack 6 identical layers high -> 80 × 6 = 480 cubes.

Block Way 2 (+20 pts): For the 2½ × 1½ × 2 ft prism, pack eighty-four ½-ft cubes (84 × 1/8 ft³ = 10½ ft³) to verify V = lwh!

Both Ways Proof: Stacking Base-Area layers (V = Bh) and packing fractional unit cubes both produce the exact same volume!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the volume of a shipping crate measuring 12 in × 8 in × 5 in, and an aquarium 3 1/2 ft × 1 1/2 ft × 2 ft.

METHOD 1 (+10 pts — 1st Token): Layer Stacking V = B × h = l × w × h

Crate: Base B = 12 × 8 = 96 in²; stack 5 layers -> 96 × 5 = 480 in³. Aquarium: (7/2) × (3/2) × 2 = 21/2 = 10 1/2 ft³.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Decimal Conversion & [f] [PRISM]

Convert aquarium edges to decimals: 3.5 × 1.5 × 2 = 3.5 × 3 = 10.5 ft³ (10 1/2 ft³).

SC-6 Keystrokes: 12 ENTER 8 ENTER 5 f PRISM

Answer: Crate = 480 in³; Aquarium = 10 1/.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 8.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Base Area × Height (V = Bh) | Method 2: Unit-Cube..

1 Find the volume of a rectangular prism measuring 10 cm long, 6 cm wide, and 4 cm high.

Ans: _____

2 Find the volume of a cube with edge length s = 5 inches.

Ans: _____

3 Find the volume of a crate measuring 15 ft by 8 ft by 6 ft.

Ans: _____

4 A prism has Base Area B = 32 m² and height h = 7 m. Find its volume.

Ans: _____

5 Fractional Edges: Find the volume of a fish tank 3 1/2 ft long, 1 1/2 ft wide, and 2 ft tall.

Ans: _____

6 Fractional Edges: Find the volume of a block 4 1/2 in by 3 in by 2 in.

Ans: _____

7 Missing Height: A box has Volume = 270 in³, length = 9 in, and width = 5 in. Find height h.

Ans: _____

8 How many 1/2-inch mini-cubes fit inside 1 cubic inch? (Compute 2^3!)

Ans: _____

9 Modeling Real Life: A planter box has base area B = 45 in² (9 in × 5 in) and volume V = 360 in³. Find the depth h of the box.

Ans: _____

10 How many 1/2-inch cubes fit inside a box with volume 27 in³? (Multiply 27 by $2^3 = 8$!)

Ans: _____

Section 8.3 — Practice B: Modeling Real Life (Volumes of Rectangular Prisms)

Textbook Methods: Method 1: Base Area × Height (V = Bh) | Method 2: Unit-Cube Packing & Fractional Multiplication

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: FINDING MISSING DEPTH OF A PLANTER BOX (V = 360 IN³, B = 45 IN²) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A cedar planter box has a base area of B = 45 square inches (9 in × 5 in) and holds 360 cubic inches of soil. How deep (h) is the soil?

METHOD 1 (+10 pts — 1st Token): Substitute into V = Bh and Divide

Write 360 = 45h. Divide both sides by the base area 45: h = 360 ÷ 45 = 8 inches deep.

COUNT-01 Way 1: Build one 15 × 12 = 180-cube base layer; divide 1,440 total cubes by 180 per layer -> 8 layers high.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Divide by Length First, Then by Width

Divide 360 ÷ 9 in = 40, then divide 40 ÷ 5 in = 8 inches! Check: 9 [ENTER] 5 [ENTER] 8 [f] [PRISM] = 360 in³.

ALGB-01 Way 2: Solve 15 × 12 × h = 1,440 by dividing 1,440 ÷ 15 = 96, then 96 ÷ 12 = 8 in!

SC-6 Keystrokes: 360 ENTER 45 ÷

Answer: h = 8 inches deep

SECTION 8.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS 1 Way = +10 pts | 2 Ways = +20 pts Double

1 Swimming Lap Pool Volume: A rectangular pool is 25 meters long, 10 meters wide, and 2 meters deep. Find its volume in cubic meters (V = lwh).

Ans: _____

2 Raised Garden Soil: A cedar garden bed is 8 ft long, 4 ft wide, and 1 1/2 ft deep. How many cubic feet of soil are needed to fill it?

Ans: _____

3 3D Printer Build Volume: A 3D printer has a build volume measuring 25 cm × 25 cm × 25 cm. How many cubic centimeters is its build volume?

Ans: _____

4 Fractional Lunchbox Volume: A bento box measures 6 1/2 inches long, 4 inches wide, and 2 1/2 inches tall. Find its volume in cubic inches.

Ans: _____

5 Unit-Cube Packing (1/2-inch Cubes): How many 1/2-inch cubes fit inside a 1-cubic-inch space? How many fit inside a 3 in × 2 in × 2 in box (V = 12 in³)?

Ans: _____

6 Missing Water Depth: A rectangular fish tank is 20 inches long and 10 inches wide (Base B = 200 in²). You pour in 3,000 in³ of water. How deep is the water?

Ans: _____

7 Shipping Container Capacity: A cargo box is 5.5 ft long, 4 ft wide, and 3.5 ft tall. Find its volume.

Ans: _____

8 Concrete Patio Slab: A patio is 12 ft long, 9 ft wide, and 1/3 ft (4 inches) thick. How many cubic feet of concrete are needed?

Ans: _____

9 Two-Box Stack Composite Volume: A trophy base is a 10 cm × 10 cm × 3 cm bottom block topped by a 6 cm × 6 cm × 5 cm upper block. Find the combined volume.

Ans: _____

10 Gallons in an Aquarium: A tank measures 24 in × 12 in × 16 in. Find its volume in cubic inches, then divide by 231 in³/gal to find how many gallons it holds!

Ans: _____

Section 8.4 — Practice A: Circles, Cones, and the Pythagorean Theorem

Key Vocabulary: circle • radius (r) • diameter (d = 2r) • circumference (C = 2πr) • area (A = πr²) • cone (V = ⅓πr²h) • hypotenuse

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)





M1: ½(b×h) | M2: (b÷2)×h

DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Build a 5×5 square (25 tiles) and 12×12 square (144 tiles) on the legs of a right triangle -> they fill a 13×13 square (169)!

Block Way 2 (+20 pts): Compare the 3D Cone to a matching Cylinder of radius r and height h—3 cones fill 1 cylinder (V = ⅓πr²h)!

Both Ways Proof: Physical tile squares prove a² + b² = c² (25 + 144 = 169), and 3-to-1 cylinder filling proves V_cone = ⅓Bh!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the area of a pizza with radius r = 6 in, and the diagonal of a 12 ft by 5 ft ramp.

METHOD 1 (+10 pts — 1st Token): Step-by-Step Formula Evaluation

Pizza: A = πr² = π × 6² = 36π ≈ 113.10 in². Ramp diagonal: c = √(12² + 5²) = √(144 + 25) = √169 = 13 ft.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Direct SC-6 Geometry Operators

Press 6 [f] [CIRC] -> 113.10 in², and 12 [ENTER] 5 [g] [HYP] -> 13 ft.

SC-6 Keystrokes: 6 | f | CIRC | 12 | ENTER | 5 | g | HYP

Answer: Pizza = 113.10 in²; Ramp diagonal.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 8.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Exact Formula with π (≈ 3.1416) | Method 2: SC-6 Geometry Keys ([f][CIRC],..

1 Find the area of a circle with radius r = 5 cm (round to the nearest hundredth).

Ans: _____

2 Find the area of a circle with radius r = 10 inches (round to the nearest hundredth).

Ans: _____

3 Find the hypotenuse c of a right triangle with legs a = 3 units and b = 4 units.

Ans: _____

4 Find the hypotenuse c of a right triangle with legs a = 6 ft and b = 8 ft.

Ans: _____

5 Find the slanted length (hypotenuse) of a skate ramp with horizontal base 12 ft and back height 5 ft.

Ans: _____

6 A circle has diameter d = 16 meters. Find its radius r and its area (round to the nearest hundredth).

Ans: _____

7 Find the volume of a cone with radius r = 3 cm and height h = 12 cm (V = ⅓πr²h).

Ans: _____

8 Find the volume of a cone with radius r = 6 in and height h = 5 in (V = ⅓πr²h).

Ans: _____

9 Modeling Real Life: Compare the volume of a cone (r = 3 cm, h = 10 cm) to a cylinder with the same radius and height.

Ans: _____

10 TV screens are measured along the diagonal. A screen is 32 in wide and 24 in tall. Find its diagonal size 2 ways!

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 8.4 — Practice B: Modeling Real Life (Circles, Cones, and the Pythagorean Theorem)

Textbook Methods: Method 1: Exact Formula with π (≈ 3.1416) | Method 2: SC-6 Geometry Keys ([f][CIRC], [g][CONE], [g][H..

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: WAFFLE CONE VOLUME (R = 3 CM, H = 10 CM) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: An ice cream waffle cone has radius r = 3 cm and height h = 10 cm. Find its volume rounded to the nearest hundredth.

METHOD 1 (+10 pts — 1st Token): One-Third of a Cylinder (V = ⅓πr²h)
Cylinder volume = π × 3² × 10 = 90π ≈ 282.74 cm³. Divide by 3 for the cone: 90π ÷ 3 = 30π ≈ 94.25 cm³.

COUNT-01 Way 1: Find the cylinder volume π × 3² × 10 = 90π, then take 1/3 for the cone -> 30π ≈ 94.25 cm³.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Direct Cone Key ([g] [CONE])
Enter radius 3 [ENTER] height 10 [g] [CONE] -> 94.25 cm³ (30π)!

FRAC-01 Way 2: Three 30π cones stack to match the 90π cylinder (3 × 94.25 = 282.74 cm³)!

SC-6 Keystrokes: 3 | ENTER | 10 | g | CONE

Answer: 94.25 cm³ (exact: 30π cm³)

SECTION 8.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Giant Trampoline Mat: A circular trampoline has a radius of r = 7 feet. Find its jumping area (A = πr², rounded to the nearest hundredth).**

Ans: _____

2 **Ferris Wheel Diameter: A Ferris wheel has a diameter of d = 40 meters. First divide by 2 to find radius r = 20 m, then find the area of the circle.**

Ans: _____

3 **Bike Tire Rotation (Circumference): A bicycle wheel has radius r = 14 inches. Find its circumference C = 2πr (how far the bike rolls in 1 turn).**

Ans: _____

4 **Zipline Cable Length (Hypotenuse): A zipline tower is 15 meters tall and anchored 20 meters away across a creek. Find the diagonal cable length c!**

Ans: _____

5 **TV Screen Diagonal: A monitor screen is 16 inches wide and 12 inches tall. Use the Pythagorean hypotenuse formula to find its diagonal screen size.**

Ans: _____

6 **Snow Cone Cup Volume: A paper snow cone cup has radius r = 4 cm and height h = 9 cm. Find its volume (V = ⅓πr²h).**

Ans: _____

7 **Traffic Cone Volume: A highway safety cone has base diameter d = 12 in (so r = 6 in) and height h = 15 in. Find its volume.**

Ans: _____

8 **Pizza Value Showdown: One 16-inch pizza (r = 8 in) vs. Two 10-inch pizzas (r = 5 in each). Compare their total areas—which gives you more pizza?**

Ans: _____

9 **Semicircle Basketball Court Top: The top of the basketball key is a semicircle with radius r = 6 feet. Find the area of the full circle and divide by 2!**

Ans: _____

10 **Ladder Against a Wall: An 8-foot base and 15-foot wall form a right triangle. How long must a ladder be to reach the top of the 15-foot wall?**

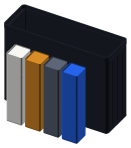
Ans: _____

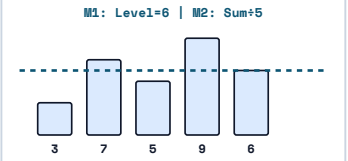
Section 9.1 — Practice A: Statistical Questions, Dot Plots, and the Mean

Key Vocabulary: statistical question • dot plot • peak • cluster • gap • measure of center • mean (fair share)

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & ARRAY TILES)





DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Build 5 COUNT-01 towers [18, 22, 15, 25, 25] and move top blocks from taller to shorter towers until all 5 equal 21!

Block Way 2 (+20 pts): Snap all 5 towers into one long bar of 105 blocks and deal equally into 5 towers -> 105 ÷ 5 = 21!

Both Ways Proof: Leveling towers to equal height (Fair Share) and dividing Total Blocks ÷ 5 both prove Mean = 21!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the sum ([f] [Σx]), count ([N]), and Mean ([g] [x-bar]) of 18, 24, 15, 27, 21.

METHOD 1 (+10 pts — 1st Token): Sum Divided by Count (Σx ÷ n)

Add the 5 values: 18 + 24 + 15 + 27 + 21 = 105. Divide by n = 5 values: Mean = 105 ÷ 5 = 21.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Fair-Share Leveling & [Σ+] -> [g][x-bar]

Shift 6 from 27 to 15 (both become 21) and 3 from 24 to 18 (both become 21) -> all 5 towers equal 21!

SC-6 Keystrokes: g CLΣ 18 Σ+ 24 Σ+ 15 Σ+ 27 Σ+ 21 Σ+ g x̄

Answer: Sum = 105, Count = 5, Mean = 21 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 9.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Method 1: Fair-Share Tower Leveling / Balance Point | Method 2: Sum Divided by Count..

1 Find the sum and the mean (fair-share average) of the data set: 2, 4, 6, 8.

Ans: _____

2 Find the sum and Mean of: 12, 15, 18, 11, 14.

Ans: _____

3 Find the Mean of your 5 gaming scores: 18, 24, 15, 27, 21.

Ans: _____

4 Find the Mean of these 4 decimal weights (g): 4.2, 3.8, 4.5, 3.5.

Ans: _____

5 Error Analysis: A student added 10, 20, and 35 instead of 10, 20, and 25. Correct the sum and find the true mean.

Ans: _____

6 In a dot plot, the value 10 appears 4 times and 20 appears 2 times. Find the mean of all 6 data values.

Ans: _____

7 Find the Mean daily high temperature (°F): 68, 72, 75, 71, 64, 70.

Ans: _____

8 If 5 towers have a Mean height of 12 blocks, how many total blocks are in all 5 towers combined?

Ans: _____

9 You need a Mean of 90 across 4 quizzes (360 pts). Your first 3 are 85, 92, 88. Find Quiz 4 score 2 ways!

Ans: _____

10 6 players have a Mean score of 15 pts (90 pts). A 7th player scores 29 pts. What is the new Mean of all 7?

Ans: _____

BLOCK KIT: COUNT-01 • Counting & Array Tiles

Section 9.1 — Practice B: Modeling Real Life (Statistical Questions, Dot Plots, and the Mean)

Textbook Methods: Method 1: Fair-Share Tower Leveling / Balance Point | Method 2: Sum Divided by Count (x-bar = Σx ÷ n)

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: TARGET SCORE FOR A 90 AVERAGE (QUIZZES: 86, 92, 88, ?) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: You want a mean score of 90 across 4 math quizzes. Your first 3 quiz scores are 86, 92, and 88. What score do you need on Quiz 4?

METHOD 1 (+10 pts — 1st Token): Total Points Needed Minus Current Sum

4 quizzes × 90 mean = 360 total points needed. Current sum = 86 + 92 + 88 = 266. Needed on Quiz 4: 360 - 266 = 94!

COUNT-01 Way 1: Four towers of mean 85 require 4 × 85 = 340 blocks; subtract the first 3 towers (246) -> 94 blocks.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Balance Deviations Around 90

Compare each score to 90: 86 is -4, 92 is +2, 88 is -2. Net balance so far = -4 + 2 - 2 = -4. Quiz 4 needs +4 above 90 -> 94!

RACK-01 Way 2: Deviations from 85 are -3, +5, -11 (net -9), so Quiz 4 must be +9 above 85 -> 85 + 9 = 94!

SC-6 Keystrokes: 4 ENTER 90 × 86 ENTER 92 + 88 + -

Answer: 94 on Quiz 4

SECTION 9.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Basketball Points Per Game: In 5 games, Maya scored 14, 18, 12, 22, and 19 points. Find the sum of her points and her mean (average) points per game.

Ans: _____

2 3D Print Batch Weights: Six printed parts weigh 41g, 43g, 42g, 40g, 44g, and 42g. Find the mean weight of the parts.

Ans: _____

3 Dot Plot Frequency Table: On a quiz out of 10, three students scored 8, five students scored 9, and two students scored 10. Find the mean of all 10 students.

Ans: _____

4 Target Grade Goal: You want a mean of 85 across 5 tests. Your first 4 test scores are 80, 88, 82, and 84. What score do you need on Test 5?

Ans: _____

5 Daily Step Counter: Over 4 days you walked 8,200, 9,500, 7,800, and 10,500 steps. Find your total steps and your daily mean.

Ans: _____

6 Removing a Dropped Score: A student's 4 quiz scores are 60, 88, 92, and 90 (Mean = 82.5). If the teacher drops the lowest quiz (60), what is the new mean of the 3 quizzes?

Ans: _____

7 Fair-Share Block Leveling: Four towers have heights 6, 10, 14, and 18 blocks. Move blocks from the tall towers to the short towers—what is the leveled height?

Ans: _____

8 Science Plant Growth: Five bean seedlings grew 4.2 cm, 5.1 cm, 4.8 cm, 5.5 cm, and 4.4 cm in a week. Find the mean growth.

Ans: _____

9 Statistical Question Check: Explain why 'How many minutes did students in our class read last night?' IS a statistical question, while 'How many minutes are in 1 hour?' is NOT.

Ans: _____

10 Combined Team Average: Group A (4 students) has a mean of 20 points (sum = 80). Group B (6 students) has a mean of 30 points (sum = 180). Find the mean of all 10 students!

Ans: _____

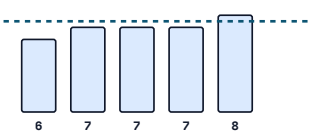
Section 9.2 — Practice A: Measures of Center: Median, Mode, and Range

Key Vocabulary: median (middle value) • mode (most frequent) • range (max - min) • outlier • best measure of center

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (PROB-01: PROBABILITY & STATS STUDIO)





DEMONSTRATE BOTH METHODS WITH PROB-01 BLOCKS:

Block Way 1 (+10 pts): Line up the 8 COUNT-01 towers shortest-to-tallest [6,7,7,7,8,8,9,10]; middle pair 7 & 8 gives Median = 7.5, Mode = 7.

Block Way 2 (+20 pts): Stand the tallest tower (10) beside the shortest tower (6) to measure Range = 10 - 6 = 4 blocks!

Both Ways Proof: Ordering towers reveals the middle Median & most frequent Mode, while comparing end towers shows Range!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: For shoe sizes 7, 9, 6, 7, 10, 8, 9, 7, find Mean, Median, Mode, Min, Max, and Range.

METHOD 1 (+10 pts — 1st Token): Sort Data from Least to Greatest

Ordered (n=8): 6, 7, 7, [7, 8], 9, 9, 10. Median = (7+8)/2 = 7.5; Mode = 7 (most frequent); Range = 10 - 6 = 4.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Enter Once with [Σ+] & Read Keys

Enter all 8 values with [Σ+], then press [g][x-bar] = 7.875, [f][MED] = 7.5, [g][MODE] = 7, and [RNG] = 4.

SC-6 Keystrokes: g CLΣ 7 Σ+ 9 Σ+ 6 Σ+ 7 Σ+ 10 Σ+ 8 Σ+ 9 Σ+ 7 Σ+ f MED g MODE RNG

Answer: Mean = 7.875, Median = 7.5, Mode.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 9.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Order Least-to-Greatest by Hand | Method 2: One-Pass [Σ+] Statistical Regist..

1 Order the data set 4, 9, 2, 7, 8 from least to greatest. Find the median and the range.

Ans: _____

2 For the data set 12, 15, 12, 18, 20, 13, find the median, the mode, and the range.

Ans: _____

3 Find the Mean, Median, Mode, and Range of: 5, 8, 8, 10, 14.

Ans: _____

4 Find the minimum, maximum, and range of the data set: 23, 45, 19, 38, 51.

Ans: _____

5 Even Count Median: Find the Median of the 6 test scores 70, 80, 84, 90, 92, 98.

Ans: _____

6 Outlier Check: Earnings are $15, $18, $15, $20, $17, $90. Find Mean and Median. Which represents a typical day?

Ans: _____

7 What happens to the Mean and Median of {10, 12, 14, 16, 18} if 18 changes to 100?

Ans: _____

8 Mystery Range: The minimum score on a quiz is 64 and the Range is 31. What was the maximum score?

Ans: _____

9 Find the mean, median, and range of the data set: 14, 8, 21, 11, 16.

Ans: _____

10 Home prices on a block (in $1,000s): 210, 220, 215, 225, 980. Find Mean and Median—which should a buyer trust?

Ans: _____

Section 9.2 — Practice B: Modeling Real Life (Measures of Center: Median, Mode, and Range)

Textbook Methods: Method 1: Order Least-to-Greatest by Hand | Method 2: One-Pass [Σ+] Statistical Registers

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: HOW AN OUTLIER SKEWED THE MEAN ($15, $18, $15, $20, $17, $90) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Daily dog-walking earnings are $15, $18, $15, $20, $17, and $90 (holiday bonus!). Find the Mean and Median, and tell which is better.

METHOD 1 (+10 pts — 1st Token): Compute Mean vs. Ordered Median

Sum = $175 -> Mean = 175 ÷ 6 = $29.17 (higher than 5 of the 6 days!). Ordered: 15, 15, [17, 18], 20, 90 -> Median = $17.50.

COUNT-01 Way 1: Order the 6 price towers [$15,$15,$15,$20,$20,$90] -> middle two ($15 & $20) average to Median = $17.50.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Compare With and Without the $90 Outlier

Without the $90 outlier, Mean = $85 ÷ 5 = $17.00 and Median = $17.00. The Median ($17.50) resists the outlier and represents a typical day!

COUNT-01 Way 2: Leveling all 6 towers pulls the Mean up to $29.17 because of the $90 outlier tower!

SC-6 Keystrokes: g CLΣ 15 Σ+ 18 Σ+ 15 Σ+ 20 Σ+ 17 Σ+ 90 Σ+ g x̄ f MED

Answer: Mean = $29.17, Median = $17.50 (Median is best)

SECTION 9.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Roller Coaster Speeds (Odd Count n=7): Seven coasters have top speeds (mph) of 55, 62, 48, 70, 62, 50, 66. Order them to find the Median, Mode, and Range.

Ans: _____

2 Coding Club Ages (Even Count n=6): Six members are ages 11, 13, 12, 14, 12, 16. Order them and average the two middle values to find the Median!

Ans: _____

3 Video Game Prices & Outlier: Five games cost $20, $25, $20, $30, and $105 (collector's edition!). Find the Mean ($40) and Median ($25). Which better represents a typical game?

Ans: _____

4 Bimodal Data Set: In the shoe size list 5, 6, 6, 6, 7, 8, 8, 8, 9, which two sizes tie for the highest frequency (making the data bimodal)?

Ans: _____

5 No-Mode Sprint Times: Five runners finish in 11.2, 11.5, 11.8, 12.1, and 12.4 seconds. Find the Median, Range, and explain why there is NO mode.

Ans: _____

6 Weather High-Low Range: This week's high temperatures (°F) were 72, 68, 75, 81, 64, 70, 74. Find the Minimum, Maximum, Range, and Median.

Ans: _____

7 Pizza Shop Orders: Daily orders of mushroom pizza were 8, 12, 9, 15, 12, 10. Find the Mean, Median, Mode, and Range.

Ans: _____

8 Create a Data Set Puzzle: Write a list of 5 numbers that has a Median of 10, a Mode of 8, and a Range of 7.

Ans: _____

9 Adding a Constant to Every Score: Every student on a quiz (scores: 70, 75, 80, 85, 90) gets +5 bonus points (75, 80, 85, 90, 95). How do the Median and Range change?

Ans: _____

10 Backpack Weights (lb): Eight backpacks weigh 9, 12, 10, 14, 11, 15, 10, 13 lb. Find the Mean, Median, Mode, and Range.

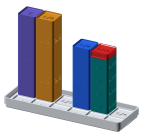
Ans: _____

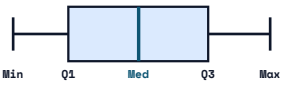
Section 9.3 — Practice A: Quartiles, Interquartile Range (IQR) & Box Plots

Key Vocabulary: quartiles • first quartile (Q1) • third quartile (Q3) • interquartile range (IQR = Q3 - Q1) • box-and-whisker plot •..

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (FRAC-01: FRACTION & STATS STUDIO)





M1: Peel Ends | M2: IQR=Q3-Q1

DEMONSTRATE BOTH METHODS WITH FRAC-01 BLOCKS:

Block Way 1 (+10 pts): Order the 8 towers on COUNT-01 and place a central divider (Med = 82), then split the lower half (Q1 = 73.5) & upper half (Q3 = 90).

Block Way 2 (+20 pts): Frame a box around the middle 4 towers (middle 50% of data) from Q1 to Q3 -> IQR = 90 - 73.5 = 16.5!

Both Ways Proof: Three quartile dividers split the ordered towers into 4 equal 25% quarters, and the middle two quarters form the IQR box!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the 5-Number Summary and IQR of: 68, 72, 75, 78, 82, 85, 88, 92, 96.

METHOD 1 (+10 pts — 1st Token): Median of Lower Half (Q1) & Upper Half (Q3)

Median = 82. Lower half (68, 72, 75, 78) -> Q1 = 73.5. Upper half (85, 88, 92, 96) -> Q3 = 90. IQR = 90 - 73.5 = 16.5.

METHOD 2 (+20 pts DOUBLE — 2nd Token): 5-Number Summary on SC-6 ([IQR])

Enter scores with [Σ+], then read [f][MIN]=68, [f][1Q]=73.5, [f][MED]=82, [g][3Q]=90, [g][MAX]=96, and [IQR]=16.5.

SC-6 Keystrokes: f MIN f 1Q f MED g 3Q g MAX IQR

Answer: Min=68, Q1=73.5, Med=82, Q3=90, M.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 9.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE) Methods: Method 1: Split Ordered Data into Lower & Upper Halves | Me..

1 For the ordered data set 10, 20, 30, 40, 50, find the 5-number summary (Min, Q1, Median, Q3, Max) and the IQR.

Ans: _____

2 For the ordered data set 4, 6, 8, 10, 12, 14, 16, find Q1, the median, Q3, and the interquartile range (IQR).

Ans: _____

3 If a box plot has Q1 = 24 and Q3 = 58, what is the Interquartile Range (IQR)?

Ans: _____

4 What percent of data in a Box Plot falls above Q1 (the 25th percentile)?

Ans: _____

5 Find the 5-number summary and interquartile range (IQR) of the 9 test scores: 68, 72, 75, 78, 82, 85, 88, 92, 96.

Ans: _____

6 Find Q1, the median, Q3, and the IQR of the 8 speeds (mph): 42, 45, 48, 50, 52, 55, 58, 60.

Ans: _____

7 Consistency Check: Player A has IQR = 4 pts and Player B has IQR = 19 pts. Who is more consistent?

Ans: _____

8 Outlier Rule (1.5 × IQR): If Q1 = 40, Q3 = 60 (IQR = 20), any score above Q3 + 1.5(IQR) is an outlier. Find that cutoff!

Ans: _____

9 For the data set 12, 15, 18, 22, 25, 28, 34, find the first quartile (Q1), third quartile (Q3), and IQR (Q3 - Q1).

Ans: _____

10 200 students take a test. The box plot has Min=50, Q1=70, Med=80, Q3=90, Max=100. How many students scored between 70 and 90?

Ans: _____

Section 9.3 — Practice B: Modeling Real Life (Quartiles, Interquartile Range (IQR) & Box Plots)

Textbook Methods: Method 1: Split Ordered Data into Lower & Upper Halves | Method 2: [1Q], [MED], [3Q] & [IQR] Keys

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: INTERPRETING QUARTILE PERCENTAGES ON A BOX PLOT (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A box-and-whisker plot of 40 test scores has Min = 60, Q1 = 72, Med = 80, Q3 = 88, Max = 98. How many students scored between 72 and 88?

METHOD 1 (+10 pts — 1st Token): Four 25% Quartile Sections

A box plot divides data into four 25% sections. The box (Q1 to Q3) holds the middle 2 sections: 25% + 25% = 50% of 40 = 20 students.

COUNT-01 Way 1: Split 40 student pegs into 4 equal quarters of 10 pegs each -> the box holds 2 quarters = 20 students.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Count 10 Students per Quartile Quarter

40 students ÷ 4 quartiles = 10 students per section. From Q1 (72) to Med (80) is 10 students + Med (80) to Q3 (88) is 10 = 20 students!

COUNT-01 Way 2: Measure the box width from Q1 = 72 to Q3 = 88 -> IQR = 88 - 72 = 16 points!

SC-6 Keystrokes: 25 ENTER 2 ×

Answer: 50% of the data = 20 students (IQR = 16)

SECTION 9.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Archery Tournament Scores (n=7): Ordered scores are 40, 45, 50, 55, 60, 65, 70. Find the 5-Number Summary (Min, Q1, Med, Q3, Max) and the IQR.

Ans: _____

2 Battery Life Test (hours, n=8): Ordered battery times are 6, 7, 8, 9, 11, 12, 14, 15. Find Q1, Median, Q3, and the Interquartile Range (IQR).

Ans: _____

3 Robotics Autonomous Points (n=9): Unordered scores are 30, 15, 45, 25, 35, 20, 50, 40, 30. Sort them first, then find Q1, Median, Q3, and IQR.

Ans: _____

4 Reading a Box-and-Whisker Plot: A box plot of 60 tree heights has Min = 12 ft, Q1 = 18 ft, Med = 24 ft, Q3 = 31 ft, Max = 40 ft. Find the Range and the IQR.

Ans: _____

5 Quartile Percentage Rule: In the tree height box plot from Problem 4 (60 trees total), what percent (and how many trees) are taller than Q1 (18 ft)?

Ans: _____

6 Top 25% Whisker: In the same 60-tree forest (Q3 = 31 ft, Max = 40 ft), how many trees are in the top quartile between 31 ft and 40 ft?

Ans: _____

7 Comparing Two Classes' IQRs: Class A has IQR = 8 points and Class B has IQR = 19 points. Which class had more consistent scores in the middle 50%?

Ans: _____

8 1.5 × IQR Outlier Fence Test: A data set has Q1 = 20, Q3 = 30, so IQR = 10. An upper outlier is any value above Q3 + 1.5(IQR). Calculate this upper cutoff!

Ans: _____

9 Daily Push-Up Challenge (n=6): 18, 22, 25, 29, 34, 40. Find the 5-number summary and IQR.

Ans: _____

10 Why IQR Beats Range: In {10, 12, 14, 16, 18, 20, 100}, the outlier 100 makes Range = 90. Find Q1, Q3, and IQR to see how IQR ignores the extreme ends!

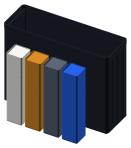
Ans: _____

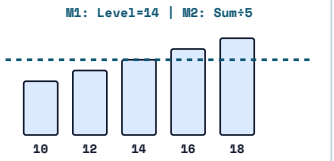
Section 9.4 — Practice A: Mean Absolute Deviation (MAD)

Key Vocabulary: mean absolute deviation (MAD) • deviation from the mean (|x - x-bar|) • measure of variation • consistency

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (COUNT-01: COUNTING & STATS STUDIO)





DEMONSTRATE BOTH METHODS WITH COUNT-01 BLOCKS:

Block Way 1 (+10 pts): Set a horizontal bar at Mean = 14 across COUNT-01 towers [10,12,14,15,19]; count gap blocks [4,2,0,1,5] -> 12 ÷ 5 = 2.4.

Block Way 2 (+20 pts): Check balance on COUNT-01: gap blocks below 14 (4 + 2 = 6) equal gap blocks above 14 (1 + 5 = 6) -> 12 ÷ 5 = 2.4!

Both Ways Proof: Summing all 5 absolute gap towers (12) and doubling the one-sided balance (6 + 6 = 12) both give MAD = 2.4!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the Mean and Mean Absolute Deviation (MAD) of 10, 12, 14, 16, 18.

METHOD 1 (+10 pts — 1st Token): 3-Step MAD Table by Hand

1. Mean = 70÷5 = 14. 2. Distances from 14: |10-14|=4, |12-14|=2, |14-14|=0, |16-14|=2, |18-14|=4. 3. MAD = 12 ÷ 5 = 2.4.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Symmetric Tower Pairs & [g] [MAD]

Sum of distances below mean (4+2 = 6) equals sum above mean (2+4 = 6) -> 12 ÷ 5 = 2.4. Confirm with [g] [MAD]!

SC-6 Keystrokes: `g` `CLΣ` `10` `Σ+` `12` `Σ+` `14` `Σ+` `16` `Σ+` `18` `Σ+` `g` `x̄` `g` `MAD`

Answer: Mean = 14, MAD = 2.4

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 9.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: 3-Step Distance Table (Mean -> |x - x-bar| -> Average) | Method 2: [Σ+] & [g]..

1 Find the mean and the mean absolute deviation (MAD) of the data set: 2, 4, 6.

Ans: _____

2 Find the mean and the mean absolute deviation (MAD) of the data set: 10, 12, 14, 16, 18.

Ans: _____

3 Find the mean and the mean absolute deviation (MAD) of the data set: 5, 8, 8, 10, 14.

Ans: _____

4 What is the MAD of 7, 7, 7, 7, 7 (5 identical towers)?

Ans: _____

5 Find the Mean and MAD of: 20, 24, 28, 32, 36.

Ans: _____

6 Find the Mean and MAD of the 6 sprint times: 12.4, 13.0, 12.6, 13.2, 12.8, 14.0.

Ans: _____

7 Find the Mean and MAD of quiz scores: 82, 85, 88, 90, 95.

Ans: _____

8 Team A has Mean = 50, MAD = 2.1. Team B has Mean = 50, MAD = 8.4. Which team's scores are more spread out?

Ans: _____

9 For the data set 6, 10, 11, 13, find the mean (10) and calculate the mean absolute deviation (MAD).

Ans: _____

10 Data Analysis Capstone: For the data set {5, 8, 8, 10, 14}, find the mean, median, mode, range, IQR, and MAD.

Ans: _____

Section 9.4 — Practice B: Modeling Real Life (Mean Absolute Deviation (MAD))

Textbook Methods: Method 1: 3-Step Distance Table (Mean -> |x - x-bar| -> Average) | Method 2: [Σ+] & [g] [MAD] Verification

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: TRACK SPRINT CONSISTENCY (12.4, 13.0, 12.6, 13.2, 12.8, 14.0 SEC) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A sprinter's 100m times are 12.4, 13.0, 12.6, 13.2, 12.8, and 14.0 seconds. Find the Mean and Mean Absolute Deviation (MAD).

METHOD 1 (+10 pts — 1st Token): Find Mean (13.0 s) & Average the 6 Distances

Mean = 78.0 ÷ 6 = 13.0 s. Distances from 13.0: 0.6, 0.0, 0.4, 0.2, 0.2, 1.0. Sum of distances = 2.4 s -> MAD = 2.4 ÷ 6 = 0.40 s.

COUNT-01 Way 1: From Mean = 13.0, sum the 5 gap heights [0.6, 0.2, 0.0, 0.4, 0.8] = 2.0 -> 2.0 ÷ 5 = 0.40 s.

METHOD 2 (+20 pts DOUBLE — 2nd Token): One-Pass Statistical Entry ([g] [MAD])

Enter the 6 sprint times with [Σ+], press [g] [x-bar] -> 13.0 s, and press [g] [MAD] -> 0.40 s!

COUNT-01 Way 2: Gaps below 13.0 (0.6 + 0.2 = 0.8) match gaps above 13.0 (0.4 + 0.4 = 0.8); total 2.0 ÷ 5 = 0.40 s!

SC-6 Keystrokes: g | CLΣ | 12.4 | Σ+ | 13.0 | Σ+ | 12.6 | Σ+ | 13.2 | Σ+ | 12.8 | Σ+ | 14.0 | Σ+ | g | x̄ | g | MAD

Answer: Mean = 13.0 s, MAD = 0.40 s

SECTION 9.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 3D Print Nozzle Temps (°C): Four readings are 206, 208, 212, 214. Find the Mean (210°C), each distance from 210, and the Mean Absolute Deviation (MAD).

Ans: _____

2 Basketball Free-Throw Streaks: In 5 practices you made 7, 9, 10, 11, and 13 shots. Find the Mean and the MAD.

Ans: _____

3 Zero MAD Puzzle: Five identical blocks each weigh 50 grams: 50, 50, 50, 50, 50. Without calculating, what is the Mean and what is the MAD? Why?

Ans: _____

4 Comparing Two Archers' Consistency: Archer A's scores are 8, 9, 9, 10 (Mean=9, MAD=0.5). Archer B's scores are 6, 8, 10, 12 (Mean=9). Find Archer B's MAD. Who is more consistent?

Ans: _____

5 Quiz Score Distances: Five scores are 76, 82, 85, 88, 94. Find the Mean and the MAD.

Ans: _____

6 Drone Hover Altitude (m): Six hover readings are 15, 16, 14, 15, 17, 13 meters. Find the Mean and MAD.

Ans: _____

7 Summer Daily Highs (°F): 80, 84, 86, 90. Find the Mean (85°F), list the 4 absolute deviations |x - 85|, and compute the MAD.

Ans: _____

8 Why Absolute Value Bars Matter in MAD: For {10, 15, 20}, the mean is 15. What happens if you add the signed deviations (-5) + 0 + (+5) WITHOUT absolute value bars?

Ans: _____

9 Coffee Shop Wait Times (min): 2, 3, 4, 5, 11. Find the Mean and the MAD.

Ans: _____

10 Full Statistical Profile: For the 5 robotics scores {12, 15, 15, 18, 20}, find the Mean, Median, Mode, Range, IQR, and MAD!

Ans: _____

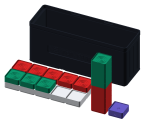
BLOCK KIT: BASE-10 • Ten-Frame Zero-Pair Counters

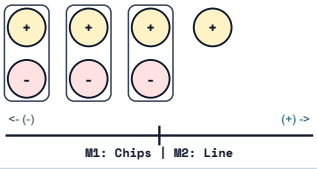
Section 10.1 — Practice A: Adding and Subtracting Rational Numbers

Key Vocabulary: additive inverse • zero pair • Same-Sign Sum Rule • Different-Sign Difference Rule • Add-the-Opposite (a - b..

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (BASE-10: TEN-FRAME ZERO-PAIR COUNTERS)





DEMONSTRATE BOTH METHODS WITH BASE-10 BLOCKS:

Block Way 1 (+10 pts): Place 8 Red (-) and 5 Yellow (+) counters on RACK-01; 5 zero pairs cancel to 0, leaving 3 Red (-3).

Block Way 2 (+20 pts): For 10 - (-4) on RACK-01, add 4 zero pairs and remove 4 Red (-) counters -> leaves 10 + 4 = 14 Yellow (+)!

Both Ways Proof: Removing 4 Red negative counters from zero pairs physically proves why subtracting -4 is adding +4!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Evaluate -8 + 5 and -6 + (-4) using counters and the SC-6 [+/-] key.

METHOD 1 (+10 pts — 1st Token): Zero-Pair Counter Cancellation

In -8 + 5, five (-1, +1) zero pairs cancel to 0, leaving 3 negatives (-3). In -6 + (-4), 6 negatives + 4 negatives = -10.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Big Ideas Sign Rules

Different signs (-8 + 5): subtract |8| - |5| = 3, keep sign of -8 -> -3. Same signs (-6 + -4): add |6| + |4| = 10, keep '-' -> -10.

SC-6 Keystrokes: `8` `+/-` `ENTER` `5` `+` `6` `+/-` `ENTER` `4` `+/-` `+`

Answer: -8 + 5 = -3; -6 + (-4) = -10

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 10.1 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Zero-Pair Counters / Number Line Jumps | Method 2..

1 Solve side-by-side: (a) -15 + 9 and (b) -15 + (-9).

Ans: _____

2 Solve: 14 + (-23).

Ans: _____

3 Taking Away a Penalty: Solve (a) 12 - (-8) and (b) -12 - (-8).

Ans: _____

4 Solve: -9 - 16. (Start at -9 and go 16 steps further negative!)

Ans: _____

5 Solve with decimals: -4.8 + (-3.5).

Ans: _____

6 Solve with decimals: 6.4 - 11.2.

Ans: _____

7 Solve with decimals: -7.5 - (-10.2).

Ans: _____

8 Solve with signed fractions: -3/4 - (-5/8).

Ans: _____

9 Temperature in Fairbanks is -18°F at dawn and rises to +7°F by noon. Find the change (7 - (-18)) 2 ways!

Ans: _____

10 Evaluate -2 1/2 + 4 1/4 - (-1 1/8) by rewriting subtraction as adding the opposite and using a common denominator.

Ans: _____

BLOCK KIT: BASE-10 • Ten-Frame Zero-Pair Counters

Section 10.1 — Practice B: Modeling Real Life (Adding and Subtracting Rational Numbers)

Textbook Methods: Method 1: Zero-Pair Counters / Number Line Jumps | Method 2: Absolute Value Rules & Add-the-Oppos..

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: SUBTRACTING NEGATIVES (10 - (-4) AND -3/4 - (-5/8)) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Evaluate 10 - (-4) and -3/4 - (-5/8) by rewriting subtraction as adding the opposite.

METHOD 1 (+10 pts — 1st Token): Add the Opposite (Keep-Change-Flip Sign)
10 - (-4) = 10 + (+4) = 14. For -3/4 - (-5/8): rewrite as -6/8 + (+5/8) = -1/8.

RACK-01 Way 1: Count steps on the vertical number rail from -4°C up to 0°C (+4) and up to +10°C (+10) -> +14°C.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Number Line Distance Between (End - Start)
How far is it from -4 up to +10 on a thermometer? 4 degrees to 0 plus 10 degrees above 0 = +14 degrees!

FRAC-01 Way 2: Convert -3/4 to -6/8 and add +5/8 -> 5 zero pairs cancel, leaving -1/8!

SC-6 Keystrokes: 10 ENTER 4 +/- - . 3 . 4 +/- ENTER . 5 . 8 +/- -

Answer: 10 - (-4) = 14; -3/4 - (-5/8) = -1/8

SECTION 10.1 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 Arctic Temperature Swing: At dawn the temperature is -14°F. By noon it warms up by 22°F (-14 + 22). What is the noon temperature?

Ans: _____

2 Submarine Double Dive: A research sub is at -45 meters and dives another 38 meters (-45 + (-38)). What is its new elevation?

Ans: _____

3 Add-the-Opposite Elevation Difference: Mt. Blanc is at +4,800 m and a nearby valley cave is at -200 m. Evaluate 4,800 - (-200) to find the vertical difference!

Ans: _____

4 Game Penalty Reversal: You have -15 points, and the referee cancels a -25 point penalty: -15 - (-25). Rewrite as addition and find your score!

Ans: _____

5 Signed Decimal Bank Ledger: Your account is at -$12.50. You deposit $30.00 and then buy a $8.75 lunch. What is your new balance?

Ans: _____

6 Signed Fraction Addition: Evaluate -5/8 + 3/4 using a common denominator of 8.

Ans: _____

7 Signed Fraction Subtraction: Evaluate -2/3 - 5/6 by rewriting as -4/6 + (-5/6) and simplifying.

Ans: _____

8 Scuba Ascent & Descent: A diver starts at -18.4 m, rises 7.9 m, and dives 5.5 m. Find her final depth.

Ans: _____

9 Thermometer Drop: The temperature drops from +6.5°C to -8.5°C. Evaluate -8.5 - 6.5 to find the signed temperature change.

Ans: _____

10 Three-Term Signed Challenge: Evaluate -14 - (-20) + (-9) - 5.

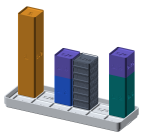
Ans: _____

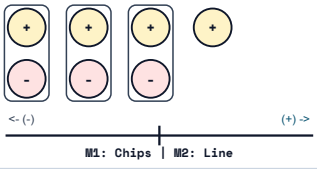
Section 10.2 — Practice A: Multiplying and Dividing Rational Numbers

Key Vocabulary: multiplicative sign rules • same signs (positive +) • different signs (negative -) • even/odd negative factors

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (BASE-10: ZERO-PAIR & FRACTION STUDIO)





DEMONSTRATE BOTH METHODS WITH BASE-10 BLOCKS:

Block Way 1 (+10 pts): Build 6 groups of 4 Red (-) counters on RACK-01 -> 24 Red counters (-24).

Block Way 2 (+20 pts): For (-6) × (-4), take 6 groups of 4 Red counters (-24) and flip every counter to its opposite color -> +24 Yellow!

Both Ways Proof: A negative multiplier flips counter color on RACK-01—one flip gives Red (-), two flips restore Yellow (+)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Compute 6 × (-4), (-6) × (-4), and (-5)² on your SC-6.

METHOD 1 (+10 pts — 1st Token): Same Signs = (+), Different Signs = (-)

6 × (-4) has different signs -> -24. (-6) × (-4) has same signs -> +24. (-5)² = (-5) × (-5) -> +25.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Count Negative Factors (Even vs. Odd)

1 negative factor (odd) -> negative (-24). 2 negative factors (even) -> positive (+24 and +25).

SC-6 Keystrokes: `6` `ENTER` `4` `+/-` `×` `5` `+/-` `x²`

Answer: 6×(-4) = -24, (-6)×(-4) = +24, (-.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 10.2 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Repeated Signed Addition / Sign-Flip Pattern | Me..

1 Compute: (a) 7 × (-8) and (b) (-7) × (-8).

Ans: _____

2 Compute: (a) -45 ÷ 9 and (b) -45 ÷ (-9).

Ans: _____

3 Evaluate (-5)². Explain why the square of a negative number is always positive.

Ans: _____

4 Evaluate (-2)³. Explain why the cube of a negative number is negative.

Ans: _____

5 Multiply signed decimals: (-3.5) × (-2.4).

Ans: _____

6 Divide signed decimals: -14.4 ÷ 1.2.

Ans: _____

7 Multiply signed fractions: (-2/3) × (9/10).

Ans: _____

8 Divide signed fractions: -5/6 ÷ (-5/12).

Ans: _____

9 Evaluate the three-factor product (-4) × (-5) × (-3). How does counting negative signs determine the sign?

Ans: _____

10 A submarine dives 360 ft in 8 minutes (-45 ft/min). Confirm the total change in 8 min and the change per minute!

Ans: _____

Section 10.2 — Practice B: Modeling Real Life (Multiplying and Dividing Rational Numbers)

Textbook Methods: Method 1: Repeated Signed Addition / Sign-Flip Pattern | Method 2: Count Negative Factors (Even = +,..

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: DIVING RATE DIVISION (-5/6 M ÷ -5/12 M/MIN) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: A submersible descends to -5/6 kilometer at a rate of -5/12 kilometer per minute. How many minutes did the descent take?

METHOD 1 (+10 pts — 1st Token): Same-Sign Division Rule + Multiply by Reciprocal

Negative ÷ Negative = Positive (+). Multiply by the reciprocal: (-5/6) × (-12/5) = +(5 × 12)/(6 × 5) = +12/6 = 2 minutes.

RACK-01 Way 1: Count how many (-45 m) steps make a (-90 m) change -> 2 positive steps (+2 min).

METHOD 2 (+20 pts DOUBLE — 2nd Token): Common Denominator Twelfths

Rewrite -5/6 as -10/12 km. Divide: (-10 twelfths) ÷ (-5 twelfths/min) = (-10) ÷ (-5) = +2 minutes!

RACK-01 Way 2: Even number of negative signs in (-90) ÷ (-45) flips to positive -> +2 minutes!

SC-6 Keystrokes: [.] [5] [.] [6] [+/-] [ENTER] [.] [5] [.] [12] [+/-] [÷]

Answer: +2 minutes

SECTION 10.2 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Submarine Descent Rate: A submarine descends at -15 feet per second for 8 seconds. Multiply 8 × (-15) to find its elevation change.**

Ans: _____

2 **Reversing a Loss (Negative × Negative): Losing 4 penalties of -10 points each is (-4) × (-10). Find the point change and explain the sign rule.**

Ans: _____

3 **Mean Temperature Drop: Over 6 hours, the temperature changed by -27°F in all. Divide -27 ÷ 6 to find the average change per hour.**

Ans: _____

4 **Signed Square vs. Cube: Evaluate $(-4)^2$ and $(-4)^3$. Why is $(-4)^2$ positive while $(-4)^3$ is negative?**

Ans: _____

5 **Signed Fraction Multiplication: Multiply (-3/4) × (-8/9) by cross-canceling common factors and applying the Same-Sign Rule.**

Ans: _____

6 **Signed Fraction Division: Divide (-7/8) ÷ (7/16) by multiplying by the reciprocal.**

Ans: _____

7 **Signed Decimal Product: Evaluate (-2.5) × (-1.6) × (-3). First count the 3 negative factors—will the answer be positive or negative?**

Ans: _____

8 **Glacier Retreat Rate: A glacier retreated -14.4 meters over 4.5 years. Divide -14.4 ÷ 4.5 to find the average yearly change.**

Ans: _____

9 **Four-Factor Sign Rule: Evaluate (-2) × (-3) × (-4) × (-5). Since there are 4 (even) negative factors, what is the sign and product?**

Ans: _____

10 **Signed Order of Operations: Evaluate $(-6)^2 - 4 \times (-5) + (-18) \div (-3)$.**

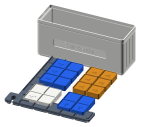
Ans: _____

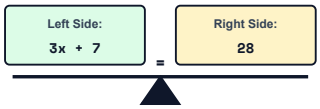
Section 10.3 — Practice A: Solving Two-Step Equations (ax + b = c)

Key Vocabulary: two-step equation (ax + b = c) • reverse order of operations • undo addition/subtraction first • undo multiplic..

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (ALGB-01: ALGEBRA BOXES & STAGE)





M1/M2: Step 1: 28-7=21 -> Step 2: 21÷3..

DEMONSTRATE BOTH METHODS WITH ALGB-01 BLOCKS:

Block Way 1 (+10 pts): On the ALGB-01 Balance Stage, remove 7 unit cubes from both pans of 3x + 7 = 28 (3x = 21), then split into 3 x-boxes (x = 7).

Block Way 2 (+20 pts): Load 7 cubes into each of the 3 x-boxes + 7 unit cubes on ALGB-01 -> 21 + 7 = 28 (Balances both pans!).

Both Ways Proof: Unwrapping constant cubes first then dividing boxes (Method 1) is verified by refilling the 3 x-boxes (Method 2)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Solve 3x + 7 = 28 using inverse operations and check your solution.

METHOD 1 (+10 pts — 1st Token): Undo Add/Subtract First, Multiply/Divide Second

In 3x + 7 = 28: Step 1: Subtract 7 from both sides -> 3x = 21. Step 2: Divide both sides by 3 -> x = 7.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Substitute x = 7 Back to Prove

Plug x = 7 into 3x + 7: 3(7) + 7 = 21 + 7 = 28 (Matches Right Pan!).

SC-6 Keystrokes: 28 | ENTER | 7 | - | 3 | ÷

Answer: x = 7 (Step 1: 3x = 21)

1 Method = +10 pts | Both Methods = +20 pts!

SECTION 10.3 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: 2-Step Inverse Operations | Method 2: Algebra Til..

1 Solve 4x + 9 = 45. Write the Step 1 value (4x = ?) and final x.

Ans: _____

2 Solve the two-step equation 5x - 12 = 38 by undoing subtraction first and multiplication second. Check your solution.

Ans: _____

3 Solve 6y + 11 = 59.

Ans: _____

4 Solve 8n - 15 = 49.

Ans: _____

5 Solve m / 4 + 6 = 15. (Subtract 6 first, then multiply by 4!)

Ans: _____

6 Solve k / 3 - 5 = 7.

Ans: _____

7 Solve -3x + 14 = 2.

Ans: _____

8 Solve 2.5w + 8 = 23.

Ans: _____

9 A repair tech charges $35 flat plus $25/hr (25h + 35 = 160). Find hours h by unwrapping AND plug-in check!

Ans: _____

10 You save $15/week toward a $120 skateboard and already have $45 (15w + 45 = 120). How many weeks w?

Ans: _____

BLOCK KIT: ALGB-01 • Algebra Boxes & Stage

Section 10.3 — Practice B: Modeling Real Life (Solving Two-Step Equations (ax + b = c))

Textbook Methods: Method 1: 2-Step Inverse Operations | Method 2: Algebra Tile Balance Scale & Substitution Proof

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: TEMPERATURE DROP EQUATION (-3X + 14 = 2) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: The temperature starts at 14°F and drops 3°F per hour (-3x + 14 = 2). After how many hours x does it reach 2°F?

METHOD 1 (+10 pts — 1st Token): Two-Step Inverse Unwrapping with Negative Coefficient

Step 1: Subtract 14 from both sides: -3x = 2 - 14 = -12. Step 2: Divide both sides by -3: x = (-12) ÷ (-3) = +4 hours.

ALGB-01 Way 1: Remove 35 setup cubes from both pans of 30x + 35 = 155 (30x = 120), then divide into 30 boxes -> x = 4.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Hour-by-Hour Countdown Table & Substitution Check

Start 14°F -> Hr 1: 11°F -> Hr 2: 8°F -> Hr 3: 5°F -> Hr 4: 2°F! Check: -3(4) + 14 = -12 + 14 = 2°F.

COUNT-01 Way 2: From a 155-total bar, cut off the 35 fixed block (120 left) and partition into 30-unit blocks -> 4 blocks!

SC-6 Keystrokes: 2 ENTER 14 - 3 +/- ÷

Answer: x = 4 hours

SECTION 10.3 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS

1 Way = +10 pts | 2 Ways = +20 pts Double

1 **Go-Kart Track Admission: Admission is $8 plus $6 per lap r. You spend $38 in all (6r + 8 = 38). Solve the two-step equation to find how many laps r you drove!**

Ans: _____

2 **3D Print Service Quote: A print shop charges a $12 setup fee plus $4 per hour h. Your bill is $48 (4h + 12 = 48). Solve for h and check!**

Ans: _____

3 **Savings Goal Countdown: You have $35 saved and save $15 per week w to buy a $140 bike (15w + 35 = 140). How many weeks w will it take?**

Ans: _____

4 **Coupon Discount Two-Step: You buy 4 identical books and use a $10 gift card, paying $46 out of pocket (4b - 10 = 46). Find the price b of each book.**

Ans: _____

5 **Negative Coefficient Equation: Solve -4x + 18 = -6 by subtracting 18 first and then dividing by -4. Check by substituting back!**

Ans: _____

6 **Division & Addition Two-Step: Solve (n ÷ 5) + 9 = 16 by subtracting 9 first and then multiplying by 5.**

Ans: _____

7 **Scuba Air Tank Pressure: A tank starts at 3,000 psi and loses 40 psi per minute m until 600 psi remains (3000 - 40m = 600). How many minutes m was the dive?**

Ans: _____

8 **Fractional Coefficient Two-Step: Solve (2/3)x - 5 = 11 by adding 5 first (to get 16) and multiplying by the reciprocal 3/2.**

Ans: _____

9 **Rectangle Perimeter Two-Step: A rectangle has length 18 cm and perimeter 56 cm (2w + 36 = 56). Solve for the width w!**

Ans: _____

10 **Decimal Two-Step Taxi Fare: A ride costs $3.50 base fare plus $2.25 per mile m. The total fare is $21.50 (2.25m + 3.50 = 21.50). How many miles m was the ride?**

Ans: _____

Section 10.4 — Practice A: Theoretical Probability, Combinations & Permutations

Key Vocabulary: experiment • outcomes • event • sample space • theoretical probability P(E) • complement P(not E) • factoria..

Name: _____ Date: _____ Score: ____ / 10

BLOCK KIT: 2 WAYS TO BUILD IT (PROB-01: PROBABILITY DICE & SPINNER STUDIO)





M1: Grid/Tree | M2: Count Rule

DEMONSTRATE BOTH METHODS WITH PROB-01 BLOCKS:

Block Way 1 (+10 pts): Draw from 10 PROB-01 tiles (4 Blue / 10 total = 2/5 = 40%); arrange 5 distinct tiles in 5! = 120 ordered rows.

Block Way 2 (+20 pts): Compare ordered 3-tile podiums (10×9×8 = 720) vs. unordered 3-tile groups (divide by 3! = 6 rearrangements -> 120)!

Both Ways Proof: Grouping identical 3-tile sets on the PROB-01 tray shows why every 1 combination has 3! = 6 permutations!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: A 10-sector spinner has 4 Blue, 3 Gold, and 3 Red sectors. Find P(Blue) in 3 forms, and find how many ways to line up 5 trophies (5!).

METHOD 1 (+10 pts — 1st Token): P(E) = Favorable / Total & Complement 1 - P(E)

P(Blue) = 4/10 = 2/5 = 0.4 = 40%. Complement P(Not Blue) = 1 - 2/5 = 3/5 (60%). For 5 trophies: 5! = 5×4×3×2×1 = 120.

METHOD 2 (+20 pts DOUBLE — 2nd Token): SC-6 Probability Keys ([g][SIMP], [g][x!])

Enter [.] 4 [.] 10 [g][SIMP] -> 2/5 ([FDISP] -> 0.4), and 5 [g][x!] -> 120 arrangements.

SC-6 Keystrokes: `.` `4` `.` `10` `g` `SIMP` `5` `g` `x!`

Answer: P(Blue) = 2/5 = 0.4 = 40%; 5! = 1.. 1 Method = +10 pts | Both Methods = +20 pts!

SECTION 10.4 PRACTICE A — SOLVE USING METHOD 1 (+10 PTS) OR BOTH METHODS (+20 PTS DOUBLE)

Methods: Method 1: Favorable ÷ Total & Complement Rule | Method 2: F..

1 Roll a fair 6-sided die. Find P(rolling a 5) as a fraction, and P(rolling an even number) in simplest form.

Ans: _____

2 A 10-sector spinner has 4 Blue, 3 Gold, 3 Red. Find P(Blue) as a simplified fraction, decimal, and percent.

Ans: _____

3 Using the same 10-sector spinner, find P(NOT Red) as a fraction and percent.

Ans: _____

4 In how many different orders can 4 students line up at the classroom door? Evaluate 4! (4 factorial).

Ans: _____

5 In how many different ways can 6 trophies be arranged in a row on a shelf? Evaluate 6!.

Ans: _____

6 In how many ways can a coach choose a 3-player group from 8 players when order does NOT matter (combinations)?

Ans: _____

7 In how many ways can 1st, 2nd, and 3rd place ribbons be awarded to 8 finalists when order DOES matter (permutations)?

Ans: _____

8 Two Independent Events: You flip a coin (1/2) AND roll a 6-sided die (P(>4) = 2/6 = 1/3). Multiply to find P(Heads and >4)!

Ans: _____

9 Compare choosing 3 pizza toppings from 10 options (combinations) vs. awarding 1st, 2nd, and 3rd place to 10 runners (permutations).

Ans: _____

10 Probability & Counting Capstone: Evaluate 5! and find how many 2-person teams can be chosen from 5 friends.

Ans: _____

Section 10.4 — Practice B: Modeling Real Life (Theoretical Probability, Combinations & Permutations)

Textbook Methods: Method 1: Favorable ÷ Total & Complement Rule | Method 2: Fundamental Counting Principle, [nPr] & [n..

Name: _____ Date: _____ Score: ____ / 10

MODELING REAL LIFE WORKED EXAMPLE — MODELING REAL LIFE: PIZZA TOPPINGS (COMBINATIONS) VS. RACE PODIUM (PERMUTATIONS) (SOLVE 2 WAYS = +20 PTS DOUBLE)

Problem: Compare choosing 3 pizza toppings from 10 options (order doesn't matter) vs. awarding 1st, 2nd, and 3rd place to 10 runners (order matters).

METHOD 1 (+10 pts — 1st Token): Permutations (Order Matters -> 10 × 9 × 8)

Podium (Order matters): 10 choices for 1st × 9 for 2nd × 8 for 3rd = 720 permutations (10 [ENTER] 3 [g] [nPr] = 720).

PROB-01 Way 1: Fill 3 ordered podium slots from 10 tiles: 10 × 9 × 8 = 720 ordered permutations.

METHOD 2 (+20 pts DOUBLE — 2nd Token): Combinations (Divide Out 3! = 6 Duplicate Orders)

Pizza (Order doesn't matter): Divide 720 by 3! (3×2×1 = 6 ways to arrange the same 3 toppings): 720 ÷ 6 = 120 combinations ([f] [nCr])!

PROB-01 Way 2: Since any 3 chosen tiles can be rearranged in 3! = 6 ways, 720 ÷ 6 = 120 unordered combinations!

SC-6 Keystrokes: 10 ENTER 3 f nCr 10 ENTER 3 g nPr

Answer: 120 combinations vs. 720 permutations

SECTION 10.4 PRACTICE B — MODELING REAL LIFE & MULTI-METHOD APPLICATIONS 1 Way = +10 pts | 2 Ways = +20 pts Double

1 **20-Sided Die (D20) Probability: You roll a fair 20-sided die numbered 1 to 20. Find the theoretical probability P(rolling a multiple of 5) as a fraction, decimal, and percent.**

Ans: _____

2 **Complement of an Event: If the probability of rain tomorrow is 35% (7/20), what is the probability that it does NOT rain P(not rain)?**

Ans: _____

3 **Marble Bag Draw: A bag holds 6 blue, 5 green, and 4 gold marbles (15 total). Find P(green) and P(not gold) in simplest form.**

Ans: _____

4 **Fundamental Counting Principle: A custom skateboard offers 5 deck designs, 4 truck colors, and 6 wheel sets. Multiply to find how many different skateboards you can build!**

Ans: _____

5 **Playlist Song Shuffle (Factorial): You put 5 favorite songs on a mini-playlist. In how many different orders (5!) can the shuffle play all 5 songs?**

Ans: _____

6 **Relay Batting Order (Permutations): A coach picks 3 batters in order (1st, 2nd, 3rd) from 7 players. Since order matters, compute 7 × 6 × 5 (7 nPr 3).**

Ans: _____

7 **Science Lab Partner Trio (Combinations): You choose a 3-person lab group from 7 students. Since order does NOT matter, compute 7 nCr 3.**

Ans: _____

8 **Smoothie Fruit Pairs: A smoothie bar has 8 fruits and you pick any 2 fruits to blend (order doesn't matter: 8 nCr 2). How many 2-fruit combos are possible?**

Ans: _____

9 **Experimental Prediction: A spinner lands on Gold 3/8 of the time. If you spin it 160 times, how many times do you expect it to land on Gold?**

Ans: _____

10 **Two-Coin & Die Compound Event: You flip a fair coin (2 outcomes) and roll a 6-sided die (6 outcomes). How many total outcomes are in the sample space, and what is P(Heads and 6)?**

Ans: _____